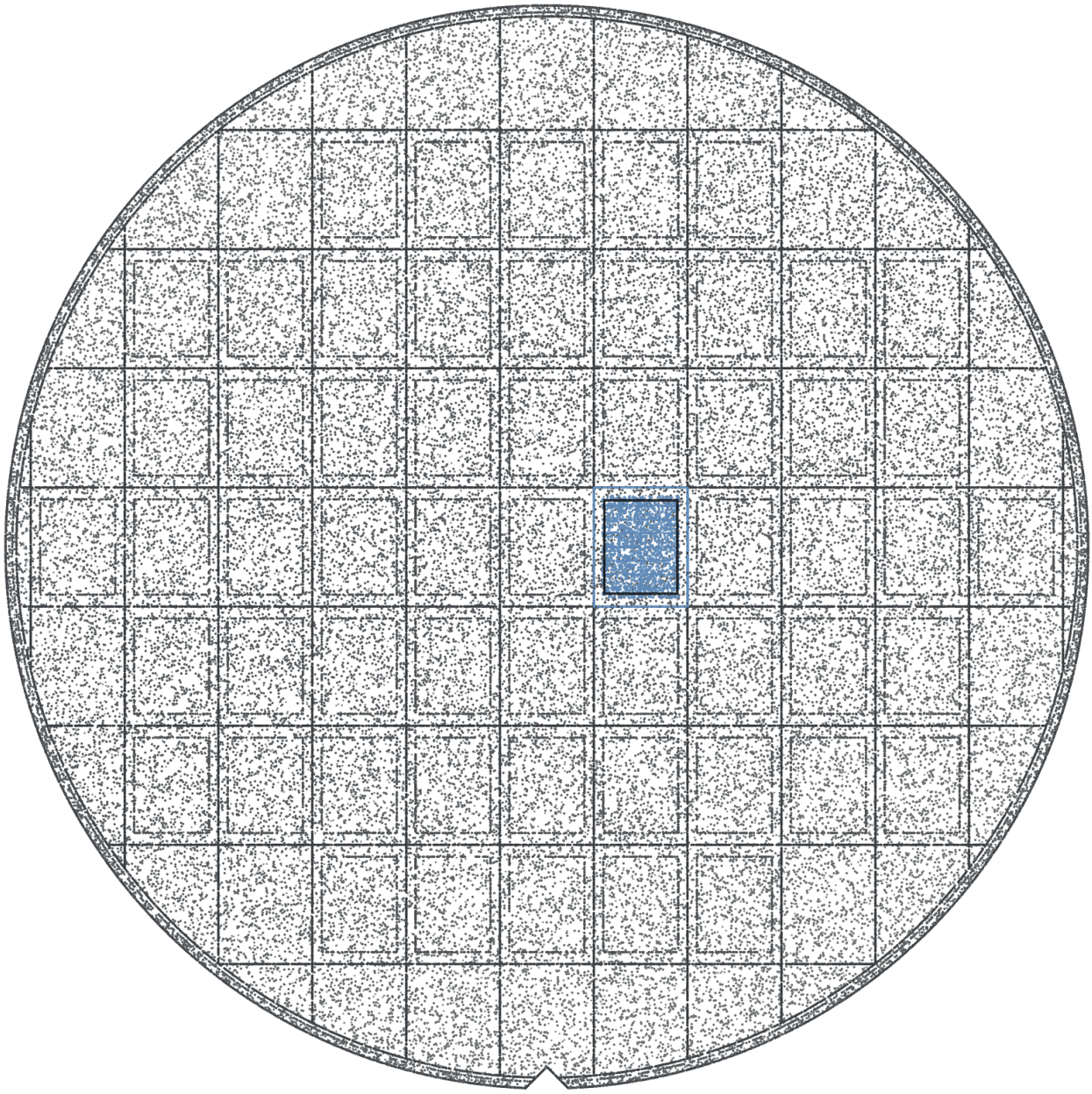


CHIP SUPPLY CHAIN

How AI chips are made

Every stage of the supply chain, and who controls it.

Yashvardhan Sharma with Claude

The Chain: seventeen chapters.

Printed from chipsupplychain.org · Updated 21 September 2026

HOW TO READ A CHAPTER

Each of the fifteen stage chapters opens with a chokepoint card.

Concentration. How many firms credibly supply this stage: Extreme, High, Moderate or Low.

Substitutability. How hard it would be for a well-funded state or firm to build an alternative: Easy, Moderate, Hard or Very hard, rated against four tests.

Price or market size. What one unit costs, or what the stage earns in a year.

The card also gives where China and the United States stand. The chapters run in process order.

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CHAPTER 01
STAGE 1 OF 15

Chip Design, EDA and IP

The cheapest stage in the chain to fund and one of the most concentrated. Three software firms sell the tools used to design every leading-edge AI accelerator.

IN PLAIN TERMS

Chip design turns an idea for a new chip into a complete blueprint. A chip holds billions of transistors, tiny electrical switches, and the blueprint fixes where each one sits and how it is wired to the others. The chip works in ticks, and a signal has to cross those wires within one tick, less than a billionth of a second. Software from three companies, Synopsys, Cadence and Siemens, lays out a plan like that, checks it and signs it off, and a chip factory accepts only designs that have passed those tools. The factory prints each layer of the design onto silicon through a stencil, and changing the plan after that costs a fresh set of stencils and several months. Withhold the software and no new chip can be designed.

IN SHORT

Three vendors sell every complete design flow used at the leading edge [3]. Design is cheap next to what it commits: tens of millions of dollars of engineering and masks decide what tens of billions of dollars of factory capacity will make [1]. It is the control a government can impose fastest, and the one China is closest to neutralizing, though CSET still finds no complete domestic flow [17]. The largest buyers now design their own parts, with Broadcom doing the implementation and the buyer's name on the product [11].

3 EDA Market Primer. SemiAnalysis · 21 May 2026 [B] newsletter.semianalysis.com/p/eda-market-primer

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due To Photomask Sets, Verification, and Validation. SemiAnalysis · 24 July 2022 [B] newsletter.semianalysis.com/p/the-dark-side-of-the-semiconductor

17 Semiconductors: More U.S. Leverage, More Bad News For Beijing (Part 3). Center for Security and Emerging Technology (CSET) · 25 October 2021 [A] cset.georgetown.edu/article/semiconductors-more-u-s-leverage-more-bad-news-for-beijing-par…

11 OpenAI and Broadcom announce strategic collaboration to deploy 10 gigawatts of OpenAI-designed AI accelerators. Broadcom [A] investors.broadcom.com/news-releases/news-release-details/openai-and-broadcom-announce-str…

CHOKEPOINT CARD | EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Hard Only three complete sets of chip design software exist, and a fourth would have to be written from nothing and approved by a foundry.	A 3 nm photomask set costs about $40M, and startups have designed and made a 7 nm chip for $50-75M

WHO LEADS

- US **Synopsys** $7.05B revenue in the fiscal year ended 31 October 2025, Ansys included from July; 90%+ of the timing-signoff market
- US **Cadence** $5.30B revenue in calendar 2025; 55-60% of the hardware emulator market
- DE **Siemens EDA** $2.2-2.5B revenue in 2025; 85%+ of physical verification, the final layout check
- GB **Arm** About 50% of processor compute at the top cloud firms, fiscal 2026
- US **Broadcom** $16.7B of AI chip revenue in the third quarter of fiscal 2026

WHERE IT IS MADE

- US **United States** Synopsys, Cadence, Broadcom, Marvell, Nvidia, AMD; Siemens EDA's main sites
- GB **United Kingdom** Arm processor and interconnect designs, licensed from Cambridge
- TW **Taiwan** Alchip, Global Unichip and MediaTek: design services that carry a chip to the factory
- IL **Israel** Annapurna Labs, the AWS design house behind Trainium and Graviton
- IN **India** Nearly 20% of the world's chip design engineers, on the Indian government's count

WHY SUBSTITUTION IS SLOW

Three complete sets of chip design software exist today, and tools from one set cannot be combined with tools from another. The foundry that makes the chip approves exact versions of each tool. It accepts the check that every signal arrives on time only from the tool it has approved for that check. A newcomer would have to write a fourth set from nothing. It would then spend five to ten years finding the rare ways a chip can fail, which the approved tools already catch. CSET at Georgetown University judges that Chinese design tools cannot yet handle designs for the newest chips.

WHERE CHINA STANDS

China designs competitive AI accelerators, but it cannot yet complete a design for the newest chips on Chinese software alone. CSET at Georgetown University judges that Chinese design tools cannot support designs for the newest chips. The US government stopped sales of American and German design software to China for five weeks in

2025, then allowed them again.	WHERE THE US STANDS The United States has the two largest design-software vendors and the leading custom-chip design houses. Siemens EDA is German-owned, but the American technology inside it still needs US export licenses.

Design is the only stage in this chain with no factory, and the one a foreign government can shut off fastest. In 2025 one did. For five weeks, every Chinese chip designer lost the software that turns a design into something a factory can print.

How it works

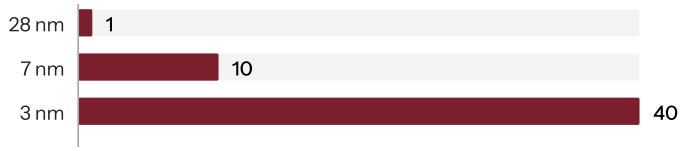
Design runs in five stages, each one's output the next one's input, all of it in software.

- **Architecture.** Engineers fix the shape of the chip: how wide the block that multiplies numbers is, how much fast on-chip memory sits beside it, how many high-bandwidth memory stacks line the edges. The output is a plan.
- **RTL.** Engineers write the plan as register-transfer-level code, RTL, in the hardware language SystemVerilog. A register is a small store on the chip that holds one number, and the clock is the tick that moves every register forward together. The code says what every register holds on every tick, and the output is a text description of the whole chip.
- **Verification.** The biggest team on most projects checks that the code does what the plan says. They run it in simulation, prove parts of it correct with mathematics, and load it into emulators, cabinets of reprogrammable chips that behave like the design, so the chip's own software can boot before any silicon exists.
- **Physical design.** Software swaps each piece of the code for a logic cell, a small ready-made circuit the foundry has already proved it can make, then places the cells on the chip, wires them together and proves that every signal arrives in time.
- **Tape-out.** The handover. Software turns the placed layout into pattern files for the mask shop. First it pre-distorts each shape so that it prints as drawn, which is optical proximity correction; then it cuts the layout into the files a mask writer reads, which is mask data preparation.

All that checking exists because a mistake costs a whole new mask set. One set costs more than $1 million at 28 nm, more than $10 million at 7 nm and about $40 million at 3 nm, and a whole leading-edge chip has gone from design to tape-out on TSMC 7 nm for $50 million to $75 million, everything included [1]. The nanometer labels name generations of the process, each finer than the last, and no longer measure anything.

1 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due To Photomask Sets, Verification, and Validation. SemiAnalysis · 24 July 2022 B newsletter.semianalysis.com/p/the-dark-side-of-the-semiconductor

What one photomask set costs, by node ($M). The 28 nm and 7 nm figures are minimums $M



Source: SemiAnalysis, The Dark Side of the Semiconductor Design Renaissance, July 2022

Variants and trade-offs

Merchant GPU versus custom ASIC

Nvidia designs Blackwell once and sells it to everyone, so the buyer gets mature software and a resale market. A custom accelerator, an ASIC built for one company's own workload, is narrower and tuned to that one workload. Google's seventh-generation tensor processing unit, Ironwood, carries 192 GB of memory per chip, six times its predecessor's, and delivers twice the performance per watt [2].

The EDA flow

Three complete flows exist, and they do not mix. A team commits to one vendor's set of tools because the foundry certifies its reference recipes against specific tool versions, and timing signed off in one tool is not accepted in another. Synopsys alone holds more than 90 percent of static timing analysis, the check that proves a chip will run at its rated speed [3], so inside each step of the flow the concentration is higher still.

Each leading vendor's share of its own design-software segment, 2025 %



Source: SemiAnalysis, EDA Market Primer, May 2026

Buy the IP or build it

Design houses license the processor cores, the interface controllers and the blocks that drive high-bandwidth memory. Arm supplies most of the processor side: $4.92 billion of revenue, $2.61 billion of it royalties, and roughly 50 percent of processor compute at the top cloud firms in fiscal 2026 [4]. The hardest block to buy is the memory interface, which has to hold its timing across thousands of microscopic solder bumps into a stacked memory die. RISC-V, the open instruction set that needs no license, handles the small control tasks on the chip, and Nvidia now uses it in every GPU [5].

2 Ironwood: The first Google TPU for the age of inference. Google · 9 April 2025 [A] blog.google/innovation-and-ai/infrastructure-and-cloud/google-cloud/ironwood-tpu-age-of-in...

3 EDA Market Primer. SemiAnalysis · 21 May 2026 [B] newsletter.semianalysis.com/p/eda-market-primer

4 Arm delivers record-breaking quarter and full-year results. Arm · 6 May 2026 [A] newsroom.arm.com/news/arm-q4-fye26-results

5 Behind The Scenes of SHD Group's 2026 RISC-V Market Forecast. RISC-V International · 19 June 2026 [A] riscv.org/blog/shd-forecast-2026

Chiplets

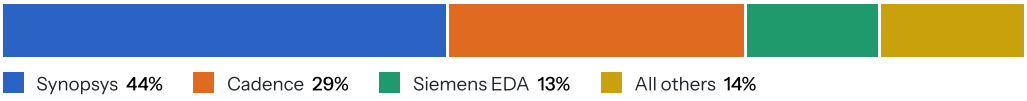
The reticle limit, the largest area a scanner can print in one exposure, ended the single-die accelerator. Modern parts are chiplets, several smaller dies wired together, and the UCIe standard lets those links come from different vendors: version 3.0 adds 48 and 64 GT/s data rates [6]. Broadcom's 3.5D XDSiP is the largest of these, packing more than 6,000 mm2 of silicon and 12 high-bandwidth memory stacks into one package [7].

Who makes it

Design software and licensed blocks were an $18 billion market in 2025, and Synopsys, Cadence and Siemens EDA took more than 85 percent of it, with no other vendor above 5 percent in any core category [3].

Synopsys is the largest. Its own results put revenue at $7.054 billion for the fiscal year to 31 October 2025, with Ansys, bought that July, contributing $756.6 million [8]. Cadence, whose year ends in December, took $5.30 billion in calendar 2025 [9]. The two fiscal years do not line up, so any total for this market depends on which twelve months are counted and on how much of Ansys it includes.

Design software and licensed-block revenue share, 2025. Synopsys includes Ansys %



Source: SemiAnalysis, EDA Market Primer, May 2026

Design services, where one firm turns another's specification into a manufacturable chip, are narrower again, and most of the announced AI deals are Broadcom's. It booked $16.7 billion of AI semiconductor revenue in the third quarter of fiscal 2026, up 221 percent [10].

In October 2025 OpenAI and Broadcom said they would deploy 10 gigawatts of OpenAI-designed accelerators, with Broadcom turning the design into silicon and supplying the Ethernet networking around it [11]. In April 2026 Broadcom extended

6 Specifications | UCIe Consortium. UCIe Consortium [A] uciexpress.org/specifications

7 3.5D XDSiP Platform Technology. Broadcom · 4 December 2024 [A] docs.broadcom.com/doc/3-5d-xdsip-platform-technology

3 EDA Market Primer. SemiAnalysis · 21 May 2026 [B] newsletter.semianalysis.com/p/eda-market-primer

8 SYNOPSYS INC, Form 8-K current report for the period ended 2025-12-10 (8-K). U.S. Securities and Exchange Commission (filing by SYNOPSYS INC) · 10 December 2025 [A] sec.gov/Archives/edgar/data/883241/000119312525314200/d29055dex991.htm

9 CADENCE DESIGN SYSTEMS INC, Form 10-K annual report for the period ended 2025-12-31 (financial statement R115). U.S. Securities and Exchange Commission (filing by CADENCE DESIGN SYSTEMS INC) · 19 February 2026 [A] sec.gov/Archives/edgar/data/813672/000081367226000016/R115.htm

10 Broadcom Inc., Form 8-K current report for the period ended 2026-09-02 (8-K). U.S. Securities and Exchange Commission (filing by Broadcom Inc.) · 2 September 2026 [A] sec.gov/Archives/edgar/data/0001730168/000173016826000076/avgo-08022026x8kxex99.htm

11 OpenAI and Broadcom announce strategic collaboration to deploy 10 gigawatts of OpenAI-designed AI accelerators. Broadcom [A] investors.broadcom.com/news-releases/news-release-details/openai-and-broadcom-announce-str...

its Meta partnership to multiple gigawatts of MTIA, Meta's in-house chip, and the first 2 nm AI accelerator [12].

Below Broadcom and Marvell sit the Taiwanese houses that carry a customer's circuit list all the way to a finished mask set on TSMC's newest nodes; Alchip reported $992 million of revenue in 2025 [13]. The engineers are spread across more countries than the firms, and India alone holds nearly 20 percent of the world's chip designers [14].

The chokepoint

In late May 2025 the US Bureau of Industry and Security, the agency that writes export licenses, told the design-software vendors to stop selling to China. Cadence got its letter on 23 May: exporting design software classified 3D991 and 3E991 to any party in China, or to a Chinese military end user anywhere, now needed a license [15]. Synopsys got its letter on 29 May and a rescinding letter on 2 July, after trade talks in which Washington eased technology restrictions [16].

Washington can impose this control alone, and at once. It also speeds up the substitution it was meant to prevent: Synopsys's China revenue fell from 16 percent of the total in fiscal 2024 to 12 percent in fiscal 2025 [3].

CSET at Georgetown University finds that Chinese designers rely on US tools for all their chip designs, that US firms alone sell the full range of software a leading-edge chip needs, and that Chinese vendors cannot yet support designs at those nodes [17]. No domestic vendor has a certified, complete flow for a 5 nm-class accelerator, and the domestic flows match on features and fall short in the rare failure cases a foundry has already validated.

China can design an accelerator. It cannot finish one at the leading edge without foreign tools, or build what it designs without foreign equipment: SMIC's 7 nm capacity is small enough that Huawei has to choose between its phone chips and its AI chips [18].

Key evaluation criteria

- **Flow completeness.** How many steps one vendor covers, from the first line of code to final signoff.
- **Foundry certification.** Whether the foundry has validated that exact tool version against its reference flow and process design kit, the file set describing what the factory can print. Without that, the foundry refuses signoff.

12 Broadcom Announces Extended Partnership with Meta to Deploy Technology to Support Multi-Gigawatts of Meta's Custom Silicon, MTIA. Broadcom · 14 April 2026 [A] investors.broadcom.com/news-releases/news-release-details/broadcom-announces-extended-part…

13 Financials - Alchip. Alchip Technologies [A] alchip.com/en/Investors/financials

14 India's Semiconductor Vision Gathers Momentum with 3nm Chip Design and Large-Scale Talent Development Initiatives. Press Information Bureau, Government of India [A] pib.gov.in/PressReleasePage.aspx

15 CADENCE DESIGN SYSTEMS INC, Form 8-K current report for the period ended 2025-07-02 (8-K). U.S. Securities and Exchange Commission (filing by CADENCE DESIGN SYSTEMS INC) · 3 July 2025 [A] sec.gov/Archives/edgar/data/813672/000081367225000093/cdns-20250702.htm

16 SYNOPSYS INC, Form 8-K current report for the period ended 2025-07-02 (8-K). U.S. Securities and Exchange Commission (filing by SYNOPSYS INC) · 3 July 2025 [A] sec.gov/Archives/edgar/data/883241/000119312525155294/d80081d8k.htm

3 EDA Market Primer. SemiAnalysis · 21 May 2026 [B] newsletter.semianalysis.com/p/eda-market-primer

17 Semiconductors: More U.S. Leverage, More Bad News For Beijing (Part 3). Center for Security and Emerging Technology (CSET) · 25 October 2021 [A] cset.georgetown.edu/article/semiconductors-more-u-s-leverage-more-bad-news-for-beijing-par…

18 Pushing the Limits: Huawei's AI Chip Tests U.S. Export Controls. Center for Security and Emerging Technology (CSET) · 17 June 2024 [A] cset.georgetown.edu/publication/pushing-the-limits-huaweis-ai-chip-tests-u-s-export-controls

- **Verification throughput.** Simulation cycles and emulator capacity per day, which set the schedule more than anything else.
- **IP availability at node.** Whether the licensed blocks, the processor cores, the high-speed serial links, PCIe and the memory interface, have been proven in silicon on that process.
- **Design services depth.** Whether a partner can carry a customer's circuit list to a finished mask set on the newest node.
- **Legal exposure.** How much of the flow is US-origin technology, and so subject to export licensing.

CHAPTER
02
STAGE 2 OF 15

Silicon and Wafers

China makes almost all of the world's polysilicon and almost none of the 300 mm wafers. Five firms in Japan, Taiwan, Germany and South Korea make the 300 mm wafer every AI accelerator starts on.

IN PLAIN TERMS

Every chip is built on a wafer, a thin round slice of silicon about 30 centimeters across that carries hundreds of chips at once. Making one starts with sand, which is silicon bound to oxygen. The sand is refined until, of every hundred billion atoms, fewer than one is anything other than silicon. The silicon is melted, a small seed crystal is lowered to the surface and drawn slowly back up, and the melt freezes onto it as one long cylinder, a single crystal with its atoms in one unbroken orderly pattern from end to end. Saws cut the cylinder into discs, and each disc is polished until its surface is flat to within a few atoms, right out to the edge. Only five companies in the world make these wafers to that standard, and none of them is American or Chinese.

IN SHORT

The wafer is a cheap input and a concentrated one: five firms and $11.4 billion of revenue sit under every leading-edge chip [5] [1]. China's polysilicon dominance is real but in the wrong grade, and its 300 mm position is small [6]. The epitaxial mix matters more than the square inches, because that is what AI demand buys and where the incumbents make money. The failed move to 450 mm showed that this stage of the chain changes over decades.

5 **FOUNDATION OF DIGITAL LIFE Investor Presentation.** Siltronic · 29 April 2026 A
siltronic.com/fileadmin/investorrelations/2026/Q1/20260429_Siltronic_InvestorPresentation_...

1 **SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results.** PR Newswire · 10 February 2026 A
prnewswire.com/news-releases/semi-reports-2025-annual-worldwide-silicon-wafer-shipments-an...

6 **The Semiconductor Supply Chain - Issue Brief.** Center for Security and Emerging Technology (CSET) · 21 January 2021 A
cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION

High

SUBSTITUTABILITY

Moderate

Five firms already make 300 mm wafers that fabs have approved, but a fab has to test and approve a new supplier separately for every chip it makes.

PRICE OR MARKET SIZE

The world wafer market was $11.4bn in 2025, on 12,973 million square inches shipped

WHO LEADS

- JP **Shin-Etsu Handotai** Largest supplier; the parent's electronics materials arm sold ¥750.3bn from April to December 2025
- JP **SUMCO** ¥409.7bn of 2025 net sales, but an ¥11.8bn loss
- TW **GlobalWafers** NT$60.6bn revenue in 2025
- KR **SK Siltron** One of the five firms that supply the whole wafer market
- DE **Siltronic** €1,346.7m revenue in 2025

WHERE IT IS MADE

- JP **Japan** Shin-Etsu and SUMCO crystal growth and polishing
- TW **Taiwan** GlobalWafers headquarters and plants
- DE **Germany** Siltronic Burghausen and Freiberg; Wacker semiconductor-grade polysilicon
- KR **South Korea** SK Siltron, being sold to Doosan
- US **United States** Hemlock polysilicon in Michigan; GlobalWafers Sherman, Texas and St Peters, Missouri

WHY SUBSTITUTION IS POSSIBLE

Five firms make 300 mm wafers good enough for the newest chips, so losing one leaves four. A new wafer plant can still be built: GlobalWafers of Taiwan spent $3.5 billion on a plant in Texas, the first wafer production line of its kind in the United States in over twenty years. The slow part is approval. A fab tests a new wafer supplier separately for each chip it makes, and each test runs for months. Supply contracts are also signed years ahead. Switching wafer suppliers therefore takes two to five years.

WHERE CHINA STANDS

China made about 93 percent of the world's polysilicon in 2023, but 98 percent of it was solar grade, too impure for chips. It held under 1 percent of the 300 mm wafer market as of 2021, and it supplies only 12 percent of the wafers its own 300 mm fabs use.

WHERE THE US STANDS

No American firm makes wafers in volume. Hemlock makes polysilicon in Michigan, and GlobalWafers of Taiwan took a CHIPS Act award of up to $406 million toward a $3.5 billion Texas plant that opened in May 2025.

The whole world market for silicon wafers came to $11.4 billion of revenue on 12,973 million square inches in 2025 [1]. That is a small fraction of Nvidia's data center revenue, and every leading-edge chip behind it was built on one of these discs.

How it works

Silicon starts as quartzite, a rock that is silicon bound to oxygen. An arc furnace melts it with carbon, and the carbon takes the oxygen away as gas, leaving rough silicon metal. Refiners turn the metal into a gas, trichlorosilane, and distill it, because the impurities boil at other temperatures and stay behind. The clean gas then flows over silicon filaments heated inside a bell jar. On the hot surface the gas breaks apart, its silicon settles onto the filaments, and over days the filaments thicken into gray rods of polysilicon, the Siemens process. Everything downstream is made from those rods. The same process serves solar cells and chips, but only the ultra-pure semiconductor grade is any use for chips [2], and the two grades are separate industries with separate suppliers.

A grower melts the polysilicon at about 1,420 degrees Celsius, hotter than lava, and pulls a single crystal out of it by the Czochralski method. The grower lowers a small seed crystal on the end of a rod to the surface of the melt and draws it back up, slowly turning. Silicon freezes onto the seed in the seed's own atomic pattern, and a cylinder grows beneath it, the ingot. The pull starts with a thin neck so that the flaws formed where the seed first touched the melt end in the neck and never reach the ingot. Boron or phosphorus stirred into the melt, a step called doping, sets how readily the wafer conducts and whether the current in it is carried by negative charges or positive ones.

A wire saw, a web of fine wire running through abrasive, slices the ingot into discs. The maker then laps each disc, grinding both faces flat between rotating plates, etches off the damaged surface with acid, and polishes it to a mirror with colloidal silica, a slurry of silica particles far too small to see. Epitaxial wafers get one more layer of near-perfect silicon grown on top at about 1,200 degrees Celsius [3], still hot enough to glow orange, and almost every step throws material away.

Variants and trade-offs

Prime versus epitaxial

A prime wafer is a uniformly doped, front-polished single crystal: cheapest, and good enough for most memory and mature logic. An epitaxial wafer adds a thin, lightly doped, defect-controlled layer on top. The transistors live in that layer, so its quality sets the lowest leakage the chip can reach and its risk of latch-up, a parasitic short between power and ground. SEMI, the industry association, credits the 2025 rise in shipments to AI demand for advanced epitaxial wafers and high-bandwidth memory [1].

1 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire · 10 February 2026 A prnewswire.com/news-releases/semi-reports-2025-annual-worldwide-silicon-wafer-shipments-an...

2 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 A pubs.usgs.gov/periodicals/mcs2026/mcs2026-silicon.pdf

3 SUMCO, production processes page. SUMCO A sumcosi.com/english/products/process

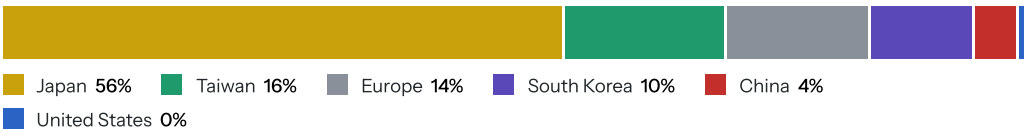
Silicon on insulator

A buried layer of oxide, which is glass, separates a thin working layer of silicon from the bulk beneath, and Soitec's Smart Cut process makes nearly all of it [4]. It matters for radio and low-power chips. AI GPUs run on plain epitaxial silicon.

Who makes it

Siltronic's own map of the chain puts five major suppliers behind the whole wafer market [5]. CSET at Georgetown names them: Shin-Etsu, SUMCO, GlobalWafers, Siltronic and SK Siltron, headquartered in Japan, Taiwan, Germany and South Korea [6].

Silicon wafer market by supplier headquarters. Five firms, four countries, no US producer %



Japan 56% Taiwan 16% Europe 14% South Korea 10% China 4%
United States 0%

Source: CSET, The Semiconductor Supply Chain, January 2021

Their 2025 results split between an AI boom and a slump in the older, mature nodes.

- **Shin-Etsu.** Its electronics materials arm sold ¥750.3 billion in the nine months to December 2025, on strong AI-related wafer demand [7].
- **SUMCO.** Sold ¥409.7 billion and still lost ¥11.8 billion, as operating profit fell from ¥36.9 billion to ¥1.3 billion while it added leading-edge 300 mm capacity and reorganized its weak 200 mm lines [8].
- **Siltronic.** Lost €77.9 million on €1,346.7 million of sales while spending €369.1 million on its Singapore fab [9].
- **GlobalWafers.** NT$60.6 billion of revenue, down 3.24 percent in local currency [10].

Area shipped rose 5.8 percent in 2025 while revenue fell 1.2 percent [1]. Wafers sell on multi-year agreements that fix price and volume years ahead, so a jump in

4 Soitec | Key figures. Soitec A soitec.com/home/group

5 FOUNDATION OF DIGITAL LIFE Investor Presentation. Siltronic · 29 April 2026 A siltronic.com/fileadmin/investorrelations/2026/Q1/20260429_Siltronic_InvestorPresentation_...

6 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 A cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

7 Consolidated Financial Results for the First Three Quarters Ended December 31, 2025. Shin-Etsu Chemical · 26 January 2026 A shinetsu.co.jp/wp-content/uploads/2025/07/20260127_con_E.pdf

8 Note: This document has been translated from the Japanese original for reference purposes only. In the event of any. SUMCO · 3 March 2026 A sumcosi.com/english/pdf/ir/library/shareholders/27/pdf/nc_e_27.pdf

9 Siltronic AG: Robust business performance in 2025 demonstrates resilience despite challenging conditions. Siltronic · 27 August 2026 A siltronic.com/en/press/press-releases/siltronic-ag-robust-business-performance-in-2025-dem...

10 GlobalWafers Reports Full Year 2025 Results - GlobalWafers Co., Ltd. All rights reserved.. GlobalWafers · 3 March 2026 A sas-globalwafers.com/en/gwc_news_en_20260303

1 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire · 10 February 2026 A prnewswire.com/news-releases/semi-reports-2025-annual-worldwide-silicon-wafer-shipments-an...

demand does not change the price in the year it happens.

China made 1.50 million metric tons of polysilicon, the raw material, in 2023, about 93 percent of world output [11]. Solar-grade material was 98 percent of that output and electronic-grade 2 percent [11], so almost none of it can go into a chip.

China's polysilicon output by grade, 2023. China made about 93% of the world's 1.5 million metric tons that year %



Solar-grade 98% Electronic-grade 2%

Source: USGS Minerals Yearbook, China, 2023

Siltronic sizes electronic-grade silicon at $1.4 billion, with five major suppliers, none of them Chinese: Wacker, Hemlock, OCI, Tokuyama and Mitsubishi [5].

Crystal growth and polishing sit in Japan, Taiwan, Germany and South Korea, with no major US-headquartered producer as of 2021 [6]. GlobalWafers opened a $3.5 billion plant in Sherman, Texas in May 2025, the first production line of its kind built in the United States in over twenty years, and announced another $4 billion for phases three and four the same day [12]. Washington had awarded it up to $406 million under the CHIPS incentives program [13].

The chokepoint

China dominates polysilicon and has almost no share in wafers. As of 2021 Chinese firms held under 5 percent of the wafer market and under 1 percent of 300 mm [6]. Chinese producers supplied 12 percent of the wafers consumed by China's own 300 mm fabs, against 18 percent at 200 mm and below [6].

Leading-edge 300 mm wafers depend on know-how that was never written down and that is not built into any machine a rival could buy, and 300 mm carried 99.7 percent of world fab capacity at 45 nm and below as of 2021 [6]. A fab has to qualify

11 The Mineral Industry of China in 2020-2021. U.S. Geological Survey · 17 February 2026 [A] pubs.usgs.gov/myb/vol3/2023/myb3-2023-china.pdf

5 FOUNDATION OF DIGITAL LIFE Investor Presentation. Siltronic · 29 April 2026 [A] siltronic.com/fileadmin/investorrelations/2026/Q1/20260429_Siltronic_InvestorPresentation_...

6 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 [A] cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

12 GlobalWafers America Officially Opens for Business - GlobalWafers Co., Ltd. All rights reserved.. GlobalWafers · 15 May 2025 [A] sas-globalwafers.com/en/gwc_news_en_20250516

13 Biden-Harris Administration Announces CHIPS Incentives Awards with GlobalWafers to Support Domestic Production of Silicon Wafers. U.S. Department of Commerce · 17 December 2024 [A] commerce.gov/news/press-releases/2024/12/biden-harris-administration-announces-chips-incen...

each new wafer source for every chip it makes, which takes quarters, while the contracts already in place run for years.

The last increase in wafer size, to 300 mm, happened in 2002 [6]. Bigger discs make printing a bigger share of the processing cost: photolithography takes half the cost of processing a 300 mm wafer against a quarter of a 150 mm one [6]. The attempt to move to 450 mm never reached production, and every leading-edge chip is still made on 300 mm.

One fatal defect ruins far more silicon on a GPU die that fills a whole printed field than on a small chip, and extreme-ultraviolet printing stays in focus over only a very small range of heights (see Lithography), so flatness has to hold at every point on the disc.

Key evaluation criteria

- **Grade.** Electronic against solar polysilicon separates a Chinese commodity market from a German, American, Japanese and Korean one; solar was 98 percent of China's 2023 output [11].
- **Defect density** in the epitaxial layer. A big AI die loses more good silicon per defect than a small chip does, which is why AI demand shows up first in advanced epitaxial wafers [1].
- **Flatness and nanotopography**, the fine ripple across the surface, because extreme-ultraviolet exposure stays in focus over only a very small range of heights, and that has to hold across a 300 mm wafer.
- **Diameter**, where 300 mm carries every leading-edge node and 99.7 percent of capacity at 45 nm and below [6].
- **Contract structure**, since long-term agreements set price and volume years ahead and keep a jump in demand from moving the price.

6 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 A cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

11 The Mineral Industry of China in 2020-2021. U.S. Geological Survey · 17 February 2026 A pubs.usgs.gov/myb/vol3/2023/myb3-2023-china.pdf

1 SEMI Reports 2025 Annual Worldwide Silicon Wafer Shipments and Revenue Results. PR Newswire · 10 February 2026 A prnewswire.com/news-releases/semi-reports-2025-annual-worldwide-silicon-wafer-shipments-an...

CHAPTER 03
STAGE 3 OF 15

Chemicals, Gases and Photoresist

A few hundred chemicals separate a blank wafer from a working chip. Japan makes about four-fifths of the photoresist. Twice the supply of these materials has been restricted, and no fab stopped, because the countries affected stockpiled and developed substitutes instead.

IN PLAIN TERMS

Photoresist is a coating that changes wherever light hits it. A machine spins a coat of it, thinner than a soap bubble, across the wafer. Light shines through a stencil onto the coat, and a liquid wash then takes away the parts the light reached. What remains is a pattern, and the next machine cuts that pattern into the wafer beneath, a step called etching. A few hundred other gases and liquids lay down layers, strip them off and clean the wafer between steps. All of them have to be clean to about one stray metal atom in every trillion. Each formula is approved for one factory and one product at a time, and a new supplier has to earn that approval from scratch, so a second source takes years. Nine tenths of the world's photoresist comes from Japan, and no factory can swap it out quickly.

IN SHORT

Photoresist is the most concentrated input at this stage: about 90 percent Japanese in 2021 and 78 percent in 2023 [3] [4], with one of the largest suppliers now 84 percent owned by a Japanese state fund [5]. Individual molecules are far more concentrated than the gas industry that sells them: neon and nitrogen trifluoride come from a handful of plants. China's gallium, germanium and antimony controls affect non-silicon chips and optics but not a silicon GPU, and the gallium suspension runs for one year from November 2025 [14]. The qualification cycle matters more than the spot price, because every disruption so far was absorbed within a year and each one added a second source somewhere.

3 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 [A] cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

4 METI, Semiconductor and digital industry strategy: future direction (半導体・デジタル産業戦略の今後の方向性), 23 December 2025. Ministry of Economy, Trade and Industry (Japan) · 23 December 2025 [A] meti.go.jp/policy/mono_info_service/joho/conference/semicon_digital/0014/handeji14-4.pdf

5 April 17, 2024 Japan Investment Corporation. Japan Investment Corporation · 17 April 2024 [A] jiccapital.co.jp/en/news/.assets/E_20240417_JIC_JICC_PressRelease.pdf

14 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 [A] pubs.usgs.gov/periodicals/mcs2026/mcs2026-gallium.pdf

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Rated on photoresist, the hardest input in this stage: Japan makes about four-fifths of it, and Korea needed five years to develop substitutes.	**Wafer fabrication materials were a $45.8bn market in 2025 and packaging materials $27.4bn, $73.2bn in total**

WHO LEADS

- JP **JSR** Owner of Inpria's metal-oxide resist for extreme-ultraviolet printing; 84% held by Japan Investment Corporation, a state-backed fund, since April 2024
- JP **Tokyo Ohka Kogyo** Top-five resist maker; ¥139.7bn of sales in the first half of fiscal 2026
- FR **Air Liquide** €2,465m of electronics revenue in 2025, 9.1% of group sales
- US **Entegris** $3.20bn of 2025 revenue in filters, polishing slurries, gas delivery and wafer carriers
- JP **JX Advanced Metals** About 65% of sputtering targets, the metal plates that chip wiring is made from

WHERE IT IS MADE

- JP **Japan** Photoresist, ultra-pure hydrofluoric acid, sputtering targets, polishing slurries
- US **United States** Entegris filters and wafer carriers, DuPont polishing slurries, Air Products gases
- FR **France** Air Liquide electronics gases and precursors
- KR **South Korea** SK Materials nitrogen trifluoride, Dongjin resist, on-site gas for memory fabs
- CN **China** Gallium, germanium and antimony refining; growing wet chemicals and polishing slurries

WHY SUBSTITUTION IS POSSIBLE

Photoresist is the light-sensitive coating that holds the printed pattern on the wafer. Each formula is approved for one layer of one product at one fab. A new supplier therefore has to develop its version together with the fab, one layer at a time. Korea shows what switching takes. When Japan required an individual export license for each shipment of resist in 2019, Korea kept its fabs running and spent five years funding its own substitutes. Switching takes two to five years, and closer to five for the newest chips.

WHERE CHINA STANDS

China cannot make photoresist for extreme-ultraviolet printing or for the finest 193 nm printing. It held under 5 percent of the photoresist market as of 2021. It refines 99 percent of the world's primary low-purity gallium. Since August 2023 it has required licenses to export gallium and germanium.

WHERE THE US STANDS

American firms are strong in filters, containers, polishing slurries and gas delivery: Entegris, DuPont and Air Products. They are nearly absent from advanced photoresist since JSR of Japan bought Inpria, an American startup, in 2021.

A modern fab is a chemical plant with lithography attached. Air Liquide alone supplies more than 200 molecules and roughly 50,000 cylinders a year [1]. None of it costs much by the metric ton, and any one of them missing stops the fab.

How it works

The resist has to change only where the light lands, and change fast enough that the scanner, the machine that projects the pattern, can expose each patch of the wafer in milliseconds. Since the 1990s the standard resist has been the chemically amplified resist. Each photon that lands frees one acid molecule inside the film, and nothing more happens until the wafer is baked. In the bake, each acid molecule sets off hundreds of reactions around itself that turn the film soluble, so a little light does a lot of chemistry. The wash then takes away the exposed film and leaves the rest. The light comes at 248 nm from a krypton fluoride laser or 193 nm from an argon fluoride laser shining through a layer of water, which sharpens the image.

That chemistry works less well with extreme ultraviolet (see Lithography). A 13.5 nm photon carries about fourteen times the energy of a 193 nm one, so the same dose of light arrives as far fewer photons. With fewer photons, where each one happens to land matters, and that randomness shows up as ragged line edges and missing contact holes, the small wells that join one layer of wiring to the next. Resist makers answered by absorbing more of the light: metal-oxide resists, built on clusters of tin and oxygen, take up extreme ultraviolet far better than carbon films, which is why JSR bought Inpria in 2021 [2].

Everything else puts material on the wafer, takes it off, or keeps it clean.
Bulk gases arrive by the metric ton, specialty gases in grams. Silane carries silicon into the chamber for deposition. A plasma tears nitrogen trifluoride into charged fragments, which scrub the chamber walls clean between wafers.
Hydrofluoric acid cleans the wafer itself. Polishing slurries grind each new layer flat before the next goes on. Sputtering targets are slabs of pure metal; a plasma knocks atoms off them, and the atoms settle on the wafer as the metal for the wiring. Filters and carriers keep contamination out.

Variants and trade-offs

RESIST CLASS	WAVELENGTH	WHERE IT IS USED	LIMIT
i-line novolak	365 nm	Packaging, mature nodes	Resolution
KrF chemically amplified	248 nm	Implant layers, thick films	Resolution
ArF immersion chemically amplified	193 nm	Most critical layers below 28 nm	Needs several exposures per layer
EUV chemically amplified	13.5 nm	Leading-edge single-exposure layers	Shot noise from too few photons

1 **Electronics | Air Liquide.** Air Liquide [A] airliquide.com/group/activities/electronics

2 **JSR Agrees to Acquire EUV Pioneer Inpria Corporation | 2021 | News.** JSR Corporation · 17 September 2021 [A] jsr.co.jp/jsr_e/news/2021/20210917.html

EUV metal-oxide	13.5 nm	High-NA tools, tightest pitches	Cost, defects, outgassing

Each resist class is tied to one scanner and one layer [3].

Who makes it

Japan produced about 90 percent of the world's semiconductor photoresist as of 2021, the rest mostly in the United States and South Korea [3]. A later survey by Fuji Keizai, relayed in the December 2025 strategy deck of Japan's Ministry of Economy, Trade and Industry, puts Japan at 78 percent in 2023, the United States at 13 and South Korea at 6, with China inside the small remainder [4].

Semiconductor photoresist production share by country, 2021 %



Japan 90% United States, South Korea and others 10%

Source: CSET, The Semiconductor Supply Chain

A vehicle of Japan Investment Corporation, a state-backed fund, took 84.36 percent of JSR's shares at ¥4,350 each in a tender offer that closed on 16 April 2024, then delisted it [5].

Tokyo Ohka Kogyo sold ¥139,668 million in the first half of fiscal 2026, up 25.1 percent on AI server demand [6]. Fujifilm lifted its polishing-slurry capacity at Kumamoto by 30 percent in January 2025 [7] and opened extreme-ultraviolet resist lines at Shizuoka and Pyeongtaek that October [8].

Wafer fabrication materials sold $45.8 billion in 2025 and packaging materials $27.4 billion [9].

Gases are a Western-led oligopoly: Linde, Air Liquide, Air Products, Nippon Sanso and Messer. Air Liquide's electronics arm alone sold €2,465 million in 2025, 9.1 percent of group revenue [1]. Smaller firms specialize in single molecules: SK Materials in nitrogen trifluoride, Kanto Denka in fluorine chemistry, Stella Chemifa and Morita in ultra-high-purity hydrofluoric acid.

3 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 [A] cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

4 METI, Semiconductor and digital industry strategy: future direction (半導体・デジタル産業戦略の今後の方向性), 23 December 2025. Ministry of Economy, Trade and Industry (Japan) · 23 December 2025 [A] meti.go.jp/policy/mono_info_service/joho/conference/semicon_digital/0014/handeji14-4.pdf

5 April 17, 2024 Japan Investment Corporation. Japan Investment Corporation · 17 April 2024 [A] jiccapital.co.jp/en/news/.assets/E_20240417_JIC_JICC_PressRelease.pdf

6 Consolidated Financial Results for the Second Quarter (Interim) of the Fiscal Year Ending December 31, 2026 [J-GAAP]. Tokyo Ohka Kogyo · 5 August 2026 [A] tok.co.jp/application/files/7417/8598/0069/q2_2612_en.pdf

7 Fujifilm to Enhance its Production Capacity of Advanced Semiconductor Material CMP Slurries at the Kumamoto Site. Fujifilm · 5 December 2024 [A] fujifilm.com/jp/en/news/hq/11942

8 Fujifilm Launches EUV Resist and EUV Developer. Fujifilm · 29 October 2024 [A] fujifilm.com/jp/en/news/hq/11842

9 Global Semiconductor Materials Market Revenue Reaches Record $73.2 Billion in 2025, SEMI Reports. PR Newswire · 12 May 2026 [A] prnewswire.com/news-releases/global-semiconductor-materials-market-revenue-reaches-record-...

1 Electronics | Air Liquide. Air Liquide [A] airliquide.com/group/activities/electronics

As of 2021 five firms held more than 60 percent of wet chemicals, and DuPont and Cabot Microelectronics about 56 percent of a $790 million market in polishing slurries [3]. Sputtering targets are more concentrated than any of those, with JX Advanced Metals holding about 65 percent of the world market [10].

The chokepoint

Supply at this stage has been disrupted twice since 2019, once by a government and once by a war.

In July 2019 Japan put photoresist, hydrogen fluoride and fluorinated polyimide under individual export licenses for South Korea. It had supplied more than 90 percent of Korean imports of two of the three, and hydrogen fluoride exports fell 87.9 percent [11]. Korea kept its fabs running and spent five years funding domestic substitutes, which is what materials controls usually produce.

In 2022 Russia invaded Ukraine, which supplied about 70 percent of the world's neon and 90 percent of the semiconductor-grade neon used by US industry [12]. Buyers had stockpiled since the 2014 annexation of Crimea, and Korean and Japanese chipmakers reported adequate neon from China instead [12].

China's export controls fall on the metals. Beijing announced export licenses for gallium and germanium in July 2023, in force from August, then banned both, plus antimony and superhard materials, to the United States in December 2024 [13]. In November 2025 it lifted the gallium ban for one year [14].

Neither metal goes into a silicon logic chip. Germanium goes into fiber optics, infrared optics, solar cells for satellites and radiation detectors [15]; gallium into the non-silicon wafers behind radio front ends, LEDs and lasers [14]. They move the price instead. Germanium metal averaged $4,100 a kilogram in 2025 against $1,392 in 2023, and US imports of the metal fell 67 percent [15]. The US Geological Survey put the cost to US GDP of a complete ban at about $3.4 billion [13].

3 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 [A] cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

10 Quick Guide to JX Advanced Metals | Corporate Overview. JX Advanced Metals [A] jx-nmm.com/english/company/glance

11 The impact of export controls on international trade: evidence from the Japan–Korea trade dispute in the semiconductor industry. Research Institute of Economy, Trade and Industry (Japan) · 8 May 2023 [A] rieti.go.jp/en/columns/v01_0201.html

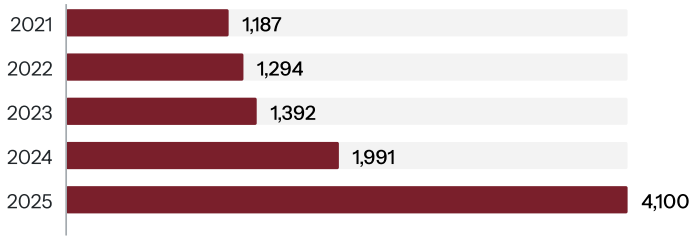
12 Russia's Invasion of Ukraine Impacts Gas Markets Critical to Chip Production | Perspectives on Innovation. Center for Strategic and International Studies [A] csis.org/blogs/perspectives-innovation/russias-invasion-ukraine-impacts-gas-markets-critic…

13 China’s Germanium and Gallium Export Restrictions: Consequences for the United States. Stimson Center · 19 March 2025 [A] stimson.org/2025/chinas-germanium-and-gallium-export-restrictions-consequences-for-the-uni…

14 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 [A] pubs.usgs.gov/periodicals/mcs2026/mcs2026-gallium.pdf

15 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 [A] pubs.usgs.gov/periodicals/mcs2026/mcs2026-germanium.pdf

Germanium metal, annual average price ($/kg). China licensed exports from August 2023 and banned US shipments in December 2024 $/KG



Source: USGS Mineral Commodity Summaries, February 2026 (Argus Media prices)

The slower risk is the European Union's proposed restriction on PFAS, the long-lived fluorinated chemicals. The five member states behind it published a revised text on 20 August 2025, after working through more than 5,600 comments [16]. The committees of ECHA, the EU chemicals agency, now back an EU-wide restriction with targeted exemptions, and the scientific evaluation is due to finish by the end of 2026 [16]. These chemicals are everywhere in a fab: photoacid generators, etch gases, chamber liners, pump seals and filter membranes are all fluorinated. An exemption is likely, but nothing is law yet.

Key evaluation criteria

- **Purity, in parts per trillion** of metallic contamination, a million times finer than parts per million, which is the spec that keeps entrants out.
- **Qualification lock-in**: a resist or slurry is approved for one layer of one product at one fab, so switching costs are counted in tape-outs.
- **Photon efficiency** for EUV resists: how much light the resist absorbs sets the dose, the dose sets how many wafers an hour the scanner prints, and too few photons per dose set the defect rate, which is the case for metal-oxide chemistry.
- **Transportability**: bulk gases are made on site at the fab, and only cylinder-shipped specialty gases can be embargoed.
- **Byproduct exposure**: gallium is recovered mostly from bauxite refining and germanium from zinc concentrates, so supply depends on another industry's economics [14].

16 ECHA supports PFAS restriction with targeted derogations. European Chemicals Agency · 26 March 2026 [A] echa.europa.eu/-/echa-supports-pfas-restriction-with-targeted-derogations

14 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 [A] pubs.usgs.gov/periodicals/mcs2026/mcs2026-gallium.pdf

CHAPTER 04
STAGE 4 OF 15

Lithography

The most concentrated stage in the chain. One firm in the Netherlands builds every extreme ultraviolet scanner in the world, the machine that prints the finest circuit layers. It shipped 48 of them in 2025, at around $200 million each.

IN PLAIN TERMS

Lithography prints the pattern of a circuit onto the wafer. A machine holds a stencil of one layer of the circuit up to a lamp and projects the image, four times smaller, onto the silicon. Light cannot draw a line much finer than its own wave, so the finer the lines, the shorter the wavelength of light needed to print them. Air soaks up the shortest light now in use, so the machine has to print in a vacuum. Only one company has ever built a machine that prints with it, and it is Dutch. Export controls are government rules on who a company may sell to. Every advanced chip passes through that firm's machines, so a rule aimed at it reaches all advanced chipmaking.

IN SHORT

ASML, in Veldhoven, using Zeiss mirrors from Oberkochen, sets the resolution of every leading-edge AI chip. ASML booked 48 EUV systems in 2025, four of them High-NA [1]. Export controls cut China's share of ASML system sales from 41 percent in 2024 to 33 percent in 2025 [3], and China's answer is domestic immersion DUV plus an EUV prototype that has not yet made a chip. No announced program changes this bottleneck before 2030.

1 **ASML 2025 Annual Report.** ASML · 24 February 2026 [A] ourbrand.asml.com/m/419103cb23dfeaa4/original/asml-2025-annual-report-financial-performanc...

3 **Presentation Investor Relations Q4 2025.** ASML · 27 January 2026 [A] ourbrand.asml.com/m/3136300aa4999bc1/original/2026_01_28_Presentation-Investor-Relations-Q...

CHOKEPOINT CARD | EXTREME CONCENTRATION | VERY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Very hard ASML is the only maker of EUV scanners, only Zeiss can polish their mirrors, and a state-funded Chinese team has made the light but has not printed a chip.	About $200M for a standard EUV scanner, $350-400M for the newer High-NA version and about $60M for a DUV tool, according to Reuters. ASML does not publish per-system prices

WHO LEADS

- NL **ASML** 100% of EUV (extreme ultraviolet) and above 80% of DUV (deep ultraviolet); EUV was 48% of its own 2025 system revenue and immersion DUV 42%
- JP **Nikon** 22 new chipmaking scanners in the year to March 2026, in a market Nikon estimates at 570 units
- JP **Canon** i-line and krypton fluoride steppers for older, coarser layers, plus the only commercial nanoimprint tool

WHERE IT IS MADE

- NL **Netherlands** ASML design and final assembly, Veldhoven
- DE **Germany** Zeiss SMT optics, Oberkochen; Trumpf carbon dioxide lasers that drive the light source, Ditzingen
- US **United States** ASML light-source research and manufacturing, San Diego
- JP **Japan** Nikon and Canon deep ultraviolet scanners and steppers

WHY SUBSTITUTION IS SLOW

Nobody else has ever built an EUV scanner. A newcomer would first have to make mirrors a meter wide, polished smooth to within tens of picometers, a picometer being a trillionth of a meter. It would also need a light source that hits tin droplets with a laser 50,000 times a second. The best-funded attempt so far, in China, has made extreme-ultraviolet light but has printed no chip. CSIS judges that China cannot yet build a working EUV scanner, and no announced program would change that before 2030.

WHERE CHINA STANDS

No EUV scanner has ever been sold to a customer in China. CSIS judges that China cannot build the technology despite state investment. Shanghai Aishengna Electronic Technology Group, a state-owned firm registered in 2023, is reported to have started producing Chinese immersion DUV scanners in 2026. SMEE, the older Chinese supplier, holds about 4 percent of the world market for i-line scanners, the coarsest kind, and no share of any finer kind.

WHERE THE US STANDS

No American firm makes scanners. The US government still controls who may buy them, through Dutch export licenses, the American-made parts inside each machine, and its entity list of firms that may not be supplied.

Every transistor in an AI accelerator gets its shape from a machine one company builds. ASML sold 327 lithography systems in 2025, 48 of them extreme ultraviolet , and nobody else sells EUV [1].

How it works

The stencil is a photomask, the film is photoresist and the machine is a scanner (see Photomasks and pellicles).

The wafer is coated with photoresist, a film that changes wherever light touches it. The mask holds one layer of the circuit as a pattern of clear and dark areas, drawn four times larger than life. The machine shines light through the mask, shrinks the image four times with a lens, and lands it on the film. A wash then removes the film where the light hit, and the pattern is left standing on the wafer for the next machine to etch in or fill with metal. Then the film is stripped and the next mask goes in. A chip takes dozens of masks, one per layer.

The image from one mask covers a patch about 26 by 33 mm, so the wafer moves under the lens one patch at a time, close to a hundred patches per wafer. Before each one, the machine finds marks printed in earlier layers and lines the new layer up on them to within a few nanometers.

Finer lines need shorter light. The finest light now in use, at 13.5 nm, is stopped by glass and by air, so an EUV scanner works in a vacuum and uses mirrors instead of lenses, and its masks are mirrors too. The older machines, which use light fourteen times longer, gain extra sharpness by filling the gap between lens and wafer with water [2].

When one exposure cannot draw a pattern finely enough, the layer is printed with two, three or four masks instead. Each extra pass costs a mask, machine time and a chance to misalign, so a single EUV exposure usually works out cheaper.

Variants and trade-offs

Deep ultraviolet

Deep ultraviolet (DUV) light comes from gas lasers: 248 nanometers from krypton fluoride (KrF), 193 from argon fluoride (ArF), shone through air or water. These tools still dominate by volume: of ASML's 327 systems in 2025, 131 were ArF immersion, 78 KrF, 54 the older, coarser i-line and 16 ArF dry [3]. A DUV scanner costs roughly $60 million. DUV is the one tier with named competitors, Nikon and Canon, and ASML still holds above 80 percent of it [4].

Immersion is the contested tier, because without EUV a fab needs NXT:2000i-class scanners and several passes per layer to reach leading-edge logic, and that is the tier Dutch export licensing covers.

1 ASML 2025 Annual Report. ASML · 24 February 2026 [A] ourbrand.asml.com/m/419103cb23dfeaa4/original/asml-2025-annual-report-financial-performanc…

2 Lenses & mirrors - Lithography principles. ASML [A] asml.com/en/technology/lithography-principles/lenses-and-mirrors

3 Presentation Investor Relations Q4 2025. ASML · 27 January 2026 [A] ourbrand.asml.com/m/3136300aa4999bc1/original/2026_01_28_Presentation-Investor-Relations-Q…

4 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters · 28 January 2026 [B] reuters.com/world/asia-pacific/250-million-asml-printer-behind-nvidias-chips-2026-07-28

Extreme ultraviolet

The 13.5 nm light comes from dropping molten tin into a vacuum vessel 50,000 times a second and hitting each droplet twice, with a low-intensity pulse to flatten it and then one that vaporizes it into plasma [5]. Zeiss mirrors carry it to the wafer, and the largest are a meter across, polished flat to within tens of picometers, thousandths of a nanometer [2].

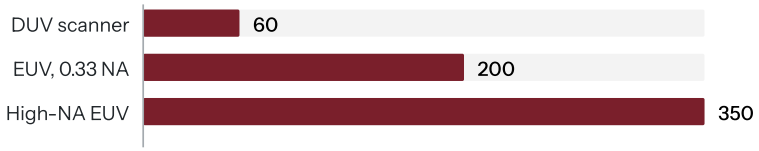
Brighter light means more wafers an hour, and ASML ran the first 1,000-watt EUV source in April 2025 [6]. Its TWINSCAN NXE:3800E now ships at its full 220 wafers an hour, 37 percent better than the NXE:3600D [6].

High-NA EUV

High numerical aperture (High-NA) raises the aperture from 0.33 to 0.55 and cuts the smallest printable feature from 13 nm to 8 nm [7]. Its optics shrink the mask image four times in one direction and eight in the other, halving the patch of wafer one exposure covers and doubling the exposures per wafer [7]. The second-generation TWINSCAN EXE:5200B runs at 175 wafers an hour, 60 percent above the EXE:5000 [6].

Reuters puts a standard EUV tool at around $200 million and a High-NA machine at $350 to $400 million, prices ASML does not publish [4].

What one lithography system costs. The High-NA figure is quoted as a $350-400M range; the bar shows the low end $M



Source: Reuters, The $400 million ASML 'printers', July 2026

Intel and TSMC have chosen differently on High-NA:

- **Intel Foundry** ships the first high-volume logic product made with High-NA, on Intel 18A, and installed the first EXE:5200B [8], after more than a million High-NA wafers [9].

5 All about light and lasers in lithography. ASML [A] asml.com/en/technology/lithography-principles/light-and-lasers

2 Lenses & mirrors - Lithography principles. ASML [A] asml.com/en/technology/lithography-principles/lenses-and-mirrors

6 ASML 2025 Annual Report. ASML · 24 February 2026 [A] ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-sect…

7 5 things you should know about High NA in EUV. ASML · 25 January 2024 [A] asml.com/en/company/stories/2024/5-things-high-na-euv

4 Explainer: The $400 million ASML 'printers' key for the AI chip boom. Reuters · 28 January 2026 [B] reuters.com/world/asia-pacific/250-million-asml-printer-behind-nvidias-chips-2026-07-28

8 High NA EUV reaches new readiness milestone with first high-volume Logic product. ASML · 15 July 2026 [A] asml.com/en/news/press-releases/2026/high-na-euv-reaches-new-readiness-milestone

9 Intel Foundry and ASML Accelerate Industry Readiness for High-NA EUV. Intel · 7 September 2026 [A] intel.com/content/www/us/en/newsroom/news/intel-foundry/intel-foundry-asml-accelerate-indu…

- **TSMC** waited, and now says it will use High-NA in high-volume manufacturing from 2030 [10].

ASML booked four EXE systems as revenue in 2025 against two in 2024 [1], and expects the platform to carry high-volume manufacturing from 2027 [6]. On 8 September 2026 it announced with TSMC a move to 12-inch photomasks to lift the field-size limit, with a pilot mask line in 2031 and production systems by 2033 [10].

EUV systems ASML booked as revenue each year TOOLS



Source: ASML annual reports 2024 and 2025

Who makes it

ASML took EUR 32.7 billion in net sales in 2025 at a 52.8 percent gross margin [1], and in July 2026 raised its 2026 forecast to EUR 43 to 45 billion at 54 to 56 percent [11]. It assembles more than it makes, listing 5,100 suppliers [12]. Four of them make the main parts of the EUV machine:

- **Zeiss SMT**, Oberkochen, makes the illumination system and the six-mirror projection optics [13].
- **Trumpf**, Ditzingen, makes the carbon dioxide drive laser, which amplifies a few watts to 40 kilowatts [14].
- **ASML San Diego**, the former Cymer, designs the source where the laser turns tin into plasma, and builds the droplet generator [15].
- **VDL ETG** builds the frames that suspend and position the mirrors [16].

10 ASML and TSMC Announce Initiative to Pioneer Industry Transition to Large-Format Photomasks for High NA EUV. TSMC · 8 September 2026 [A] pr.tsmc.com/english/news/3338

1 ASML 2025 Annual Report. ASML · 24 February 2026 [A] ourbrand.asml.com/m/419103cb23dfeaa4/original/asml-2025-annual-report-financial-performanc…

6 ASML 2025 Annual Report. ASML · 24 February 2026 [A] ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-sect…

11 Press Release Financial Results Q2 2026. ASML · 14 July 2026 [A] ourbrand.asml.com/asset/c8dbf3fc-4c5e-4406-83f6-27694b138245/Press-Release-Financial-Resul…

12 ASML HOLDING NV, Form 6-K report of foreign private issuer for the period ended 2025-12-31 (6-K). U.S. Securities and Exchange Commission (filing by ASML HOLDING NV) · 25 February 2026 [A] sec.gov/Archives/edgar/data/937966/000162828026011377/asml-2025xannualxreportx.htm

13 EUV lithography and technology. Carl Zeiss [A] zeiss.com/semiconductor-manufacturing-technology/inspiring-technology/euv-lithography.html

14 Good things come in ever-smaller packages. TRUMPF [A] trumpf.com/en_US/products/lasers/euv-drive-laser

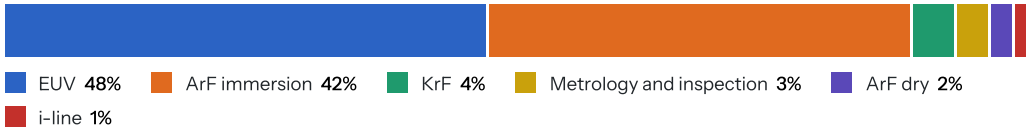
15 Explore ASML San Diego. ASML [A] asml.com/en/company/about-asml/locations/san-diego

16 VDL Groep, VDL ETG builds complex frames for Zeiss, innovation project page. VDL Groep [A] vdlgroep.com/en/vdl-groep/innovation-projects/vdl-etg-builds-complex-frames-for-zeiss

Zeiss mirror polishing and coating caps EUV output. Final assembly in Veldhoven waits on the optics.

Nikon and Canon are specialists now. Nikon sold 22 new scanners in the year to March 2026, into a market it sizes at 570 units, and lost money on Precision Equipment [17]. Canon's alternative is nanoimprint, where the FPA-1200NZ2C stamps the pattern instead of projecting it, down to a 14 nm linewidth [18]. No leading-edge logic customer has taken it up.

ASML net system sales by technology, 2025 %



Source: ASML Q4 2025 investor presentation

Controls and China's answer

Export controls bar EUV from Chinese foundries, and CSIS judges that China cannot yet build it despite state spending [19]. Below EUV, the restrictions arrived in this order:

- **Dutch national licensing** of advanced DUV took effect on 1 September 2023, and The Hague widened it on 6 September 2024 [20].
- **A partial license revocation** disclosed on 1 January 2024 stopped NXT:2050i and NXT:2100i shipments to a few Chinese customers [21].
- **The BIS Affiliates Rule** extended entity-list controls to 50 percent-owned subsidiaries from 29 September 2025 [22], then BIS suspended it to 9 November 2026 [23].

China fell from 41 percent of ASML's net system sales in 2024 to 33 percent in 2025 [3].

On DUV, Reuters reported in July 2026, on one unnamed source, that Shanghai Aishengna Electronic Technology Group has started producing home-grown

17 FINANCIAL RESULTS The Year Ended March 31,2026. Nikon · 21 May 2026 [A] nikon.com/content/dam/web-assets/nikoncom/company/local/global/en/ir/ir_library/result/pdf...

18 Canon delivers FPA -1200NZ2C nanoimprint lithography system for semiconductor manufacturing to the Texas Institute for Electronics. Canon · 26 September 2024 [A] global.canon/en/news/2024/20240926.html

19 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements | Strategic Technologies Blog. Center for Strategic and International Studies [A] csis.org/blogs/strategic-technologies-blog/breakthroughs-or-boasts-assessing-recent-chines...

20 The Netherlands expands export control measure for advanced semiconductor manufacturing equipment. Government of the Netherlands · 6 September 2024 [A] government.nl/latest/news/2024/09/06/the-netherlands-expands-export-control-measure-advanc...

21 Statement regarding partial revocation export license. ASML · 1 January 2024 [A] asml.com/en/news/press-releases/2023/statement-regarding-partial-revocation-export-license

22 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register (Commerce Department; Industry and Security Bureau) · 30 September 2025 [A] federalregister.gov/documents/2025/09/30/2025-19001/expansion-of-end-user-controls-to-cove...

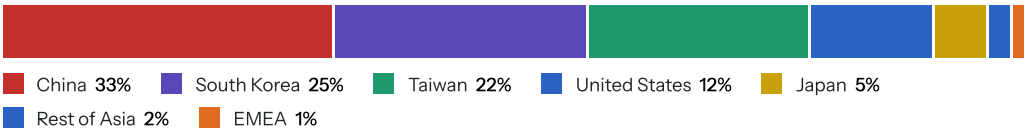
23 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain Listed Entities. Federal Register (Commerce Department; Industry and Security Bureau) · 12 November 2025 [A] federalregister.gov/documents/2025/11/12/2025-19846/one-year-suspension-of-expansion-of-en...

3 Presentation Investor Relations Q4 2025. ASML · 27 January 2026 [A] ourbrand.asml.com/m/3136300aa4999bc1/original/2026_01_28_Presentation-Investor-Relations-Q...

immersion DUV tools. The firm was registered in August 2023 with 7 billion yuan of capital and two state shareholders, Shanghai Electric Holding and a Shanghai International Trust subsidiary. It has no website, and has taken in teams from SMEE and the startup Yuliangsheng [24]. SMEE, the incumbent, is strongest in i-line and even there holds about four percent of the world market [19].

On EUV there is less evidence, and claims of Huawei mass production in 2026 remain unverified. Reuters reported in December 2025 that a Shenzhen team of former ASML engineers had built a prototype that makes extreme ultraviolet light but no chip [25]. Between that light and a printed chip sit the mirrors, and only Zeiss can polish them.

ASML net system sales by the region tools shipped to, 2025 %



China 33% South Korea 25% Taiwan 22% United States 12% Japan 5%
Rest of Asia 2% EMEA 1%

Source: ASML Q4 2025 investor presentation

The chokepoint

If ASML stops shipping, nothing at the leading edge gets printed, and even ASML could not rebuild itself quickly, since Zeiss makes its optics and Trumpf its drive laser.

For AI accelerators, EUV is the point of control. It prints the finest layers, the ones that set how densely transistors and wires pack, and TSMC expects the number of layers needing High-NA to rise as AI designs grow more complex [10]. Every added layer is more time on a tool nobody else builds.

Key evaluation criteria

- **Resolution.** Set by wavelength divided by numerical aperture. 13 nm at 0.33 NA, 8 nm at 0.55 NA [7].
- **Throughput.** Wafers an hour at a stated dose. 220 on the NXE:3800E, 175 on the EXE:5200B [6].
- **Overlay.** How precisely one layer lands on the one below; ASML credits the EXE:5200B's gain to new Zeiss optics [6].

24 China starts production of home-grown immersion DUV chipmaking tools, source says. Reuters · 28 July 2026 [B] reuters.com/world/china/china-starts-production-home-grown-immersion-duv-chipmaking-tools-...

19 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements | Strategic Technologies Blog. Center for Strategic and International Studies [A] csis.org/blogs/strategic-technologies-blog/breakthroughs-or-boasts-assessing-recent-chines...

25 How China built its 'Manhattan Project' to rival the West in AI chips. Reuters · 17 December 2025 [B] reuters.com/world/china/how-china-built-its-manhattan-project-rival-west-ai-chips-2025-12-17

10 ASML and TSMC Announce Initiative to Pioneer Industry Transition to Large-Format Photomasks for High NA EUV. TSMC · 8 September 2026 [A] pr.tsmc.com/english/news/3338

7 5 things you should know about High NA in EUV. ASML · 25 January 2024 [A] asml.com/en/company/stories/2024/5-things-high-na-euv

6 ASML 2025 Annual Report. ASML · 24 February 2026 [A] ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-sect...

- **Field size.** The patch of wafer one exposure covers. High-NA halves it, so big AI dies must be stitched until 12-inch masks arrive [7].
- **Source power.** Brighter light, more wafers an hour, which is why the 1,000-watt demonstration matters [6].

7 5 things you should know about High NA in EUV. ASML · 25 January 2024 A asml.com/en/company/stories/2024/5-things-high-na-euv

6 ASML 2025 Annual Report. ASML · 24 February 2026 A ourbrand.asml.com/m/8ab959d4926657b/original/asml-2025-annual-report-strategic-report-sect…

CHAPTER
05
STAGE 5 OF 15

Photomasks and Pellicles

Every leading-edge chip depends on a business worth tens of billions of yen. Two Japanese firms make almost all the blank plates that EUV masks are built on, one Japanese firm inspects them, and the dust cover meant to protect them is still not in production.

IN PLAIN TERMS

A photomask is the stencil that lithography prints from, one for each layer of a chip. The same plate prints its layer on wafer after wafer, so one plate shapes millions of chips. No glass lets the short light now in use pass through, so instead of a window the plate is a mirror, built up from about forty pairs of very thin layers, and the light bounces off it. A dark pattern drawn on top of the mirror soaks up the light wherever no line should print. Any speck of dust on the plate repeats on every wafer it prints. Two Japanese firms make almost all of those mirrors. A business almost nobody has heard of can hold up every advanced chip in the world.

IN SHORT

Photomasks are the cheapest chokepoint in this chain, which is why nobody has funded an entrant. Three blank suppliers in the world on AGC's count [6], one of them describing its own share as exceptionally high [8], and a single-vendor actinic inspection tool [10] are needed for every EUV layer of every leading-edge accelerator. Pellicles remain unproven in production, so fabs trade throughput against defect risk product by product [11]. And mask-set costs of $10 million to $40 million decide which AI chips get designed [14].

6 The Japanese glass behind next-generation chips. AGC [A] agc.com/en/hub/pr/the-japanese-glass-behind-next-generation-chips.html

8 Information Technology Business | Review of Operations | HOYA REPORT 2025. HOYA [A] hoya.com/ir/2025/en/review/it.html

10 Business Report | Lasertec Corporation. Lasertec [A] lasertec.co.jp/en/ir/plan/message.html

11 Indistinguishable from magic: the EUV pellicle. ASML · 14 September 2022 [A] asml.com/en/news/stories/2022/the-euv-pellicle-indistinguishable-from-magic

14 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due To Photomask Sets, Verification, and Validation. SemiAnalysis · 24 July 2022 [B] newsletter.semianalysis.com/p/the-dark-side-of-the-semiconductor

CHOKEPOINT CARD | EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Hard Only three firms make the blank plates for EUV masks, and only Lasertec sells the tool that checks a finished mask in EUV light.	**A mask set for the newest chips costs roughly $10M to $40M, depending on the node and on who is estimating**

WHO LEADS

- JP **AGC** Ranks itself No.1 worldwide in EUV mask blanks and counts three suppliers in all; targeted over ¥40bn of blank sales by 2025
- JP **Hoya** Says its share of mask blanks is exceptionally high, and expects to keep its lead as customers add a second supplier
- JP **Lasertec** The only supplier of EUV mask inspection at the wavelength the scanner itself uses; JPY 230.5bn revenue in the year to June 2026
- AT **IMS Nanofabrication** Intel calls it the established industry leader in multi-beam mask writers, the machines that draw the pattern onto the mask
- US **Photronics** Largest mask maker selling to outside customers; $849.3M revenue in fiscal 2025

WHERE IT IS MADE

- JP **Japan** Mask blanks (AGC, Hoya), DUV blanks (Shin-Etsu), inspection (Lasertec), writers (NuFlare), pellicles (Mitsui Chemicals)
- AT **Austria** IMS Nanofabrication multi-beam mask writers, Vienna
- DE **Germany** Zeiss mask repair and metrology
- US **United States** Photronics merchant mask shops, KLA inspection
- TW **Taiwan** TSMC's in-house mask shop, and a 10% stake in IMS

WHY SUBSTITUTION IS SLOW

An EUV mask starts as a blank plate coated with forty layers. AGC and Hoya took two decades to learn how to make one. Three firms have now done it and the structure of the plate is published, so a newcomer knows what to build. Reaching a plate that fabs accept would take five to ten years. The harder part is inspection, because Lasertec sells the only tool that checks a mask in the same EUV light the scanner uses. Other tool makers stayed out of EUV mask inspection because the market was too small to be worth the cost, and a large budget removes that barrier.

WHERE CHINA STANDS

SMIC runs two mask shops of its own in Shanghai, but only for the older g-line, i-line, KrF and ArF scanners. No Chinese firm makes an EUV mask blank, a tool that checks a mask in EUV light, or a multi-beam writer, the machine that draws the finest patterns onto a mask.

WHERE THE US STANDS

Photronics is the largest mask maker that sells to outside customers, and KLA makes mask inspection tools, but no American firm makes an EUV mask blank or a tool that checks a mask in EUV light.

Every EUV layer on an AI accelerator needs a flawless mask, built on a blank plate that two Japanese firms supply and passed by an inspection tool that one Japanese firm makes. AGC, one of the two, says it is the only company in the world that takes an EUV blank all the way from the glass to the finished coating [1].

How it works

A mask starts as a blank, a polished plate with nothing on it yet. For deep ultraviolet the blank is a window of quartz carrying a light-blocking film such as molybdenum silicide, and Shin-Etsu supplies those blanks [2]. For EUV the blank is a mirror of about 40 molybdenum and silicon layer pairs, 7 nm to a pair, laid on low-expansion glass, with a dark tantalum film on top, the absorber, that will carry the pattern [3].

The mask shop coats the blank with resist, and a mask writer draws the design into the coating with a beam of electrons, one shape at a time, over many hours, because no light draws finely enough. A wash removes the drawn resist, an etch removes the blocking film or absorber wherever the resist is gone, and the rest of the resist is stripped. The plate now passes light, or reflects it, in the exact shape of one layer of the chip, four times larger than it will print. An inspection tool compares the plate to the design, and a repair tool fixes the stray defects it finds.

Making the plate a mirror creates its own problems:

- **Mask 3D effects.** Light strikes the mirror at a slant, and the absorber has height, so its edges throw small shadows that distort the printed shape; software has to correct for it in advance.
- **Write precision.** Every edge of the absorber has to sit within a few nanometers of where the design says.
- **Particles.** Anything on the reflector prints on every wafer until someone finds it.

High-NA scanners, the newest EUV machines, print finer lines with larger mirrors, and each exposure then covers half the patch of wafer, doubling the exposures needed, so a large die, one chip's rectangle, has to be stitched from two masks [4]. On 8 September 2026 ASML and TSMC announced a move to 12-inch masks, larger than today's, to remove that limit, with a pilot line in 2031 [5].

Variants and trade-offs

Blanks

Two decades of learning went into the blank: laying down forty layer pairs across a six-inch plate, flat to within less than an angstrom, a tenth of a nanometer, with no particle buried anywhere underneath. Nobody publishes a market share for EUV blanks, but AGC counts the suppliers. It says it is one of three companies

1 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC · 27 April 2023 [A] agc.com/en/news/detail/1203819_2814.html

2 Photomask blanks - Shin-Etsu Chemical Co., Ltd.. Shin-Etsu Chemical · 3 April 2019 [A] shinetsu.co.jp/en/products/electronics-materials/photomask-blanks

3 The refined EUV mask model I.A. MAKHOTKIN1*, M. WU1, V. SOLTWISCH2, F. SCHOLZE2, V. PHILIPSEN1. arXiv · 19 December 2019 [A] arxiv.org/pdf/1912.09075

4 5 things you should know about High NA in EUV. ASML · 25 January 2024 [A] asml.com/en/company/stories/2024/5-things-high-na-euv

5 TSMC and ASML Announce Initiative to Pioneer Industry Transition to Large-Format Photomasks for High NA EUV. ASML · 8 September 2026 [A] asml.com/en/news/press-releases/2026/tsmc-and-asml-announce-industry-transition-to-large-f…

worldwide that supply EUV blanks, and the only one that handles every material and process itself [6]. Its own data book lists EUV photomask blanks among the products where AGC holds the No.1 position worldwide [7].

Hoya says its share of the mask blank market is exceptionally high, and expects customers to move gradually toward multi-sourcing EUV blanks [8]. AGC raised its EUV blank capacity by about 30 percent in 2025, on a line in Fukushima funded under a METI supply-chain program, and was aiming for more than ¥40 billion of sales in the business [1].

Writers

An electron-beam writer cuts the pattern into a sensitive coating on the blank. Multi-beam writers fire many beams at once, which is what lets a dense EUV mask finish in a usable time, and two firms make them: IMS Nanofabrication of Vienna and Japan's NuFlare. Intel calls IMS the established leader, and sold roughly 10 percent of it to TSMC in September 2023 at a valuation near $4.3 billion [9].

Inspection and repair

A mask defect costs yield on every wafer that mask prints, so fabs check masks three times: as blanks, after patterning and again in production. Actinic inspection looks at 13.5 nm, the wavelength the mask itself sees, and Lasertec is the only supplier, on JPY 230.5 billion of net sales in the year to June 2026 [10]. KLA sells optical reticle inspection; Zeiss sells e-beam repair.

Pellicles

A pellicle is a thin membrane on a frame above the mask. Particles land on it, too far out of focus to print, and the pattern below stays clean.

ASML's EUV membrane is 13 nanometers thick, has to hold together at 500 degrees Celsius in vacuum, and every photon it absorbs is light that never reaches the wafer [11]. The light crosses it twice, on the way in and on the way back off the mirror, so absorption counts double.

Carbon nanotube membranes are the intended fix, still short of production in a leading-edge fab. Imec has mounted them on reticles and exposed them in an NXE:3300 scanner, measuring single-pass transmission up to 97 percent and finding the effect on imaging low and correctable [12]. Mitsui Chemicals, licensed by ASML, is building capacity at Iwakuni-Ohtake for membranes above 92 percent transmittance that can withstand sources above 1,000 W [13].

6 The Japanese glass behind next-generation chips. AGC [A] agc.com/en/hub/pr/the-japanese-glass-behind-next-generation-chips.html

7 AGC Data Book. AGC Inc. · 29 May 2026 [A] agc.com/en/ir/library/outline/pdf/reference.pdf

8 Information Technology Business | Review of Operations | HOYA REPORT 2025. HOYA [A] hoya.com/ir/2025/en/review/it.html

1 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC · 27 April 2023 [A] agc.com/en/news/detail/1203819_2814.html

9 Intel to Sell Minority Stake in IMS Nanofabrication Business to TSMC. Intel · 12 September 2023 [A] intc.com/news-events/press-releases/detail/1645/intel-to-sell-minority-stake-in-ims-nanofa…

10 Business Report | Lasertec Corporation. Lasertec [A] lasertec.co.jp/en/ir/plan/message.html

11 Indistinguishable from magic: the EUV pellicle. ASML · 14 September 2022 [A] asml.com/en/news/stories/2022/the-euv-pellicle-indistinguishable-from-magic

12 Imec demonstrates CNT pellicle utilization on EUV scanner. imec · 6 October 2020 [A] imec-int.com/en/press/imec-demonstrates-cnt-pellicle-utilization-euv-scanner

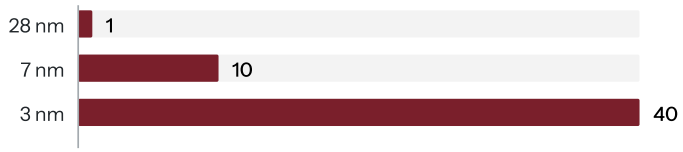
13 Mitsui Chemicals Sets up Production Facilities for CNT Pellicles to Be Used in Next-Gen EUV Lithography. Mitsui Chemicals [A] jp.mitsuichemicals.com/en/release/2024/2024_0528_1/index.htm

Cost and design economics

SemiAnalysis put mask sets beyond $1 million at 28 nm, beyond $10 million at 7 nm and near $40 million at 3 nm, as of 2022 [14].

That cost decides who can use a node at all. A GPU spread over hundreds of thousands of units absorbs a $40 million mask set easily; the same set makes a custom inference chip shipping in the thousands uneconomic [14]. That cost limits AI silicon to a handful of designs.

Cost of one mask set by node, 2022 estimates $M



Source: SemiAnalysis

Who makes it

Mask making splits between fabs that make their own and firms that sell to everyone. TSMC, Samsung and Intel run their own leading-edge mask shops, because they want to control turnaround time. Photronics is the largest outside supplier, on $849.3 million of fiscal 2025 revenue, of which $615.1 million was IC masks [15]. Toppan, Dai Nippon Printing and Hoya take most of the rest.

SMIC runs two in-house mask shops in Shanghai, for g-line, i-line, KrF and ArF tools [16]. There is no Chinese EUV blank, no Chinese actinic inspection tool and no Chinese multi-beam writer.

The chokepoint

Mask blanks are as concentrated as lithography, on a fraction of the money. Three firms supply EUV blanks worldwide, AGC says [6], one firm certifies them, and neither business is big enough to attract a funded entrant. Nobody starts from scratch at ¥40 billion of annual sales, to serve a handful of customers, with twenty years of defect learning ahead [1]. A business too small to fund a fourth entrant has kept the count at three.

A decade on, no pellicle membrane is transparent, tough and long-lived at once, so fabs trade throughput against defect risk. A production carbon nanotube pellicle would move EUV cost per wafer more than any scanner improvement.

14 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due To Photomask Sets, Verification, and Validation. SemiAnalysis · 24 July 2022 [B] newsletter.semianalysis.com/p/the-dark-side-of-the-semiconductor

15 Photronics Reports Full Year and Fourth Quarter Fiscal 2025 Results. GlobeNewswire (release by Photronics) · 10 December 2025 [A] globenewswire.com/news-release/2025/12/10/3203040/0/en/Photronics-Reports-Full-Year-and-Fo…

16 Mask Service. SMIC [A] smics.com/en/site/mask

6 The Japanese glass behind next-generation chips. AGC [A] agc.com/en/hub/pr/the-japanese-glass-behind-next-generation-chips.html

1 AGC to Boost Production Capacity of EUVL Photomask Blanks. AGC · 27 April 2023 [A] agc.com/en/news/detail/1203819_2814.html

Key evaluation criteria

- **Blank defectivity.** Particles buried in the multilayer cannot be repaired, so blank yield sets mask yield.
- **Write time and placement.** Multi-beam writers make dense, curved EUV patterns affordable, and Intel calls IMS the established industry leader in them [9].
- **Actinic inspectability.** Only a 13.5 nm inspection sees what the scanner sees, and only Lasertec sells one [10].
- **Pellicle transmission and lifetime.** Transmission counts twice, and the best demonstrated carbon nanotube membranes pass 97 percent in a single crossing [12].
- **Mask-set cost per node.** $10M to $40M at the leading edge, which decides which designs can use it [14].

9 Intel to Sell Minority Stake in IMS Nanofabrication Business to TSMC. Intel · 12 September 2023 [A] intc.com/news-events/press-releases/detail/1645/intel-to-sell-minority-stake-in-ims-nanofa...

10 Business Report | Lasertec Corporation. Lasertec [A] lasertec.co.jp/en/ir/plan/message.html

12 Imec demonstrates CNT pellicle utilization on EUV scanner. imec · 6 October 2020 [A] imec-int.com/en/press/imec-demonstrates-cnt-pellicle-utilization-euv-scanner

14 The Dark Side Of The Semiconductor Design Renaissance – Fixed Costs Soaring Due To Photomask Sets, Verification, and Validation. SemiAnalysis · 24 July 2022 [B] newsletter.semianalysis.com/p/the-dark-side-of-the-semiconductor

CHAPTER
06
STAGE 6 OF 15

Deposition and Etch

Lithography prints the pattern; deposition and etch do the rest, over a thousand steps per wafer, and it is the one kind of chipmaking tool where Chinese makers have gained real share.

IN PLAIN TERMS

Deposition and etch build the chip itself, one layer at a time. Deposition lays down a film a few atoms thick. Lithography prints a pattern on it. Etch then eats away everything the pattern does not protect, and the next film goes down on top. A chip takes more than a thousand of these rounds. Some of the holes the etch cuts are hundreds of times deeper than they are wide, the shape of a finger-wide shaft dropping through several floors, and the film has to coat them right to the bottom. The etch has to eat one material and leave the one beside it untouched. Four firms in America, Japan and the Netherlands make most of these machines, and China has come closer to matching them here than anywhere else.

IN SHORT

Deposition and etch are the volume business of a fab: over a thousand steps per wafer, four American, Japanese and Dutch firms taking most of the money, and no monopoly like ASML's in lithography. The AI roadmap makes each step harder faster than it adds steps, because gate-all-around, HBM and backside power ask each chamber for better selectivity and deeper holes. China has closed the gap fastest in deposition and etch and slowest in ALD, the tool gate-all-around logic depends on.

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Most deposition and etch steps have three or four approved tool makers; atomic layer deposition is the exception.	Tool makers publish no list prices World sales of chipmaking equipment were $135.1B in 2025, with wafer processing tools up 12%.

WHO LEADS

- US **Applied Materials** $28.4B revenue, fiscal 2025; leader in physical and chemical vapor deposition and in epitaxy
- US **Lam Research** $23.2B revenue, fiscal 2026; leader in dry etch and in metal atomic layer deposition
- JP **Tokyo Electron** 23% of dry etch and 38% of chemical vapor deposition, 2025
- NL **ASM International** €3.2B revenue, 2025; leader in single-wafer atomic layer deposition
- CN **Naura** 5th largest equipment vendor worldwide, 2025 estimate

WHERE IT IS MADE

- US **United States** Applied Materials (Santa Clara), Lam Research (Fremont)
- JP **Japan** Tokyo Electron, Kokusai Electric, Hitachi High-Tech
- NL **Netherlands** ASM International, Almere; ALD and epitaxy
- KR **South Korea** Large buyer; Semes and local suppliers in wet processing
- CN **China** Naura, AMEC, Piotech; largest equipment market at $49.3B in 2025

WHY SUBSTITUTION IS POSSIBLE

For most of the deposition and etch steps inside a fab, the tool can already be bought from three or four makers: Applied Materials, Lam, Tokyo Electron and ASM International. Switching from one maker to another means testing the new tool on the fab's own line until it matches, which takes two to five years for the newest chips. The exception is atomic layer deposition, which lays down a film one layer of atoms at a time. Chinese tool makers gained real share in etch over five years but stayed under 1 percent in atomic layer deposition, so that is the step where China is weakest.

WHERE CHINA STANDS

Of all kinds of chipmaking tool, deposition and etch is where Chinese makers are strongest. Chinese suppliers held about 11 percent of dry etch and 7 percent of deposition in 2024, but under 1 percent of atomic layer deposition.

WHERE THE US STANDS

Two of the three largest deposition and etch tool makers are American. The US government lists dry etch, atomic layer deposition and deep-hole deposition tools as controlled exports, and one rule also covers tools built outside the United States with American technology.

Applied Materials and Lam Research, the two largest deposition and etch suppliers, booked $28.4 billion and $23.2 billion of revenue in their latest fiscal years [1] [2].

How it works

Both steps happen in a vacuum chamber that holds one wafer at a time. A pump empties the chamber of air, a measured flow of gas comes in, and an electric field that reverses millions of times a second, a radio-frequency field, strips electrons off the gas molecules. The gas is now a plasma, charged fragments that react far more readily than whole molecules do.

For deposition the gas carries the material wanted. The fragments land on the wafer and stick, and the film builds from the bottom up, a few atoms at a time. To coat the floor of a deep hole as thickly as its rim, the machine can work in cycles: let one gas settle in a single layer on every surface, pump it out, admit a second gas that reacts with that layer and nothing else, pump again. Each cycle adds one layer of atoms wherever the gas reached, so the film gets to the bottom of holes hundreds of times deeper than they are wide.

For etch the gas attacks one material and leaves the one beside it. Wherever the photoresist pattern covers the wafer, nothing happens. Wherever it does not, the fragments react with the surface and turn it into a gas, which the pump carries away. A second field pulls the charged fragments straight down at the wafer, so the etch cuts vertically and a narrow hole keeps its width as it deepens. When the etch reaches the material underneath, which the gas was chosen not to touch, it slows almost to a stop, and the tool ends the step.

Change the gas, the pressure, the power and the temperature, and the same hardware either adds material or takes it away, which is why the export controls on these tools are written as process specifications.

CSET puts a single chip at more than 1,000 process steps, with the part-finished product crossing international borders 70 or more times [3].

Variants and trade-offs

Deposition families

The market sizes and leading shares below are CSET's for 2025, which it builds from TechInsights data, on a $26.2 billion deposition tool market [4] and a $21.9 billion dry etch market [5].

- **Physical vapor deposition** knocks atoms off a solid target and lands them on the wafer, the standard method for metal barriers and seed layers, $4.9 billion

1 Revenue from Contract with Customer, Excluding Assessed Tax (us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax) reported by APPLIED MATERIALS INC /DE, XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for APPLIED MATERIALS INC /DE) A
data.sec.gov/api/xbrl/companyconcept/CIK0000006951/us-gaap/RevenueFromContractWithCustomer...

2 LAM RESEARCH CORP, Form 8-K current report for the period ended 2026-07-29 (8-K). U.S. Securities and Exchange Commission (filing by LAM RESEARCH CORP) · 29 July 2026 A
sec.gov/Archives/edgar/data/707549/000070754926000033/lrcx_exhibitx991xq4x2026.htm

3 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 A
cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

4 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

5 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

in 2025 [6].

- **Chemical vapor deposition** reacts gases at the wafer surface to grow a film; the plasma-enhanced version runs cooler and is most of the $11.2 billion CVD market, $7.5 billion in 2025 [7] [8].
- **Atomic layer deposition** lays down less than one atomic layer at a time, so it coats vertical walls as evenly as horizontal floors. US regulators call it the basis for 3D scaling in 3D DRAM, 3D NAND and gate-all-around logic [9]; ASM International leads it with 54.5 percent of a $4.1 billion market in 2025 [10].
- **Epitaxy and electroplating** grow the strained silicon-germanium source and drain, and fill the copper wiring trenches [3].

Etch families

- **Conductor etch** cuts gates, metals and silicon, $12.2 billion in 2025, led by Lam at 51.2 percent [11].
- **Dielectric etch** cuts the insulating oxides and nitrides between them, $8.7 billion in 2025, led by Tokyo Electron at 55.9 percent [12].
- **High-aspect-ratio etch** decides whether memory works at all: a 3D NAND channel or DRAM capacitor is a hole far deeper than it is wide that has to come out straight. Lam chills the wafer to sharpen the profile, and says Cryo 3.0 holds critical-dimension deviation under 0.1 percent down channels 10 microns deep [13].

AI hardware asks for sideways cuts, deeper holes and a second metal stack on the wafer's back.

- **Gate-all-around** needs isotropic etch, which cuts sideways, to release the silicon nanosheets from the silicon-germanium between them, and BIS ties its dry etch controls to exactly that [9].
- **HBM** needs holes etched straight through the silicon so chips can be stacked and wired, which BIS controls at 10:1 or deeper above 7 microns a minute [14].
- **Backside power delivery** adds thinning, backside vias and a second metal stack. Applied Materials expects it to add about $1 billion per 100,000 wafer starts a month to its wiring business [15].

6 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

7 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

8 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

9 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 A govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf

10 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

3 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 A cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

11 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

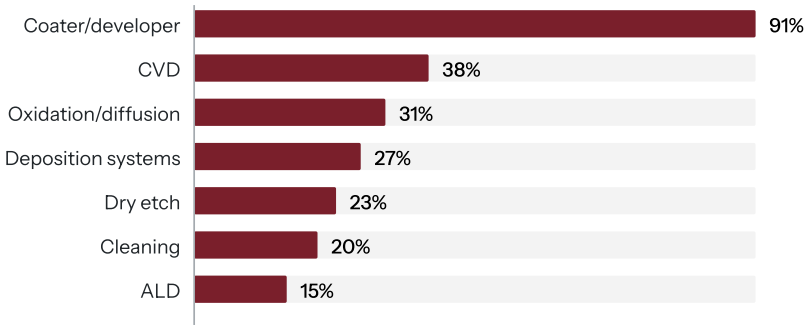
12 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

13 Lam Research Introduces Lam Cryo™ 3.0 Cryogenic Etch Technology to Accelerate Scaling of 3D NAND for the AI Era. Lam Research · 31 July 2024 A newsroom.lamresearch.com/2024-07-31-Lam-Research-Introduces-Lam-Cryo-TM-3-0-Cryogenic-Etch…

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 A govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf

15 Applied Materials Unveils Chip Wiring Innovations for More Energy-Efficient Computing. GlobeNewswire (release by Applied Materials) · 8 July 2024 A globenewswire.com/news-release/2024/07/08/2909540/0/en/applied-materials-unveils-chip-wiri…

Tokyo Electron world market share by tool type, 2025 %



Source: Tokyo Electron FY2026 Q4 results presentation

Who makes it

SEMI put global equipment billings at $135.1 billion in 2025, up 15 percent from $117.1 billion, with wafer processing equipment up 12 percent [17].

Within deposition and etch no one firm dominates as ASML does in lithography. The share chart in Tokyo Electron's own results presentation puts it at 23 percent of dry etch, 27 percent of deposition systems and 15 percent of ALD in 2025, against 91 percent of coater/developer [16].

ASM International, the atomic layer deposition specialist, reached a record EUR 3.2 billion of revenue in 2025 as customers built 2 nm gate-all-around capacity, with molybdenum ALD entering volume production [18].

What the world spent on chipmaking equipment $B



Source: SEMI, Worldwide Semiconductor Equipment Market Statistics

China and the controls

China is the tool vendors' largest customer and their fastest-improving competitor. It bought $49.3 billion of equipment in 2025, still the biggest single market, though its spending fell half a percent while Taiwan's rose 90 percent to $31.5 billion [17]. China still takes a large share of every major vendor's sales.

17 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in 2025, Up 15% Year-on-Year. SEMI · 7 April 2026 [A] semi.org/en/SEMI-Reports-Global-Semiconductor-Equipment-Billings-Reached-135-Billion-in-2025

16 Investor Relations / April 30, 2026. Tokyo Electron · 30 April 2026 [A] tel.com/ir/library/report/pjuomj00000000tf-att/fy26q4transcript-e.pdf

18 Press Release Q4, 2025 text. ASM International · 3 March 2026 [A] asm.com/media/yvxbavwe/20260303-asm-reports-q4-and-full-year-2025-results.pdf

- **Tokyo Electron**: 34.1 percent of sales in the year to March 2026, down from 41.7 percent [16].
- **Lam Research**: 26 percent in the June 2026 quarter, behind Taiwan at 27 percent [2].
- **Applied Materials**: 28 percent in the third fiscal quarter of 2026, down from 35 percent [19].
- **ASM International**: more than 30 percent of 2025 revenue [18].

Between 2019 and 2024 China-based suppliers went from 2 to 7 percent of deposition and from under 3 to about 9 percent of etch and clean, 11 percent in dry etch alone. In ALD they gained nothing and stayed under 1 percent [20]. AMEC, the biggest of them, turned over RMB 12.385 billion in 2025, up 36.6 percent, and spent RMB 3.744 billion on research, 30.2 percent of sales [21].

Washington tightened these rules four times between 2022 and 2025.

- **October 2022.** The 7 October interim final rule put manufacturing equipment and US person support under license and widened foreign-produced item rules for 28 listed Chinese entities [22].
- **October 2023.** The rule effective 17 November rewrote the equipment entries, adding isotropic and anisotropic dry etch (ECCN 3B001.c.1), spatial ALD (3B001.d.9), low-fluorine tungsten ALD and CVD (3B001.d.10) and carbon hard mask PECVD (3B001.d.5) [9].
- **December 2024.** BIS controlled 24 more types of equipment and three types of software, added 140 entities and wrote a foreign direct product rule that reaches tools built outside the United States [23]. The new entries take in TSV etch and deposition into features deeper than 200:1 [14].

16 Investor Relations / April 30, 2026. Tokyo Electron · 30 April 2026 [A] tel.com/ir/library/report/pjuomj00000000tf-att/fy26q4transcript-e.pdf

2 LAM RESEARCH CORP, Form 8-K current report for the period ended 2026-07-29 (8-K). U.S. Securities and Exchange Commission (filing by LAM RESEARCH CORP) · 29 July 2026 [A] sec.gov/Archives/edgar/data/707549/000070754926000033/lrcx_exhibitx991xq4x2026.htm

19 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials) · 13 August 2026 [A] globenewswire.com/news-release/2026/08/13/3344890/0/en/applied-materials-announces-third-q…

18 Press Release Q4, 2025 text. ASM International · 3 March 2026 [A] asm.com/media/yvxbavwe/20260303-asm-reports-q4-and-full-year-2025-results.pdf

20 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025 [A] cset.georgetown.edu/article/inside-beijings-chipmaking-offensive

21 AMEC holds its 2025 annual results briefing (中微公司成功举办2025年度业绩说明会). Advanced Micro-Fabrication Equipment (AMEC) · 1 April 2026 [A] amec-inc.com/news/708.html

22 Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification. U.S. Government Publishing Office (Federal Register) · 13 October 2022 [A] govinfo.gov/content/pkg/FR-2022-10-13/pdf/2022-21658.pdf

9 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 [A] govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf

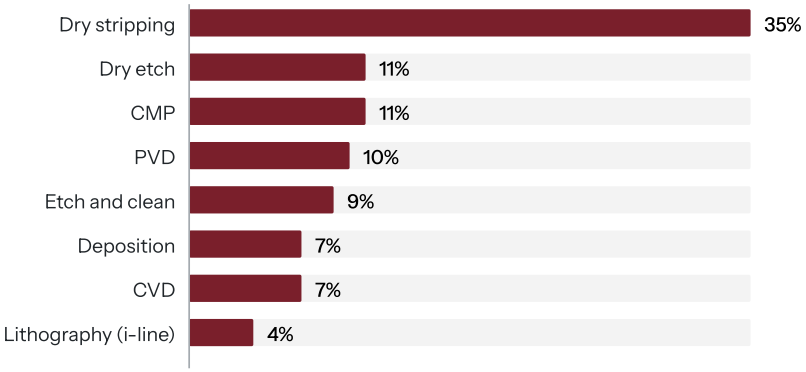
23 Commerce Strengthens Export Controls to Restrict China's Capability to Produce Advanced Semiconductors for Military Applications. U.S. Bureau of Industry and Security · 2 December 2024 [A] bis.gov/press-release/commerce-strengthens-export-controls-restrict-chinas-capability-prod…

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 [A] govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf

- **September 2025.** The Affiliates Rule extended Entity List restrictions to any company at least 50 percent owned by a listed entity [24]; BIS then stayed it to 9 November 2026 [25].

Applied Materials took a $253 million charge to settle an export controls matter [19], so the rules reach the vendors as well as their customers.

China-based suppliers' share of global tool segments, 2024 %



Source: CSET, Inside Beijing's Chipmaking Offensive

The chokepoint

The chokepoint is real but weaker than lithography's. Three or four credible vendors exist for most of the deposition and etch steps inside a fab. The exceptions are ALD, where China holds under 1 percent [20], and the extremely deep, narrow holes that HBM and 3D memory need. The December 2024 rule targets those [14].

Key evaluation criteria

- **Selectivity.** How hard the process attacks one material and spares another. BIS controls wet processing at a silicon-germanium-to-silicon ratio of 100:1 [9].
- **Conformality and aspect ratio.** Whether a film reaches the bottom of a hole, and how deep that hole is against its width. Controls start at 10:1 for TSV etch and 200:1 for 3D DRAM deposition [14].

24 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. U.S. Government Publishing Office (Federal Register) · 30 September 2025 [A] govinfo.gov/content/pkg/FR-2025-09-30/pdf/2025-19001.pdf

25 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain Listed Entities. U.S. Government Publishing Office (Federal Register) · 12 November 2025 [A] govinfo.gov/content/pkg/FR-2025-11-12/pdf/2025-19846.pdf

19 Applied Materials Announces Third Quarter 2026 Results. GlobeNewswire (release by Applied Materials) · 13 August 2026 [A] globenewswire.com/news-release/2026/08/13/3344890/0/en/applied-materials-announces-third-q…

20 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025 [A] cset.georgetown.edu/article/inside-beijings-chipmaking-offensive

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 [A] govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf

9 Export Controls on Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 25 October 2023 [A] govinfo.gov/content/pkg/FR-2023-10-25/pdf/2023-23049.pdf

- **Uniformity.** Variation across a 300 mm wafer, with under 2 percent treated as an advanced-manufacturing marker [14].
- **Throughput.** Wafers per hour, which sets cost per layer and decides how many lots a fab can run.
- **Particles and chamber matching.** How large a share of the defects a fab can tolerate comes from these tools, and whether one chamber behaves like the next, which metrology and inspection measure (see Metrology and inspection).

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 A govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf

CHAPTER 07
STAGE 7 OF 15

Metrology and Inspection

No fab is planned around measuring and inspecting wafers, and no fab works without it. One American firm holds about 57% of the market for those tools, roughly seven times its nearest rival, and its share is still growing.

IN PLAIN TERMS

Metrology and inspection are the factory's quality control. Between the steps that build a chip, another machine looks the wafer over. It checks whether the lines are the right width, whether this layer landed squarely on the last one, and whether a speck of dirt has killed a circuit. An AI accelerator is the part built to run AI, and its main chip, the one that does the calculating, is one of the largest cut from a wafer. A bigger chip is a bigger target for a speck. On a chip that size, halving the stray specks lifts the share of working chips from about half to about seven in ten. One American firm sells seven times as many of those machines as its nearest rival, so a chip factory, or fab, cut off from it can buy every other tool and still not learn why its chips fail.

IN SHORT

Yield is set by defect density and die area, and AI accelerators have the largest dies in production. KLA holds about 57 percent of the market, six to seven times its nearest rival, and is still gaining [2], with Applied Materials, Lasertec, Hitachi High-Tech, ASML, Nova, Onto and Camtek splitting the rest [3]. KLA's packaging process control revenue is heading toward $1.1 billion in 2026 [1]. No one has recorded a Chinese gain in process control, which is why the controls on it are the ones China can least easily answer with domestic tools.

2 Compounding Sustainable Outperformance. KLA Corporation · 13 March 2026 [A] d1io3yog0oux5.cloudfront.net/_7791115a123b86b3f10b1a5eb5210224/klatencor/db/1166/10653/fil…

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) [A] chipexplorer.eto.tech

1 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation · 27 July 2026 [A] d1io3yog0oux5.cloudfront.net/_b9ee755a5f60dd0fb3f9e27967aed6af/klatencor/db/1117/10668/let…

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate KLA is the largest maker of measuring and inspecting tools, but every kind has a second maker, so losing KLA would slow a fab without stopping it.	**A $15.7B process control tool market in 2025, about 57% of it KLA's** KLA's fiscal 2026 revenue was $13.58B against a wafer equipment market of roughly $120B in 2025, so process control is around a tenth of what a fab spends on tools.

WHO LEADS

- US **KLA** 56.5% of process control in 2024 on its own investor-day count, 6.5 times its nearest rival; $13.58B revenue, fiscal 2026
- US **Applied Materials** 11.2% of process control tools in 2025: electron-beam defect review, line-width measurement and film metrology
- JP **Lasertec** 6.4% of process control tools in 2025, almost all of it mask inspection
- NL **ASML** 5.9% of process control tools in 2025: YieldStar layer alignment and HMI multibeam electron inspection
- IL **Nova** 4.3% of process control tools in 2025; $880.6M revenue, 2025
- US **Onto Innovation** 2.6% of process control tools in 2025; $1.01B revenue, 2025

WHERE IT IS MADE

- US **United States** KLA (Milpitas), Onto Innovation (Wilmington, MA), Applied Materials
- NL **Netherlands** ASML YieldStar and HMI e-beam, Veldhoven and San Jose
- JP **Japan** Hitachi High-Tech CD-SEM; Lasertec mask inspection; Rigaku X-ray
- IL **Israel** Nova (Rehovot) and Camtek (Migdal Haemek)
- CN **China** Domestic entrants only; 0.7% of the process control tool market in 2025

WHY SUBSTITUTION IS POSSIBLE

Metrology and inspection tools measure and inspect wafers so a fab can find and fix faults in its process. KLA is the biggest maker, but seven other firms ship approved tools, and three of them are the leader in one kind of tool. A fab that loses KLA and buys from the others keeps running. It finds fewer flaws and fixes them more slowly, and it needs two to five years to return to its old performance. A country that is not allowed to buy from any of them would take far longer: Chinese firms held 0.7 percent of the market for these tools in 2025.

WHERE CHINA STANDS

Of all major kinds of chipmaking tool, metrology and inspection is where Chinese makers are weakest. In 2019

Chinese firms held 3.4 percent of line-width measurement, 1.3 percent of defect inspection and none of mask or packaging inspection. In 2025 they still held 0.7 percent of the whole market for these tools. CSET's 2026 survey of Chinese share gains covers deposition, etch, polishing, lithography, packaging and test, and does not cover measurement and inspection.	WHERE THE US STANDS KLA is the largest maker of metrology and inspection tools in the world. It puts its own 2024 share at 56.5 percent, or 6.5 times its nearest rival. CSET's Supply Chain Explorer puts it at 56.8 percent of the 2025 market. American firms hold 72 percent of this market and Japanese firms 14 percent. A December 2024 US rule added inspection and measurement tools for patterned 300 mm wafers to the controlled list, for tools that can find defects 21 nm across or smaller.

Process control is about a tenth of what a fab spends on tools, out of a wafer equipment market of roughly $120 billion in 2025 [1]. At its March 2026 investor day KLA put its own 2024 share at 56.5 percent, six and a half times its nearest competitor [2]; CSET's own count for 2025 is 56.8 percent [3].

How it works

An inspection tool sweeps a beam of light across the wafer, photographs it, and compares the image of each chip with the image of its neighbors. The chips on a wafer are meant to be identical, so anything that shows in one and not the others is a defect, and the tool records where it is. For faults too small for light to show, a second tool scans a beam of electrons across a small patch and builds a picture from the electrons that bounce back.

A metrology tool measures known shapes. It shines light on a test pattern printed beside the circuits and works out from the scatter how wide the lines are and how thick the film is, and it reads marks printed in this layer and the last to measure how far the two are out of line, which is overlay.

The fab pays for all this to protect yield, the share of chips on a wafer that work. A wafer loses value two ways: whole regions ruined by mishandling, misalignment or over-etching, and single circuits killed by stray material or an airborne particle [4].

A bigger chip is a bigger target for the second kind, and the loss grows faster than the area. In the standard formula, die yield is (1 + D0A/a) to the power minus a, where D0 is defects per square centimeter, A is die area and a is a clustering constant of 3 to 4, which says how much defects bunch together. At a defect

1 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation · 27 July 2026 A d1io3yog0oux5.cloudfront.net/_b9ee755a5f60dd0fb3f9e27967aed6af/klatencor/db/1117/10668/let…

2 Compounding Sustainable Outperformance. KLA Corporation · 13 March 2026 A d1io3yog0oux5.cloudfront.net/_7791115a123b86b3f10b1a5eb5210224/klatencor/db/1166/10653/fil…

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

4 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara · 11 October 2020 A web.ece.ucsb.edu/~parhami/docs_folder/f33-book-dep-comp-pt2.pdf

density of 0.8 a 1 square centimeter die yields 49 percent and a 2 by 2 centimeter die 11 percent, so the bigger die costs about twenty times as much [4].

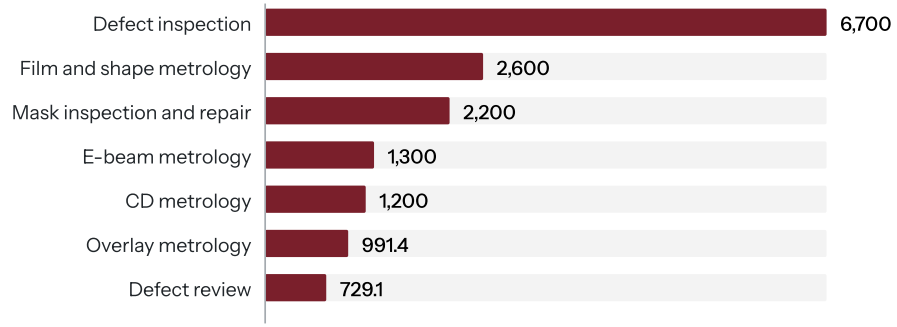
Run that equation with a clustering constant of 3 for an 8 square centimeter die, about as much as one exposure can print, and it yields about 49 percent at a defect density of 0.10 and about 69 percent at 0.05 [4]. Getting from one to the other is yield learning: find which of the thousand-odd steps is losing dies, fix it, move on.

Variants and trade-offs

The tools trade sensitivity against speed. All the market sizes and shares below are CSET's, for 2025 [3].

- **Defect inspection** scans a whole wafer with light or electrons and compares each die against its neighbors. It is by far the largest family, $6.7 billion, and the one KLA dominates, at 85.9 percent. ASML's HMI eScan 1100 runs 25 electron beams at once, up to 15 times the speed of a single beam, and finds patterning defects down to 7 nm [5].
- **Film and shape metrology** measures how thick each film is and how flat the wafer sits, without cutting it open: $2.6 billion, with KLA at 45.6 percent, Nova at 24.6 and Onto Innovation at 10.9. Onto paid about $720 million in August 2026 for 27 percent of Japan's X-ray maker Rigaku [6].
- **Mask inspection and repair** gets its own tools, because a defect on a mask repeats on every die: $2.2 billion, and the one family where KLA has a close rival, 45.9 percent against Lasertec's 42 and Zeiss's 10.1.
- **Overlay metrology** measures whether this layer sits on the one below: $991 million, split almost evenly between KLA at 51.5 percent and ASML at 47.9.
- **Critical dimension metrology** measures how wide a printed feature actually came out: $1.2 billion, and the one family Japan leads, with Hitachi at 71.9 percent and Applied Materials at 27.5.
- **Electron-beam metrology** resolves detail too fine for light: $1.3 billion, led by Applied Materials at 45.1 percent with ASML at 35.6 and KLA at 18.8.

Process control market by tool family, 2025 $M



Source: ETO Supply Chain Explorer, CSET

4 Behrooz Parhami, Dependable Computing: A Multilevel Approach, part 2. University of California, Santa Barbara · 11 October 2020 [A] web.ece.ucsb.edu/~parhami/docs_folder/f33-book-dep-comp-pt2.pdf

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) [A] chipexplorer.eto.tech

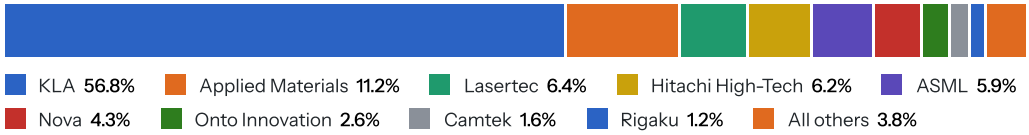
5 The HMI eScan 1100. ASML [A] asml.com/en/products/metrology-and-inspection-systems/hmi-escan-1100

6 Onto Innovation Completes Strategic Investment in Rigaku Holdings Corporation. Onto Innovation · 10 August 2026 [A] investors.ontoinnovation.com/news/news-details/2026/Onto-Innovation-Completes-Strategic-In...

Who makes it

KLA is most of the segment: 56.8 percent of a $15.7 billion market in 2025 [3], up two points since 2021 by its own figures [2]. Its fiscal 2026 revenue was $13.579 billion, after $12.156 billion in fiscal 2025 [7], and it says it gained share again in 2025 across mask, optical wafer and e-beam inspection [8].

Process control tool revenue share, 2025 %



KLA 56.8% Applied Materials 11.2% Lasertec 6.4% Hitachi High-Tech 6.2% ASML 5.9%
Nova 4.3% Onto Innovation 2.6% Camtek 1.6% Rigaku 1.2% All others 3.8%

Source: ETO Supply Chain Explorer, CSET

The rest are specialists, and nobody else reaches 11 percent.

- **Applied Materials** is the largest of them at 10.5 percent, selling e-beam defect review, CD-SEM line-width tools and film metrology next to the deposition and etch tools that create the defects.
- **Lasertec** holds 7.7 percent, almost all of it mask inspection; **Hitachi High-Tech** 5.6 percent, almost all of it CD-SEM; **ASML** 5.2 percent through YieldStar overlay and HMI e-beam inspection.
- **Nova** holds 3.7 percent and took $880.6 million in 2025 on optical film and dimensional metrology [9].
- **Onto Innovation** holds 3.0 percent on $1,005.3 million of 2025 revenue [10] and **Camtek** 1.5 percent on $496.1 million [11].

Patterning, which covers overlay and mask inspection, grew 61 percent year on year in the June 2026 quarter, when Taiwan took 31 percent of revenue and China 26 percent [1].

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) [A] chipexplorer.eto.tech

2 Compounding Sustainable Outperformance. KLA Corporation · 13 March 2026 [A] d1io3yog0oux5.cloudfront.net/_7791115a123b86b3f10b1a5eb5210224/klatencor/db/1166/10653/fil...

7 Revenue from Contract with Customer, Excluding Assessed Tax (us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax) reported by KLA CORP, XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for KLA CORP) [A] data.sec.gov/api/xbrl/companyconcept/CIK0000319201/us-gaap/RevenueFromContractWithCustomer...

8 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation · 29 April 2026 [A] d1io3yog0oux5.cloudfront.net/_a357bfc9113388e37f3bfcb2ea2f0b64/klatencor/db/1117/10655/let...

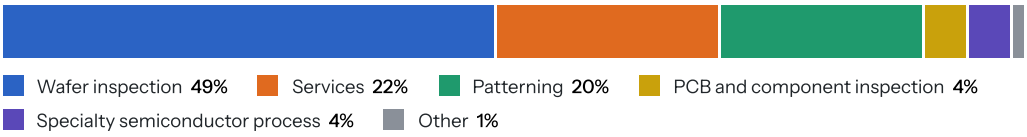
9 Revenues (us-gaap:Revenues) reported by NOVA LTD., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for NOVA LTD.) [A] data.sec.gov/api/xbrl/companyconcept/CIK0001109345/us-gaap/Revenues.json

10 Onto Innovation Reports 2025 Fourth Quarter and Full Year Results. Onto Innovation · 19 February 2026 [A] investors.ontoinnovation.com/news/news-details/2026/Onto-Innovation-Reports-2025-Fourth-Qu...

11 CAMTEK LTD, Form 6-K report of foreign private issuer for the period ended 2026-02-18 (6-K). U.S. Securities and Exchange Commission (filing by CAMTEK LTD) · 18 February 2026 [A] sec.gov/Archives/edgar/data/1109138/000117891326000549/zk2634405.htm

1 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation · 27 July 2026 [A] d1io3yog0oux5.cloudfront.net/_b9ee755a5f60dd0fb3f9e27967aed6af/klatencor/db/1117/10668/let...

KLA revenue by product line, June 2026 quarter %



Source: KLA letter to shareholders, Q4 FY2026

Process control's share of fab spending is still rising, and KLA credits tighter specifications, less built-in redundancy and more logic inside memory built for high-performance computing [8].

The advantage is analytics as much as optics: inspection produces far more candidate defects than engineers can look at, and KLA has spent five years putting AI classification through its whole line [8].

Packaging, AI and China

The AI packaging build-out is where this segment is growing fastest. KLA took the number one position in advanced wafer-level packaging process control in 2025, adding 14 points [8], and expects about $1.1 billion of packaging process control revenue in 2026, more than 70 percent growth [1]. Onto signed a volume purchase agreement worth over $240 million with an HBM maker for 2D inspection and 3D bump metrology through 2027 [10].

China has made less progress in process control than anywhere else in the tool chain. In 2019 Chinese firms held 3.4 percent of line-width measurement, 1.3 percent of defect inspection and none of photomask or packaging inspection [12]. Six years later they were at 0.7 percent of the whole process control tool market [3]. CSET's 2026 audit of where China gained runs through deposition, etch, polishing, lithography, packaging and test, and never reaches process control [13].

The December 2024 rule wrote process control into the export lists. It controls defect inspection and measurement of patterned 300 mm wafers where the tool can find defects of 21 nm or smaller using light below 400 nm, or an electron beam resolving 1.65 nm, or a cold field emission source, or two or more beams [14].

8 Letter to Shareholders Q3 Fiscal 2026. KLA Corporation · 29 April 2026 A d1io3yog0oux5.cloudfront.net/_a357bfc9113388e37f3bfcb2ea2f0b64/klatencor/db/1117/10655/let...

1 Letter to Shareholders Q4 Fiscal 2026. KLA Corporation · 27 July 2026 A d1io3yog0oux5.cloudfront.net/_b9ee755a5f60dd0fb3f9e27967aed6af/klatencor/db/1117/10668/let...

10 Onto Innovation Reports 2025 Fourth Quarter and Full Year Results. Onto Innovation · 19 February 2026 A investors.ontoinnovation.com/news/news-details/2026/Onto-Innovation-Reports-2025-Fourth-Qu...

12 The Semiconductor Supply Chain - Issue Brief. Center for Security and Emerging Technology (CSET) · 21 January 2021 A cset.georgetown.edu/wp-content/uploads/The-Semiconductor-Supply-Chain-Issue-Brief-1.pdf

3 Supply Chain Explorer: Advanced Chips. Emerging Technology Observatory (ETO) A chipexplorer.eto.tech

13 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025 A cset.georgetown.edu/article/inside-beijings-chipmaking-offensive

14 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. U.S. Government Publishing Office (Federal Register) · 5 December 2024 A govinfo.gov/content/pkg/FR-2024-12-05/pdf/2024-28270.pdf

A foreign direct product rule carries the same reach to tools built outside the United States [15].

The chokepoint

Process control is a chokepoint of concentration. Several firms can build a good film metrology tool, but one firm sets the standard across nearly the whole flow, and its lead is widening. A sanctioned fab can buy scanners and etchers on the gray market and still fail to get production going, because without sensitive inspection it cannot find out why its yield is low.

Key evaluation criteria

- **Sensitivity.** The smallest killer defect a tool can find. ASML's multibeam system is specified to 7 nm patterning defects [5].
- **Throughput.** Wafers per hour, which decides how many lots a fab can afford to look at; multibeam buys up to 15x over single-beam [5].
- **Nuisance rate.** The share of flagged events that turn out not to be real defects, which is what AI classification is meant to reduce.
- **Sampling plan.** How many wafers and sites per lot, a cost decision as much as a physics one, which sets what process control costs the fab.
- **Time to result.** How fast a measurement reaches the process engineer, which sets the speed of yield learning.

15 Commerce Strengthens Export Controls to Restrict China's Capability to Produce Advanced Semiconductors for Military Applications. U.S. Bureau of Industry and Security · 2 December 2024 [A] bis.gov/press-release/commerce-strengthens-export-controls-restrict-chinas-capability-prod…

5 The HMI eScan 1100. ASML [A] asml.com/en/products/metrology-and-inspection-systems/hmi-escan-1100

CHAPTER
08
STAGE 8 OF 15

Transistors and the Front End

Three companies can build a 2 nm-class transistor. Shrinking stopped lowering the cost per transistor about a decade ago, and every gain since has come from extra process steps.

IN PLAIN TERMS

The front end of a chip factory is the part that builds the transistors, the billions of tiny switches in every chip. Each switch is a valve. Current runs along a narrow strip of silicon called the channel, and a small voltage on a gate above the strip opens or shuts the flow. Shorter strips switch faster and more of them fit on a chip, but make the strip short enough and the valve stops sealing: current leaks through even when the gate is off, and the chip burns power doing nothing. The fix is to wrap the gate around all four sides of the strip, so it can squeeze the flow shut from every side, and that takes a long sequence of steps that have to run in exact order. Only three companies, TSMC, Intel and Samsung, can do it, a list so short that an export control can name every one of them.

IN SHORT

Every gain since the FinFET era has come from process steps and new materials, while shrinking the geometry gave less each node. TSMC's own numbers show the density gain per node falling from 1.2 times at N2P to about 1.1 at A16 [4]. Three companies can run a 2 nm-class process, and only TSMC runs it at volume for outside customers. SRAM has stopped shrinking, which is why accelerators keep their memory off the logic die. China can reach 5 nm-class geometry without EUV, and CSIS still judges that China cannot build a working EUV system, which is the gap the export controls defend [16].

4 HPC Platform – Advanced Technologies. TSMC [A] tsmc.com/english/dedicatedFoundry/technology/platform_HPC_tech_advancedTech

16 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements | Strategic Technologies Blog. Center for Strategic and International Studies [A] csis.org/blogs/strategic-technologies-blog/breakthroughs-or-boasts-assessing-recent-chines...

CHOKEPOINT CARD | EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	**Hard** Three firms make gate-all-around transistors, the newest kind, and none sells its process recipe.	**No foundry publishes prices for its newest wafers** The public measure is how many more transistors TSMC fits on each new process, and that gain falls from 1.2x at N2P to about 1.1x at A16.

WHO LEADS

- TW **TSMC** N2 in volume production since 4Q25, the first nanosheet node
- US **Intel** 18A in high-volume manufacturing; 18A-P adds 9% performance at the same power
- KR **Samsung Foundry** SF2 family; SF2Z adds backside power
- CN **SMIC** N+2 and N+3, printed in several passes on older deep-ultraviolet tools, no EUV
- JP **Rapidus** State-backed 2 nm entrant targeting mass production in 2027

WHERE IT IS MADE

- TW **Taiwan** TSMC N2 at Fab 20 Hsinchu and Fab 22 Kaohsiung
- US **United States** Intel 18A at Fab 52, Arizona; TSMC Arizona on N4 and N3
- KR **South Korea** Samsung Hwaseong and Pyeongtaek
- JP **Japan** Rapidus Chitose, Hokkaido
- CN **China** SMIC Shanghai and Beijing 300 mm lines

WHY SUBSTITUTION IS SLOW

The scarce thing is the process recipe: which steps, in which order, on which tools. Three firms have written one, so a newcomer knows it can be done but has to work out its own. Rapidus in Japan, backed by the state, is now finding out what that costs when starting with no line, on 267.6 billion yen and a 2027 target. Anyone who can buy extreme-ultraviolet scanners should expect five to ten years. China cannot buy those scanners. It has reached 5 nm-class features by printing each layer several times, and that keeps it behind the newest chips.

WHERE CHINA STANDS

SMIC ships a process it calls N+3, measured at 113.4 million transistors per square millimeter. It prints each layer in two or four passes because it cannot buy an extreme-ultraviolet scanner. Its narrowest wires sit closer together than Intel 18A's, which shows what a decade of state money buys without an EUV scanner.

WHERE THE US STANDS

Intel runs the only American-owned line that makes the newest transistors. Its 18A process brought gate-all-around transistors and backside power, where power is fed from under the transistors, to market together. Intel is also the first to use High-NA EUV scanners, the newest kind, in volume production.

A Blackwell GPU is 208 billion switches on two dies, each the largest a lithography machine can print in one shot [1]. Every switch has to turn on fully, turn off completely, and keep doing it for years. Making the switches is the front end of line; wiring them together is the back end.

How it works

The front end builds that valve in a fixed order. First it marks where each transistor will sit and fires atoms of boron or phosphorus into the silicon there, a step called ion implantation, so that those patches conduct the way the design needs.

Next it shapes the channel, in a FinFET by etching the silicon into thin fins standing up from the wafer and laying the gate over the top and both sides of each fin, three sides instead of one. In a gate-all-around transistor the fab instead grows alternating layers of silicon and silicon-germanium, cuts the stack into a narrow bar, then dissolves the silicon-germanium away, leaving thin silicon sheets held at their ends like shelves, and fills the gaps above, below and beside each sheet with gate metal, all four sides. Before the metal goes in, a film of insulator a few atoms thick is laid over every sheet, so that the gate controls the channel without touching it. Last come the contacts, small metal plugs down to the two ends of the channel and to the gate, ready for the wiring layers above.

Each of those is a deposition, an etch or an implant of its own, and they have to run in exact order, since a sheet cannot be wrapped once the metal is on, nor the insulator added once it is buried.

The leak the wrapping prevents is called short-channel leakage, and it has three fixes:

- **Better insulator.** Thicker, so fewer electrons slip straight through it, yet controlling the channel as tightly as before: Intel switched the gate insulator from silicon dioxide to hafnium at 45 nm in 2007 [2].
- **More gate.** Wrap gate metal around more sides of the channel, as fins and nanosheets do.
- **Thinner channel.** Thin the silicon until no current can flow where the gate cannot reach it.

Variants and trade-offs

Planar and FinFET

A planar transistor lays the gate on one side of a flat channel, and loses control as the channel shortens. A fin stands the channel on edge so the gate covers three sides, but current then comes in whole fins: a designer who needs more adds one, and nobody gets half a fin.

1 NVIDIA Blackwell Architecture. Nvidia · 18 December 2025 A nvidia.com/en-us/data-center/technologies/blackwell-architecture

2 Intel's Transistor Technology Breakthrough Represents Biggest Change to Computer Chips in 40 Years. Intel · 27 January 2007 A intel.com/pressroom/archive/releases/2007/20070128comp.htm

Gate-all-around nanosheets

A nanosheet lays the channel back down, stacks two to four thin sheets and threads gate metal around each one. The sheets can be drawn any width, so current comes in fine steps again, which TSMC sells as NanoFlex.

N2 entered volume production in the fourth quarter of 2025, TSMC's first nanosheet node [3]. TSMC rates the follow-on N2P at 18 percent more speed at the same power, 36 percent less power at the same speed, and 1.2 times the logic density of N3E [4]. Intel's RibbonFET shipped on 18A, and 18A-P adds another 9 percent performance at the same power [5].

Those numbers are vendor claims until someone measures the silicon. Each foundry picks its own baseline, and none publishes a density figure that can be set against a rival's.

TSMC's published density gain, node over node X VERSUS THE BASELINE NODE



Source: TSMC advanced technologies, HPC platform

Backside power delivery

Power and signal share one stack of copper wiring above the transistors, where wide power rails crowd out the signal wires. Backside power grinds the wafer thin and builds a second network underneath it.

Intel shipped it first as PowerVia: on 18A, an 11 percent cut in routed area and a tenfold cut in dynamic voltage droop, the sag in supply voltage when a block of logic switches on at once [5]. TSMC's Super Power Rail lands that network straight on each transistor's source and drain, with no buried rail between, which is harder to build and gives more: against N2P, A16 gives 8 to 10 percent more speed, 15 to 20 percent less power and up to 1.10 times the chip density [4].

3 2nm Technology. TSMC [A] tsmc.com/english/dedicatedFoundry/technology/logic/l_2nm

4 HPC Platform – Advanced Technologies. TSMC [A] tsmc.com/english/dedicatedFoundry/technology/platform_HPC_tech_advancedTech

5 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel · 16 June 2026 [A] intc.com/news-events/press-releases/detail/1772/intel-foundry-details-process-milestones-a...

TSMC's logic roadmap beyond N2, from its own announcements

- 2028-01-01
  N2U, 3-4% faster or 8-10% lower power than N2P
- 2028-01-01
  A14, up to 15% faster and more than 20% denser than N2
- 2029-01-01
  A13, a direct shrink of A14 for 6% area savings
- 2029-01-01
  A12, the A14 platform with Super Power Rail backside power

Source: TSMC, 2026 North America Technology Symposium

CFET and what comes after

A logic gate needs two transistors that switch on opposite signals, and today they sit side by side. The research institute imec expects complementary FETs to stack them and roughly halve the pair's area beyond the 1 nm nodes [7].

The back end of line

Wiring is now as hard to make as the transistor. Copper seeps into the insulator around it, so every wire needs a barrier, a liner and a cap, and once the narrowest wires sit less than 20 nm apart those wrappers take a large share of the wire's cross-section [8]. Two metals are moving into those narrow levels.

- **Ruthenium** needs no barrier, so the whole wire conducts; imec has demonstrated ruthenium lines at a 16 nm pitch [9].
- **Molybdenum** conducts worse in bulk, 5.3 against copper's 1.68 microhm-cm, but in a thin film it beats tungsten by up to 30 percent and needs no barrier either, which is moving it into the contacts on each transistor [10].

Who makes it

TSMC, Intel and Samsung build gate-all-around nodes with extreme ultraviolet lithography. SMIC builds 7 and 5 nm-class nodes without it. State-backed Rapidus targets 2 nm mass production in 2027 on 267.6 billion yen raised in February 2026 [11].

Node names describe nothing physical. SMIC's N+3, sold as 5 nm-class, sets its narrowest wires 32.5 nm apart, tighter than Intel 18A's 36 nm, and reaches 113.4

7 CFET (complementary FET). imec [A] imec-int.com/en/articles/imec-puts-complementary-fet-cfet-logic-technology-roadmap

8 Semi-damascene metallization. imec [A] imec-int.com/en/articles/semi-damascene-metallization-inflection-point-back-end-line-proce…

9 16nm Ru lines using semi-damascene integration approach. imec · 3 June 2025 [A] imec-int.com/en/press/imec-demonstrates-16nm-pitch-ru-lines-record-low-resistance-obtained…

10 Molybdenum's Role in Ultra-Fast Computing: The Metal Behind the Speed. Entegris · 18 May 2026 [A] blog.entegris.com/molybdenums-role-in-ultra-fast-computing-the-metal-behind-the-speed

11 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private Sector Companies This strategic funding plan will enable Rapidus to steadily progress from its current R&D phase to mass production of 2nm logic semiconductors by 2027 - Information - Rapidus Corporation. Rapidus [A] rapidus.inc/en/news_topics/information/rapidus-secures-267-6-billion-yen-in-funding-from-j…

million transistors per square millimeter against 107.7 for TSMC's N6 [12].

AI accelerators sit a node behind the leading edge, because a die that large comes out working only on a process that has run for years. Blackwell uses a custom TSMC 4NP [1].

SRAM, the fast memory on the logic die, is the scaling failure that hurts AI chips most. TSMC claims 22 percent more SRAM density from N3E to N2, but the gain comes from the circuits around the array, and none of it from the cell that holds one bit [13], which is much of why accelerators push their memory off the die. See Memory and HBM.

The chokepoint

No single machine makes the front end a chokepoint. The recipe does, and a recipe is which steps run in which order on which tools.

China shows what that recipe costs. The third National Integrated Circuit Fund raised 344 billion yuan, about $47 billion, in May 2024 [14], so money is not the constraint; tools are. A Shanghai state firm began building domestic immersion scanners in 2026, in single-digit annual volumes [15]. Those are deep-ultraviolet machines, a generation behind EUV, and SMIC gets fine features out of them by printing each layer in several passes. That volume keeps those lines running and comes nowhere near replacing ASML. EUV is further away: China has announced $43 billion for domestic development, and the Center for Strategic and International Studies still judges that China cannot build a working system [16].

See Lithography and Geopolitics.

Key evaluation criteria

- **Electrostatic control.** How far the gate wraps the channel. It sets how much the switch leaks when off, and the lowest supply voltage it can run on.
- **Density.** Transistors per square millimeter, comparable only inside one foundry's own numbering and for one kind of cell.
- **Interconnect resistance.** Below a 20 nm wire spacing the wire becomes the slower part and sets the delay [8].
- **Power delivery.** The voltage lost carrying power across the die. Backside power frees area and cuts the droop tenfold [5].
- **SRAM scaling.** The cell that holds one bit has stopped shrinking, which caps how much cache an accelerator can afford.

12 Is SMIC N+3's Metal Pitch Smaller than Intel 18A's?. SemiAnalysis · 14 June 2026 [B] newsletter.semianalysis.com/p/steel-smic-n3-teardown

1 NVIDIA Blackwell Architecture. Nvidia · 18 December 2025 [A] nvidia.com/en-us/data-center/technologies/blackwell-architecture

13 Clash of the Foundries: Gate All Around + Backside Power at 2nm. SemiAnalysis · 1 October 2024 [B] newsletter.semianalysis.com/p/clash-of-the-foundries

14 China launches $47bn chip fund to counter U.S. restrictions. Nikkei Asia · 27 May 2024 [B] asia.nikkei.com/spotlight/supply-chain/china-launches-47bn-chip-fund-to-counter-u.s.-restr...

15 China starts production of home-grown immersion DUV chipmaking tools, source says. Reuters · 28 July 2026 [B] reuters.com/world/china/china-starts-production-home-grown-immersion-duv-chipmaking-tools-...

16 Breakthroughs or Boasts? Assessing Recent Chinese Lithography Advancements | Strategic Technologies Blog. Center for Strategic and International Studies [A] csis.org/blogs/strategic-technologies-blog/breakthroughs-or-boasts-assessing-recent-chines...

8 Semi-damascene metallization. imec [A] imec-int.com/en/articles/semi-damascene-metallization-inflection-point-back-end-line-proce...

5 Intel Foundry Details Process Milestones and Future Innovation at VLSI Symposium. Intel · 16 June 2026 [A] intc.com/news-events/press-releases/detail/1772/intel-foundry-details-process-milestones-a...

- **Cost per transistor.** No foundry publishes wafer prices, so the shrinking density gain per node is the proxy. On that proxy, cost per transistor is rising for the first time.

CHAPTER
09
STAGE 9 OF 15

Foundries and Fabs

TSMC runs six plants that each print more than 100,000 wafers a month. All six are in Taiwan, and they hold 13 of the 17 million wafers a year the company can make.

IN PLAIN TERMS

A foundry is a fab that makes chips for other companies. Those companies send in their designs, and the foundry runs them all through the same building on the same tools. The building costs tens of billions of dollars, and the air inside is cleaner than an operating room. Even so, a wafer crosses hundreds of machines over several months inside sealed boxes, never touching that air. No chip company sells enough of one product to fill a factory like that on its own, so the world shares a handful of them. TSMC runs the biggest, and all six of its largest plants sit in Taiwan, which is why one island's politics reaches the whole chip supply.

IN SHORT

The foundry model won because leading-edge fabs are too expensive for one design house, and it concentrated because only the leader earns enough to build the next node. TSMC's six gigafabs hold 13 of the 17 million wafers a year it can make, and all six are in Taiwan [1]. Intel and Samsung are credible technically and unproven commercially, and Intel is now part-owned by its government [10]. Diversification costs $165 billion and a decade in the American case alone [9], which makes the concentration in Taiwan a fact about the 2020s and 2030s.

1 **GIGAFAB® Facilities.** TSMC [A] tsmc.com/english/dedicatedFoundry/manufacturing/gigafab

10 **Intel and Trump Administration Reach Historic Agreement to Accelerate American Technology and Manufacturing Leadership.** Intel · 22 August 2025 [A] intc.com/news-events/press-releases/detail/1748/intel-and-trump-administration-reach-histo…

9 **TSMC Intends to Expand Its Investment in the United States to US$165 Billion to Power the Future of AI.** TSMC · 4 March 2025 [A] pr.tsmc.com/english/news/3210

CHOKEPOINT CARD	EXTREME CONCENTRATION	HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	**Hard** Intel and Samsung also run lines for the newest chips, but TSMC has not built one of its largest fabs outside Taiwan, and the slow part is training the workforce.	**BCG puts the ten-year cost of owning a fab completed in 2026 at $35-43 billion** TSMC intends to spend $165 billion in the United States alone

WHO LEADS

- TW **TSMC** 77% of wafer revenue from 7 nm and below, 2Q26; six gigafabs, all in Taiwan
- US **Intel Foundry** $5.8bn revenue and a $2.089bn operating loss, 2Q26
- KR **Samsung Foundry** Hwaseong and Pyeongtaek; at least $17bn of construction at Taylor, Texas
- CN **SMIC** $9.327bn revenue in 2025 at 93.5% utilization
- JP **Rapidus** State-backed 2 nm entrant, Chitose

WHERE IT IS MADE

- TW **Taiwan** TSMC's six gigafabs and UMC; N2 and N3 for every leading AI accelerator
- KR **South Korea** Samsung Foundry, Hwaseong and Pyeongtaek
- CN **China** SMIC and Hua Hong; the largest installed capacity of any region, almost all of it on older nodes
- US **United States** Intel Foundry, GlobalFoundries, TSMC Arizona, Samsung Taylor
- JP **Japan** TSMC Kumamoto, Rapidus Chitose

WHY SUBSTITUTION IS SLOW

Money can buy the building. BCG puts the ten-year cost of a fab finished in 2026 at $35 to $43 billion. Intel and Samsung already run lines for the newest chips, so a newcomer is copying something that has been done. Money cannot quickly buy a workforce that has run one process for years. Even TSMC has yet to build one of its largest fabs anywhere but Taiwan. A working line takes five to ten years, and reaching high volume takes longer.

WHERE CHINA STANDS

SMIC and Hua Hong are China's largest foundries. SMIC had revenue of $9.327 billion in 2025 with its lines 93.5 percent full. It cannot buy an EUV scanner, so it cannot make the newest chips. China is forecast to have more fab capacity than any other region, and almost none of it can make the newest chips.

WHERE THE US STANDS

Intel Foundry is the only American-owned foundry that makes the newest chips. It lost $2.089 billion on $5.8 billion of revenue in the second quarter of 2026. The federal government owns 9.9 percent of Intel and takes no part in running it.

TSMC calls a plant a GIGAFAB once it runs more than 100,000 wafers a month, each a 300 mm disc, twelve inches across. It operates six of them, and every one is

in Taiwan [1].

How it works

Inside, the work is one loop repeated for every layer. Wafers travel in a lot, a batch of 25 in a sealed pod called a FOUP that only a machine opens. An overhead track carries the pod from tool to tool. At each tool a robot lifts one wafer out, the tool does its one job, depositing a film, printing a pattern, etching it, cleaning it or measuring it, and the robot puts the wafer back. The pod moves on to the next tool, often across the building, and the loop starts again with the next layer. A chip takes hundreds of those visits, so a wafer spends months inside.

The pod keeps the wafer clean for all of that time, because the wafer never meets the room's air, and even that air is filtered until it carries almost no dust. Every design in the building runs on the same tools because each tool is set by recipe, the list of gases, temperatures and times for one step, and a foundry offers each customer the same list of recipes on shared machines. The tools run day and night, since an idle tool earns nothing. The finished wafer is tested (see Test and assembly), then cut and packaged.

TSMC, founded in 1987, invented the pure-play model: it builds chips for other companies and promises never to compete with them. BCG puts the ten-year cost of owning a fab completed in 2026 at $35 to $43 billion, a third to two thirds above one finished three years earlier [2].

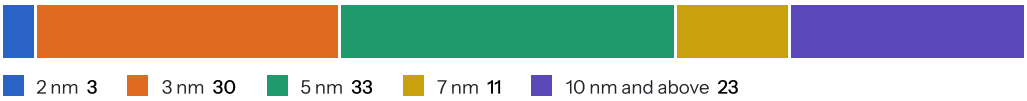
Every plant TSMC and its subsidiaries run came to more than 17 million wafers a year in 2025, counted as if each were the standard 12-inch size [3]. The six largest plants, all in Taiwan and called gigafabs, hold 13 million of that [1]; Arizona, Kumamoto, Nanjing and Washington hold the rest.

Variants and trade-offs

Pure-play foundries

TSMC and UMC build only for others. Chips at 7 nm and below, meaning the newest processes, were 77 percent of TSMC's wafer revenue in the second quarter of 2026, split 33 percent at 5 nm, 30 at 3 nm, 11 at 7 nm and 3 at 2 nm, on revenue of $40.20 billion at a 67.7 percent gross margin [4]. No other foundry earns that margin, so none can fund the next node out of cash flow.

TSMC wafer revenue by node, 2Q26 % OF WAFER REVENUE



Source: TSMC 2Q26 results, filed with the SEC

1 GIGAFAB® Facilities. TSMC [A] tsmc.com/english/dedicatedFoundry/manufacturing/gigafab

2 Navigating the Costly Economics of Chip Making. Boston Consulting Group · 28 September 2023 [A] bcg.com/publications/2023/navigating-the-semiconductor-manufacturing-costs

3 Fab Capacity. TSMC [A] tsmc.com/english/dedicatedFoundry/manufacturing/fab_capacity

4 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 6-K report of foreign private issuer for the period ended 2026-06-30 (6-K). U.S. Securities and Exchange Commission (filing by TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD) · 16 July 2026 [A] sec.gov/Archives/edgar/data/0001046179/000104617926000451/a2q26e_withguidancexfinal.htm

Integrated device makers turned foundry

Intel and Samsung design and sell their own chips and rent out capacity too. Renting out capacity is harder for them, because customers dislike handing designs to a competitor and the company's own products take the best capacity. Intel Foundry booked $5.8 billion of revenue in the second quarter of 2026, up 31 percent, against a $2.089 billion operating loss [5].

National champions

SMIC, Hua Hong and Rapidus exist because a government decided they should. SMIC's revenue rose 16.2 percent to $9.327 billion in 2025 at 93.5 percent utilization, on monthly capacity above a million wafers counted in the smaller 8-inch size, and a 21 percent gross margin [6]. Rapidus raised 267.6 billion yen in February 2026 to carry it to 2 nm mass production in 2027 [7].

Mature-node capacity

Most wafers come off older lines. The industry association SEMI expects foundries alone to reach 12.7 million wafers a month by 2026, with the fastest growth in China [8]. Those lines make power devices, analog chips and microcontrollers, and compete on price.

Capacity and cost

TSMC intends to spend $165 billion in the United States, adding $100 billion in March 2025 to the $65 billion already committed in Phoenix, for three more fabs, two packaging plants and an R&D center [9]. That is the largest foreign direct investment in US history, and it still does not put a gigafab outside Taiwan.

TSMC does not auction capacity; it books it years ahead against prepayments, so the queue for a node is set before the node exists. Through 2024 and 2025 the binding constraint was packaging. See Advanced packaging.

Who makes it

Intel is now partly state-owned. In August 2025 Washington bought 433.3 million newly issued Intel shares at $20.47, or $8.9 billion, for a 9.9 percent stake, paid out of $5.7 billion of unpaid CHIPS Act grants and $3.2 billion from the Secure Enclave program. The stake carries no control, and it came with a five-year warrant for a further 5 percent that Washington can exercise only if Intel drops below 51 percent ownership of the foundry business [10]. That makes the warrant an option on Intel keeping its fabs.

5 Intel Reports Second-Quarter 2026 Financial Results. Intel · 23 July 2026 [A] intc.com/news-events/press-releases/detail/1776/intel-reports-second-quarter-2026-financia…

6 SMIC Announces 2025 Annual Results. SMIC [A] smics.com/en/site/news_read/7951

7 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private Sector Companies This strategic funding plan will enable Rapidus to steadily progress from its current R&D phase to mass production of 2nm logic semiconductors by 2027 - Information - Rapidus Corporation. Rapidus [A] rapidus.inc/en/news_topics/information/rapidus-secures-267-6-billion-yen-in-funding-from-j…

8 Global Semiconductor Fab Capacity Projected to Expand 6% in 2024 and 7% in 2025, SEMI Reports. SEMI · 18 June 2024 [A] semi.org/en/news-media-press-releases/semi-press-releases/global-semiconductor-fab-capacit…

9 TSMC Intends to Expand Its Investment in the United States to US$165 Billion to Power the Future of AI. TSMC · 4 March 2025 [A] pr.tsmc.com/english/news/3210

10 Intel and Trump Administration Reach Historic Agreement to Accelerate American Technology and Manufacturing Leadership. Intel · 22 August 2025 [A] intc.com/news-events/press-releases/detail/1748/intel-and-trump-administration-reach-histo…

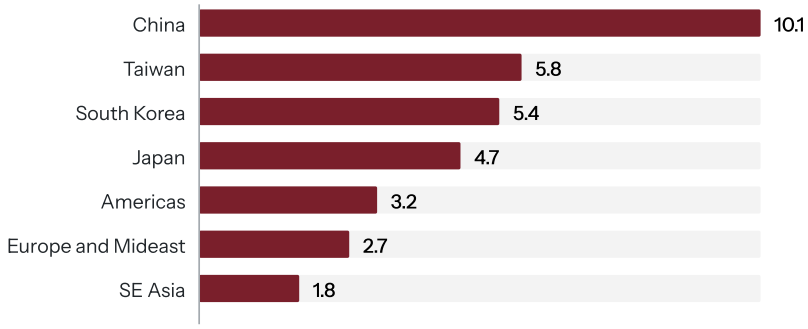
Intel's technical position improved in 2026. It reached high-volume manufacturing on part of Panther Lake using ASML's High-NA extreme ultraviolet scanners, the first company to do so, and put 18A-P into risk production, the trial runs a process goes through before it is signed off [5]. The commercial position has not caught up, since 14A depends on winning external customers first.

Samsung's main plant outside Korea is Taylor, Texas: a committed minimum of $17 billion, $6 billion of buildings and $11 billion of equipment, for leading-edge and mature logic [11].

The chokepoint

SEMI forecast China at 10.1 million wafers a month in 2025, the most of any region, against Taiwan's 5.8 million out of a world total of 33.7 million [8]. Almost none of China's is advanced. The United States had almost no capacity below 10 nm in 2022 and is projected to hold 28 percent of it by 2032, on $2.3 trillion of new fab investment [12].

Installed wafer fab capacity by region, 2025 forecast, all nodes MILLION 8-INCH-EQUIVALENT WAFERS PER MONTH



Source: SEMI World Fab Forecast

Three risks meet on one island:

- **Seismic.** Taiwan sits on an active plate boundary, and a tool knocked out of calibration scraps the work in progress across a whole fab.
- **Power.** A gigafab is a large industrial load on a small island grid. Spare generating capacity sets how fast Taiwan can add fabs.
- **Political.** A blockade or conflict would remove most sub-5 nm capacity from the world market at once, with no substitute available inside a decade. See Geopolitics.

Diversification is slow because the hard part of a fab is a workforce that has run one process for years, and TSMC has yet to build a gigafab anywhere but Taiwan [1].

5 Intel Reports Second-Quarter 2026 Financial Results. Intel · 23 July 2026 A intc.com/news-events/press-releases/detail/1776/intel-reports-second-quarter-2026-financia…

11 Taylor | US Fab | Samsung Semiconductor Global. Samsung Electronics · 11 September 2025 A semiconductor.samsung.com/sas/company/taylor

8 Global Semiconductor Fab Capacity Projected to Expand 6% in 2024 and 7% in 2025, SEMI Reports. SEMI · 18 June 2024 A semi.org/en/news-media-press-releases/semi-press-releases/global-semiconductor-fab-capacit…

12 Emerging Resilience in the Semiconductor Supply Chain. Boston Consulting Group · 8 May 2024 A bcg.com/publications/2024/emerging-resilience-in-semiconductor-supply-chain

1 GIGAFAB® Facilities. TSMC A tsmc.com/english/dedicatedFoundry/manufacturing/gigafab

Key evaluation criteria

- **Leading-edge share.** TSMC's 77 percent of wafer revenue from 7 nm and below is why it can self-fund the next node [4].
- **Yield at ramp.** How many good chips a new process gives in its first year, which separates a profitable node from a subsidized one. Intel Foundry's loss comes from yield.
- **Ten-year cost of ownership.** $35 to $43 billion for a fab finished in 2026, and rising [2].
- **Utilization.** How full the lines run. SMIC's 93.5 percent shows what a captive home market does for volume, and not for margins [6].
- **Customer concentration.** A foundry owned by a chipmaker competes with its own customers, which caps who will trust it with a flagship design.
- **Geographic exposure.** How much of a buyer's supply sits on one island, one grid, one fault system.

4 TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD, Form 6-K report of foreign private issuer for the period ended 2026-06-30 (6-K). U.S. Securities and Exchange Commission (filing by TAIWAN SEMICONDUCTOR MANUFACTURING CO LTD) · 16 July 2026 A sec.gov/Archives/edgar/data/0001046179/000104617926000451/a2q26e_withguidancexfinal.htm

2 Navigating the Costly Economics of Chip Making. Boston Consulting Group · 28 September 2023 A bcg.com/publications/2023/navigating-the-semiconductor-manufacturing-costs

6 SMIC Announces 2025 Annual Results. SMIC A smics.com/en/site/news_read/7951

CHAPTER
10
STAGE 10 OF 15

Memory and HBM

Three firms make every HBM stack in the world. The stack is the largest block of silicon in an AI package, and the cooling plate above it now limits how tall it can be.

IN PLAIN TERMS

High bandwidth memory, or HBM, is where an AI chip keeps the numbers it is working on, and it is built to hand them over fast. A memory maker stacks up to sixteen memory chips into a cube, drills holes straight down through the stack, and fills the holes with copper so that every layer is wired to the ones below. The cube then sits beside the processor, the chip that does the calculating. Stacking is the hard part, because each chip is ground so thin that the copper in its holes shows through its back, the finished cube has to fit under the plate that cools it, and one bad chip ruins the whole cube. Only SK hynix, Samsung and Micron have made it work, and a single line in an American export control reaches all three.

IN SHORT

HBM is the most expensive component in an AI accelerator, and three firms supply all of it: the two Korean ones sold 82.5 percent in 2025 [11]. The limit now is heat and height inside a 775-micrometer cube [9], and the fix, hybrid bonding, is two generations away. The export control on HBM works because one bandwidth-density threshold catches every part worth buying [15].

11 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission (filing by SK hynix Inc.) · 10 July 2026 A sec.gov/Archives/edgar/data/2120882/000119312526299963/d32785d424b4.htm

9 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for Improving Semiconductor Performance. SK hynix A news.skhynix.com/en/tech-note-series-ep2

15 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. Federal Register (Commerce Department; Industry and Security Bureau) · 5 December 2024 A federalregister.gov/documents/2024/12/05/2024-28270/foreign-produced-direct-product-rule-a...

CHOKEPOINT CARD | EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	**Hard** Three firms ship HBM, and no fourth firm has both the newest DRAM and a line that stacks it.	**An HBM stack moves data over a path 16 times wider than a DDR5 memory module's** That width costs chip area and lowers the share of stacks that come out working.

WHO LEADS

- KR **SK hynix** 63.2% of HBM revenue, 2025; first to complete HBM4, at over 10 Gbps per pin
- KR **Samsung** 19.3% of HBM revenue, 2025; Hwaseong and Pyeongtaek
- US **Micron** 17.4% of HBM revenue, 2025; HBM4 above 11 Gbps per pin

WHERE IT IS MADE

- KR **South Korea** SK hynix Cheongju and Icheon; Samsung Pyeongtaek and Hwaseong
- US **United States** Micron headquarters and R&D; SK hynix packaging plant under construction in Indiana
- TW **Taiwan** Micron DRAM and HBM packaging in Taichung; TSMC builds HBM4 logic base dies
- JP **Japan** Micron Hiroshima, 1-gamma DRAM
- SG **Singapore** Micron HBM advanced packaging

WHY SUBSTITUTION IS SLOW

Among the three HBM makers, market share shifts as soon as a customer approves a new part: Micron went from 5.8 percent of HBM revenue in 2024 to 23.1 percent by the first quarter of 2026 without adding capacity. A fourth maker would be harder. It would need the newest DRAM, a line that drills through and thins wafers, and a way to stack the thinned chips without warping them, all inside one company. Its stack would then have to pass Nvidia's speed test. That takes five to ten years, and no Chinese firm ships HBM3E or HBM4 in volume yet.

WHERE CHINA STANDS

No Chinese firm ships HBM3E or HBM4 in volume. CXMT is the only Chinese firm that could plausibly get there, and the US export controls are written to prevent that.

WHERE THE US STANDS

Micron is the only American HBM maker. BIS, the US export control office, requires a license for HBM faster than 2 GB/s per square millimeter of package or stack area, under control number ECCN 3A090.c, and every stack now in production is faster than that.

An AI accelerator is a machine for moving numbers past arithmetic units, and arithmetic got cheap faster than the moving did. High bandwidth memory now decides how fast a GPU runs.

How it works

Each layer in the cube is a DRAM chip, ground thin and bonded to the one below with microbumps, dots of solder, and the whole stack sits on a base die that manages it. The copper-filled holes are TSVs, through-silicon vias, and they let a signal reach any layer without a wire around the outside.

One cube talks to the processor over 1,024 wires at once, a 1,024-bit interface, sixteen times wider than a standard DDR5 memory module, the memory stick in a PC [1]. Ordinary memory cannot be made that wide, because the pins would not fit on a circuit board. The stack can, since its wires run down through the TSVs and across a few millimeters of silicon to the processor beside it, with no board in the way.

Answering a prompt is what the width is for. Training reuses each stored number across a huge multiplication, so a chip can fetch little and compute a lot. To generate one word of an answer, the chip reads the whole model from memory, and the whole record of the conversation so far, does a little arithmetic, and waits for the next read. That record grows with the length of the conversation and the number of users served at once. So memory size and speed usually set how many people one chip can serve.

Variants and trade-offs

GENERATION	INTERFACE	BANDWIDTH PER STACK	CAPACITY PER STACK
HBM3E	1,024-bit, 16 channels (32 pseudo-channels)	over 1.2 TB/s	24 GB (8 chips), 36 GB (12 chips)
HBM4	2,048-bit, 32 channels	over 2.0 TB/s, up to 3.3 TB/s	up to 64 GB (16 chips, 32 Gb dies)

HBM3E

Micron ships the high-volume parts, 24 GB with eight chips stacked and 36 GB with twelve [2], and HBM3E should be about two thirds of all HBM shipments in 2026 [3].

HBM4

The standards body JEDEC published JESD270-4 in April 2025, setting the 2,048-bit interface and 2 TB/s per stack at 8 Gb/s per pin [4]. Nobody ships at that baseline: SK hynix completed HBM4 at over 10 Gb/s per pin in September 2025 [5], and Micron's parts run above 11 Gb/s for more than 2.8 TB/s per stack [6].

1 High-bandwidth memory (HBM). Micron Technology [A] micron.com/products/memory/hbm

2 Micron, HBM3E product page. Micron Technology [A] micron.com/products/memory/hbm/hbm3e

3 2026 Market Outlook – “Focus on the HBM-Led Memory Supercycle”. SK hynix [A] news.skhynix.com/en/2026-market-outlook-focus-on-the-hbm-led-memory-supercycle

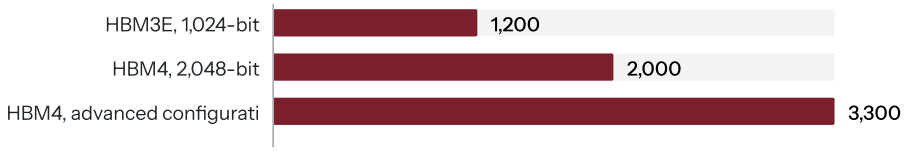
4 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard: Advancing Bandwidth, Efficiency, and Capacity for AI and HPC. JEDEC Solid State Technology Association · 16 April 2025 [A] jedec.org/news/pressreleases/jedec%C2%AE-and-industry-leaders-collaborate-release-jesd270-…

5 SK hynix Completes World’s First HBM4 Development and Readies Mass Production. SK hynix [A] news.skhynix.com/en/sk-hynix-completes-worlds-first-hbm4-development-and-readies-mass-prod…

6 Micron, HBM4 product page. Micron Technology [A] micron.com/products/memory/hbm/hbm4

The chip at the bottom of an HBM4 stack is now a logic chip, so it has to be made in a foundry, and SK hynix partnered with TSMC to build it [7]. That gives TSMC a role inside a product it does not make.

Peak bandwidth per HBM stack GB/S



Source: Siemens EDA, HBM3E and HBM4 IC design guide, April 2026

How a stack is built

- **Drill and thin.** Etch vertical holes, the through-silicon vias, into a memory wafer, fill them with copper, then grind the wafer from behind until the copper comes through.
- **Stack and join.** SK hynix uses Advanced MR-MUF, which melts the solder bumps between stacked chips and molds a filler around them in one pass, which limits heat and warping [5].
- **Stay under the height limit.** HBM4 raised the height limit on a finished cube from 720 micrometers to 775, still under a millimeter, because a stack taller than the processor beside it would hit the cooling plate [9].

Hybrid bonding is the next step and it is not ready. Pressing copper pad straight onto copper pad, with no solder and no filler, would cut the spacing between connections from about 20 micrometers to under 1, but SK hynix expects full-scale adoption only at HBM4E or HBM5, where stacks pass twenty layers [9].

Who makes it

Ordinary computer memory, DRAM, is a three-firm oligopoly, and HBM is the same three firms reordered: SK hynix, Samsung and Micron. Nobody else has both leading-edge DRAM and a stacking line inside one company.

SK hynix built its lead by being first to pass Nvidia's qualification tests, and expects roughly 70 percent of Rubin HBM4 in 2026 [3]. It started building a $4 billion packaging plant in West Lafayette, Indiana in August 2026 [10], and Micron packages in Taichung and Singapore.

7 SK hynix Partners with TSMC to Strengthen HBM Technological Leadership. SK hynix [A] news.skhynix.com/sk-hynix-partners-with-tsmc-to-strengthen-hbm-technological-leadership

5 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix [A] news.skhynix.com/en/sk-hynix-completes-worlds-first-hbm4-development-and-readies-mass-prod…

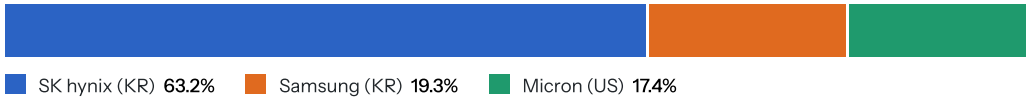
9 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for Improving Semiconductor Performance. SK hynix [A] news.skhynix.com/en/tech-note-series-ep2

3 2026 Market Outlook – "Focus on the HBM-Led Memory Supercycle". SK hynix [A] news.skhynix.com/en/2026-market-outlook-focus-on-the-hbm-led-memory-supercycle

10 SK hynix breaks ground on $4 billion advanced packaging production facility in Purdue Research Park. Purdue University · 27 August 2026 [A] purdue.edu/newsroom/2026/Q3/sk-hynix-breaks-ground-on-4-billion-advanced-packaging-product…

The split is on the record because SK hynix registered shares in the United States: of 2025 HBM revenue, SK hynix took 63.2 percent, Samsung 19.3 and Micron 17.4, which puts the two Korean suppliers at 82.5 percent of the world's HBM [11]. The filing credits the research firm IDC. These shares move fast: in 2024 the same three stood at 56.4, 37.8 and 5.8, so Samsung halved and Micron tripled inside a year on qualification wins alone. By the first quarter of 2026 Micron was at 23.1 percent and Korea down to 76.9 [11].

Global HBM revenue by supplier, 2025 %



Source: SK hynix Form 424B4 prospectus, 10 July 2026, on IDC data

HBM market revenue, Bank of America estimate $B



Source: Bank of America estimate cited in SK hynix's 2026 market outlook

Bank of America puts the 2026 HBM market at $54.6 billion, up 58 percent on the year [3]. HBM is sold under contracts signed long before the wafers exist. Every wafer turned over to HBM is a wafer not making ordinary DRAM, which is how AI demand raises the price of a laptop.

The chokepoint

Every leading accelerator depends on Korean or American HBM. Nvidia's GB300 NVL72 carries 20 TB of HBM3E at up to 576 TB/s [12], Rubin's VR200 up to 288 GB of HBM4 per GPU at up to 22 TB/s [13], and AMD's MI355X 288 GB of HBM3E at 8 TB/s [14].

The Bureau of Industry and Security rule of 5 December 2024, at 89 FR 96790, added an export classification, ECCN 3A090.c, for "High bandwidth memory (HBM) having a 'memory bandwidth density' greater than 2 gigabytes per second per square millimeter". Its technical note defines that density as the memory

11 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission (filing by SK hynix Inc.) · 10 July 2026 [A] sec.gov/Archives/edgar/data/2120882/000119312526299963/d32785d424b4.htm

3 2026 Market Outlook – "Focus on the HBM-Led Memory Supercycle". SK hynix [A] news.skhynix.com/en/2026-market-outlook-focus-on-the-hbm-led-memory-supercycle

12 NVIDIA GB300 NVL72. Nvidia · 23 July 2026 [A] nvidia.com/en-us/data-center/gb300-nvl72

13 Inside NVIDIA Rubin GPU Architecture: Powering the Era of Agentic AI. Nvidia · 21 July 2026 [A] developer.nvidia.com/blog/inside-nvidia-rubin-gpu-architecture-powering-the-era-of-agentic…

14 AMD Instinct MI355X Specs, Pricing & AI Inference Benchmarks. SemiAnalysis [B] inferencex.semianalysis.com/chips/mi355x

bandwidth in gigabytes per second divided by the area of the package or stack in square millimeters. The rule also reaches stacks made outside the United States, through a foreign direct product rule that catches goods made abroad with American technology [15]. Against the JEDEC numbers, that threshold captures every stack now in production, so there is no compliant high-end HBM to sell. Under 15 CFR 742.6(b)(10)(ii) BIS reads applications for 3A090.c into China or Macau with a presumption of approval when the buyer is not headquartered there, and a presumption of denial when it is. That split is what lets the Chinese sites of Korean and American firms, and the allied packaging houses on the mainland, go on receiving HBM [15][16].

China's HBM maker is CXMT, and it is behind the three. That gap is closing from a base that did not exist five years ago. The barrier for a new entrant is the stacking line and Nvidia's approval.

Key evaluation criteria

- **Bandwidth per stack** sets how fast a chip can serve answers: over 1.2 TB/s on HBM3E, over 2 TB/s on HBM4 [8].
- **Capacity per stack** decides how large a model fits on one accelerator, up to 64 GB with sixteen chips stacked [4].
- **Stack height**, 775 micrometers for HBM4, caps the number of layers before the cooling plate [9].
- **Stack yield** sets the price more than wafer cost does, because one bad die scraps the cube.
- **Thermal path** gets worse with every layer: the chip at the bottom runs hottest, and each one added above it makes the heat harder to get out.
- **Qualification** means passing Nvidia's data-rate test, above 10 Gb/s, against JEDEC's 8 [5].

15 Foreign-Produced Direct Product Rule Additions, and Refinements to Controls for Advanced Computing and Semiconductor Manufacturing Items. Federal Register (Commerce Department; Industry and Security Bureau) · 5 December 2024 A federalregister.gov/documents/2024/12/05/2024-28270/foreign-produced-direct-product-rule-a…

16 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations A ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-742/section-742.6

8 HBM3e and HBM4: IC design guide for next-generation high bandwidth memory. Siemens Digital Industries Software · 24 April 2026 A blogs.sw.siemens.com/semiconductor-packaging/2026/04/24/hbm3e-hbm4-ic-design-guide

4 JEDEC® and Industry Leaders Collaborate to Release JESD270-4 HBM4 Standard: Advancing Bandwidth, Efficiency, and Capacity for AI and HPC. JEDEC Solid State Technology Association · 16 April 2025 A jedec.org/news/pressreleases/jedec%C2%AE-and-industry-leaders-collaborate-release-jesd270-…

9 [Tech Note] Hybrid Bonding: Evolving into a Foundational Technology for Improving Semiconductor Performance. SK hynix A news.skhynix.com/en/tech-note-series-ep2

5 SK hynix Completes World's First HBM4 Development and Readies Mass Production. SK hynix A news.skhynix.com/en/sk-hynix-completes-worlds-first-hbm4-development-and-readies-mass-prod…

CHAPTER
11
STAGE 11 OF 15

Advanced Packaging

Lithography stopped being the limit on chip size in 2022. The limit now is how large an interposer, the silicon base plate under the chips, TSMC can build, and through 2025 how many it could build a month.

IN PLAIN TERMS

Packaging joins finished chips into one component and mounts them on a base, called a substrate, wired almost as finely as the chips themselves. A lithography machine, which prints a chip's circuits, can cover only a rectangle about the size of a postage stamp in one shot, and a modern accelerator needs more circuit than fits in one, so it cannot be one chip. It is made as several pieces, set side by side on a shared slab of silicon and wired together through it, with the stacked memory a few millimeters away. Slab, chips and base all expand at different rates when heated, so the package can warp and its joints crack. Getting that right in large numbers is hard enough that nearly every AI accelerator is packaged by TSMC, in Taiwan.

IN SHORT

Advanced packaging capped AI chip output through 2024 and 2025: the physics was solved and the lines were not built. On Epoch AI's estimate Nvidia alone took 60.3 percent of the world's CoWoS in 2025, and four American designers took around 90 percent [9]. By March 2026 the binding constraint had moved back to front-end wafers [10]. The concentration is unchanged: China can package chips but not the 2.5D structures AI accelerators need, and the American alternatives do not start until 2028.

9 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip production in 2025. Epoch AI · 12 March 2026 [A] epoch.ai/data-insights/ai-chip-supply-chain-constraints

10 The Great AI Silicon Shortage. SemiAnalysis · 12 March 2026 [B] newsletter.semianalysis.com/p/the-great-ai-silicon-shortage

CHOKEPOINT CARD

EXTREME CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	**Moderate** Intel's EMIB can do the same job as TSMC's CoWoS, and the real shortage is in making enough working packages at the volume Nvidia orders.	**No public wafer counts** TSMC told investors CoWoS capacity would grow about 60% a year, and keeps adding to that, but will not publish the total. Epoch AI estimates Nvidia alone used 60.3% of the world's CoWoS capacity in 2025

WHO LEADS

- TW **TSMC** Almost all CoWoS for leading AI accelerators
- TW **ASE Technology** NT$645.4B revenue, 2025; the largest outsourced assembly and test firm
- US **Amkor** $6.71B net sales, 2025; the largest American assembly and test firm
- CN **JCET** RMB 35.96B revenue, 2024; the largest Chinese assembly and test firm
- SG **ASMPT** $532.1M advanced packaging revenue, 2025; thermocompression bonders

WHERE IT IS MADE

- TW **Taiwan** TSMC AP fabs and the OSAT cluster; almost all 2.5D capacity for AI accelerators
- US **United States** Amkor Peoria, Arizona from 2028; two TSMC advanced packaging plants planned in Arizona
- CN **China** JCET, Tongfu and HT-Tech, strong in conventional assembly and test, weak in 2.5D
- NL **Netherlands** Besi hybrid and die bonders
- JP **Japan** Disco grinders and dicers; TEL and Shibaura bonding tools

WHY SUBSTITUTION IS POSSIBLE

None of the packaging steps is unusual, and Intel's EMIB does the same job when TSMC's CoWoS is sold out. The hard part is volume. A newcomer has to buy enough bonding machines and run enough practice lots to make the interposer, the silicon base plate under the chips, at 5.5 times the area a scanner prints in one shot. It then has to attach the memory stacks, have them work, and do all of that at the volume Nvidia alone orders. Amkor's plant in Peoria, Arizona shows the timing: production starts in early 2028, so the wait is two to five years.

WHERE CHINA STANDS

China is far stronger at packaging and testing chips than at making them. JCET alone had revenue of RMB 35.96 billion in 2024. Chinese tool makers gained share in packaging and test tools while gaining none in atomic layer deposition or lithography. Almost none of that Chinese packaging is built on a silicon interposer,

the kind AI accelerators need.

WHERE THE US STANDS

No firm yet does CoWoS-class packaging in volume in the United States. Amkor is building a $7B packaging and test plant in Peoria, Arizona, with up to $407M of CHIPS Act money, and production starts in early 2028.

A lithography scanner prints a rectangle 26 mm by 33 mm, or 858 mm2, in one exposure [1]. Every AI accelerator worth buying is bigger than that. Packaging is the stage that lets an accelerator be larger than that rectangle, and it decides how much larger.

How it works

High bandwidth memory gives a second reason to split the package: its 1,024-bit or 2,048-bit connection, that many wires at once, is too wide for a resin base board to carry, so the memory sits beside the logic on a shared silicon carrier, the interposer.

The name CoWoS says the order: chip onto wafer, then wafer onto substrate. TSMC makes a thin silicon wafer carrying wiring and vertical copper vias, metal-filled holes that join one layer of wiring to the next, but no transistors. It bonds the logic dies and memory stacks face down onto that wafer with microbumps, dots of solder 30 to 40 microns apart, about a third of a hair's width, each dot one electrical joint. Then it cuts the interposer out of the wafer, attaches it to the resin substrate, and the substrate goes onto its board.

Each of those packaging steps is ordinary on its own. But one cracked joint scraps a package holding eight qualified HBM stacks and two of the most expensive logic dies ever made.

Variants and trade-offs

FAMILY	CARRIER	SIZE LIMIT	USED FOR
CoWoS-S	Full silicon interposer	3.3x reticle, about 2,700 mm2	H100-class parts
CoWoS-R	Resin interposer with fine redistribution wiring	Above 3.3x reticle	Cost-sensitive, fewer HBM stacks
CoWoS-L	Redistribution wiring with local silicon bridges	5.5x reticle in production, 9.5x in development	Blackwell, Rubin

2.5D: silicon, RDL and bridges

TSMC has run CoWoS in volume since 2012 [2]. CoWoS-S wires most finely, but its interposer is one unbroken piece of silicon whose yield falls as the area grows, so CoWoS-L replaces the slab with fine wiring and small silicon bridges only where

1 TWINSCAN NXE:3400C. ASML [A] asml.com/en/products/euv-lithography-systems/twinscan-nxe3400c

2 TSMC, CoWoS technology page. TSMC [A] 3dfabric.tsmc.com/english/dedicatedFoundry/technology/cowos.htm

the wiring is dense. Intel's EMIB buries the same bridge idea in the resin substrate, and is a credible second source when CoWoS capacity is short.

Blackwell was the first high-volume design on CoWoS-L, and it went badly. The bridges carrying the 10 TB/s link between the two compute dies have to be placed to a tolerance that warping makes hard to hold [3].

3D: SoIC, Foveros, X-Cube

TSMC's SoIC and Intel's Foveros bond dies face to face, copper pad to copper pad with no solder, packing the connections far closer than microbumps allow. TSMC put 3 nm SoIC into volume production in 2025 [4], with an A14-on-A14 version due in 2029 carrying 1.8 times as many connections between the dies as the 2 nm generation [5]. Heat decides the layout: the bottom die overheats and the top is hard to power, so 3D designs often put cache underneath and compute on top.

Two ways off the silicon interposer

Panel-level packaging swaps the round wafer for a rectangular panel and wastes far less area; see Test and assembly. JEDEC published SPHBM4 on 13 July 2026. It puts HBM4 memory on a base chip carrying 512 data signals instead of 2,048, which spaces the connections widely enough to mount the stack on a standard resin substrate, off the silicon interposer altogether [6].

Capacity

TSMC does not publish a CoWoS wafer number, and will not. At its 2024 symposium it told investors capacity would grow at a 60 percent compound rate, and on the July 2024 earnings call C.C. Wei said he could not reach supply and demand balance [7].

In October 2025 Wei would still say only that TSMC would add capacity again in 2026 and give the real figure the year after, and that front end and back end were both, in his word, tight [8].

Epoch AI weighs each designer's component use against world supply and puts Nvidia at 60.3 percent of all CoWoS in 2025, with a 90 percent confidence interval of 56.5 to 64.3; Google took 13.5, AMD 8.4 and Amazon 7.4 [9]. Those four absorbed around 90 percent of all CoWoS and all HBM by value while taking about 12 percent of the world's advanced logic wafers, which is why packaging ran out and logic did not.

3 Nvidia's Blackwell Reworked - Shipment Delays & GB200A Reworked Platforms. SemiAnalysis · 4 August 2024 [B] semianalysis.com/2024/08/04/nvidias-blackwell-reworked-shipment

4 TSMC 2025 Annual Report, chapter 5. TSMC [A] investor.tsmc.com/static/annualReports/2025/english/pdf/2025_tsmc_ar_e_ch5.pdf

5 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC · 23 April 2026 [A] pr.tsmc.com/english/news/3302

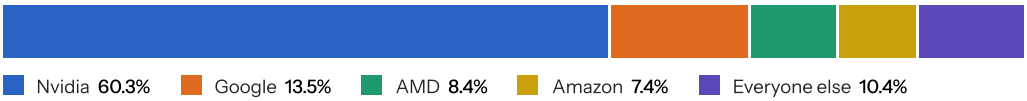
6 New JEDEC® SPHBM4 Standard Enables HBM4-Class Bandwidth on Organic Substrates. JEDEC Solid State Technology Association · 13 July 2026 [A] jedec.org/news/pressreleases/new-jedec%C2%AE-sphbm4-standard-enables-hbm4-class-bandwidth-…

7 Q2 2024 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 18 July 2024 [A] investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2024-08/5122725a56670882d777a…

8 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 October 2025 [A] investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2025-10/6860312f04fd291d0f26b…

9 Advanced packaging and HBM, not logic dies, were the bottlenecks on AI chip production in 2025. Epoch AI · 12 March 2026 [A] epoch.ai/data-insights/ai-chip-supply-chain-constraints

Who consumed the world's CoWoS capacity, 2025 %



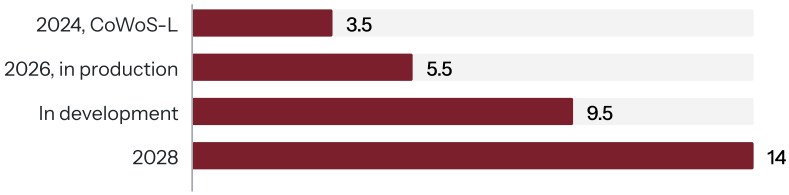
Nvidia 60.3% Google 13.5% AMD 8.4% Amazon 7.4% Everyone else 10.4%

Source: Epoch AI, advanced packaging and HBM were the bottlenecks on AI chip production in 2025

SemiAnalysis reported in March 2026 that front-end wafers are now the dominant bottleneck and CoWoS supply is easing, partly because TSMC has no reason to build interposer lines its N3 output cannot feed [10].

Who makes it

CoWoS interposer size on TSMC's roadmap RETICLES



Source: TSMC 2025 annual report and 2026 technology symposium

The word packaging covers two industries. TSMC does the 2.5D work, chips side by side on a shared carrier, in its own back-end fabs; everything else goes to outsourced assembly and test firms. ASE Technology Holding, the largest, billed NT$645.4 billion in 2025, with its assembly and test business up 23 percent, and Amkor, the largest American one, billed $6.71 billion [11] [12].

The tools are a narrower chokepoint than the assembly houses.

- **Thermocompression bonding.** Pressing dies together under heat and force. ASMPT's advanced packaging revenue reached $532.1 million in 2025, with the bonder line up about 146 percent; it targets 35 to 40 percent of a market it puts at $1.6 billion by 2028 [13].

10 The Great AI Silicon Shortage. SemiAnalysis · 12 March 2026 [B] newsletter.semianalysis.com/p/the-great-ai-silicon-shortage

11 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial Results for the Fourth Quarter and the Full Year of 2025. PR Newswire · 5 February 2026 [A] prnewswire.com/news-releases/ase-technology-holding-co-ltd-reports-its-unaudited-consolida...

12 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-02-09 (8-K). U.S. Securities and Exchange Commission (filing by AMKOR TECHNOLOGY, INC.) · 9 February 2026 [A] sec.gov/Archives/edgar/data/1047127/000104712726000007/amkr123125erex-991.htm

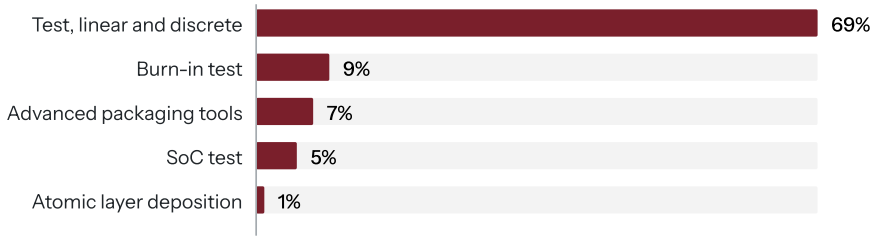
13 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group Performance. ASMPT · 4 March 2026 [A] asmpt.com/en/investor-relations/news-events/asmpt-announces-2025-annual-results

- **Hybrid bonding.** Solder disappears and copper pads meet directly. Applied Materials built its Kinex bonder with Besi, which makes the placement head [14].
- **Grinding and dicing.** Disco dominates the steps that thin wafers and cut packages apart; see Test and assembly [15].

TSMC's Arizona commitment reached $265 billion in July 2026 and covers two advanced packaging plants alongside 10 fabs [16]. Amkor started building a $7 billion campus in Peoria in October 2025, backed by up to $407 million of CHIPS money, producing from early 2028 [17], and Wei confirms it will open before TSMC's own two do [8].

The chokepoint

China-based suppliers' share of back-end tool segments, 2024 %



Source: CSET, Inside Beijing's Chipmaking Offensive (ALD shown for contrast, under 1%)

JCET, the largest Chinese assembler, billed RMB 35.96 billion in 2024, up 21 percent [19], and Chinese toolmakers took real share in packaging and test equipment over the same years they took none in atomic layer deposition [18]. Packaging tools are less export-controlled than lithography, and Huawei's most credible response to the restrictions is to build big chips from small chiplets cut on SMIC silicon and assembled at home. China still has no silicon-interposer 2.5D at volume with qualified HBM beside it.

Each generation on TSMC's roadmap needs a larger interposer.

- **2028.** A 14-reticle interposer carrying about ten large compute dies and 20 HBM stacks, roughly 12,000 mm2 of silicon, about the area of a compact disc [5].

14 Applied Materials, Kinex integrated die-to-wafer hybrid bonding system product page. Applied Materials [A] appliedmaterials.com/eu/en/product-library/kinex-integrated-die-to-wafer-hybrid-bonding-sy...

15 DISCO Corporation, The World Leader In Semiconductor Capital Equipment For Cutting, Grinding, Polishing. SemiAnalysis · 19 July 2022 [B] newsletter.semianalysis.com/p/disco-corporation-the-world-leader

16 TSMC Announces Additional $100 Billion Investment in Arizona. Arizona Commerce Authority · 16 July 2026 [A] azcommerce.com/news-events/news/2026/7/tsmc-announcement

17 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test Campus in Arizona; Expands Investment to $7 Billion. Amkor Technology [A] ir.amkor.com/news-releases/news-release-details/amkor-technology-breaks-ground-new-semicon...

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19 JCET Releases 2024 Annual Report, Achieves Record-High Revenue. JCET Group · 20 April 2025 [A] jcetglobal.com/en/site/news-detail

18 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025 [A] cset.georgetown.edu/article/inside-beijings-chipmaking-offensive

5 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC · 23 April 2026 [A] pr.tsmc.com/english/news/3302

- **2029.** More than 14 reticles, and a 40-reticle System-on-Wafer beside it [5].

Key evaluation criteria

- **Interposer area** is the limit on how much logic and memory fit, now 5.5x reticle on CoWoS-L [5].
- **Package yield** decides whether a generation ships on time, and nobody publishes it.
- **Warpage control** covers the expansion mismatch between die, bridge, interposer and substrate that broke early CoWoS-L and gets harder as packages grow [3].
- **HBM stacks supported** is about 20 on the 14-reticle package due in 2028 [5].
- **Bonder throughput** is slow by nature: thermocompression and hybrid bonding place one die at a time, and three or four firms hold the tools [13].
- **Substrate size** limits everything above it. A 14-reticle interposer needs a resin substrate that will not warp under it, which pushes ABF build-up film to its limits; see Substrates and PCBs.

5 TSMC Debuts A13 Technology at 2026 North America Technology Symposium. TSMC · 23 April 2026 [A] pr.tsmc.com/english/news/3302

3 Nvidia's Blackwell Reworked - Shipment Delays & GB200A Reworked Platforms. SemiAnalysis · 4 August 2024 [B] semianalysis.com/2024/08/04/nvidias-blackwell-reworked-shipment

13 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group Performance. ASMPT · 4 March 2026 [A] asmpt.com/en/investor-relations/news-events/asmpt-announces-2025-annual-results

CHAPTER
12
STAGE 12 OF 15

Substrates and PCBs

A Japanese food company makes 95 percent of the insulating film inside high-end chip packages, and a Japanese textile company weaves the glass cloth beneath it.

IN PLAIN TERMS

A package substrate is the adapter between a chip and the circuit board under it. The chip's connections come out finer than a human hair; the board is wired in millimeters. The substrate is built up like plywood. A stiff core of glass cloth and resin sits in the middle. Thin sheets of epoxy film are pressed onto both faces, a laser burns holes through each sheet, and copper is plated into the holes to carry signals from one layer to the next. Every added layer is another set of holes that can fail. A handful of firms build these substrates, and the film and the glass cloth each come from a single Japanese supplier, so a shortage here stops accelerators shipping.

IN SHORT

TSMC says packaging capacity already caps what its customers can ship. The two materials that make a substrate possible come from single Japanese suppliers: Ajinomoto puts its film at 95 percent of that market, and Nitto Boseki claims an overwhelming advantage in the glass cloth beneath it. Substrate plants in Japan, Taiwan and Korea are spending billions, but film and glass cloth add capacity more slowly than GPU output grows. Glass cores fix the physics eventually; on Intel's and Samsung's own dates, they do not fix 2027.

CHOKEPOINT CARD | EXTREME CONCENTRATION | HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
Extreme	Hard Rated on the insulating film inside the substrate: Ajinomoto supplies about 95 percent of it by its own count, and every substrate maker would have to re-approve a replacement layer by layer.	**Ibiden alone is spending about JPY 500 billion on high-performance substrate capacity from fiscal 2026 to 2028, JPY 220 billion of it in the first phase**

WHO LEADS

- JP **Ajinomoto** 95% of the world market for insulating film in high-end processor packages, on its own account
- JP **Nitto Boseki** Claims an overwhelming lead in low-expansion glass cloth
- JP **Ibiden** Says it wins close to 100% of each new generation of interposer substrates at launch
- TW **Unimicron** Build-up substrates were 52% of sales in Q2 2026
- KR **Samsung Electro-Mechanics** Package solution sales KRW 771.6bn in Q2 2026, up 37%

WHERE IT IS MADE

- JP **Japan** Build-up film, low-expansion glass cloth, and Ibiden and Shinko substrate plants in Gifu and Nagano
- TW **Taiwan** Unimicron, Nan Ya PCB, Kinsus; the largest cluster of substrate makers selling to all comers
- KR **South Korea** Samsung Electro-Mechanics, LG Innotek, and a glass-core joint venture with Sumitomo Chemical
- CN **China** Most of the circuit board competition, alongside Taiwan; now pushing into glass cloth and build-up film
- AT **Austria** AT&S, the only European advanced substrate maker of scale

WHY SUBSTITUTION IS SLOW

A few firms build substrates. Money can expand their plants, and every large one is expanding. The single points of failure are in the materials inside the substrate. Ajinomoto makes the thin insulating film between the substrate's wiring layers, puts its own share at about 95 percent, and has been the standard since 1999. Beneath that film is Nitto Boseki's glass cloth, woven so it barely swells with heat. A newcomer can build a line to make such a film. Every substrate maker and every packaging house then has to test the new film in each layer of each product before using it. That takes years at each of them, so the whole switch takes five to ten years.

WHERE CHINA STANDS

Most of the firms competing in circuit boards are Chinese or Taiwanese. Nitto Boseki itself calls them aggressive fast followers in special glass. Chinese firms still make almost none of the high-end insulating film.

WHERE THE US STANDS

No American firm makes these substrates in volume. TTM Technologies is the main American circuit board maker. Data center computing was 24 percent of its sales in 2025, up from 14 percent in 2023. Intel is developing substrates made of glass for the second half of the decade.

On TSMC's July 2026 earnings call, chief executive C.C. Wei said packaging capacity is now so tight that it limits customers' growth [1]. The silicon interposer that wires chips together inside a package spans a few millimeters. Past its edge the wiring turns to epoxy and glass: a build-up substrate, a printed circuit board of thirty layers, then a rack.

How it works

The substrate maker drills the core and plates it with copper before the build-up starts. Each cycle then presses on a film of epoxy about 10 micrometers thick, burns holes through it with a laser, plates copper into the holes to link the layers, and etches the wiring on top [2]. A human hair is ten times thicker than one of those films [3].

Silicon and epoxy swell at different rates when heated, so the larger the package, the more it warps. AI parts are the largest: Ajinomoto counts 18 layers of its film in a high-performance computing substrate against six in a PC part, on a body three and a half times the area [3]. The fix is a core that barely expands, which means low-expansion glass cloth and tighter control of the resin.

Variants and trade-offs

BT substrates

Bismaleimide-triazine laminate, or BT, is the cheaper and older family. It still leads on unit volume and suits memory stacks and mobile parts, but it cannot hold the fine wiring or the flatness a large logic die needs.

ABF build-up substrates

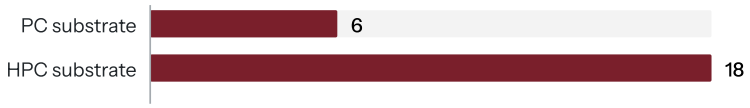
Ajinomoto Build-up Film, or ABF, is the insulating sheet between the copper layers, and it sits under every high-end CPU, GPU and AI accelerator. A major chipmaker first used it in 1999, and Ajinomoto calls it the de facto standard in package development ever since [3]. Asked on the company's own site whether its share was nearer 30 or 50 percent, the electronic materials division answered: "More like 95 percent market share", and said practically all high-performance computers and servers rely on it [2].

1 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. LSEG StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 July 2026 [A] investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897a…

2 The Ajinomoto Group's Unexpected Role in Semiconductor Manufacturing: The Insulating Film "ABF" Born from "AminoScience" | Stories. Ajinomoto · 5 March 2026 [A] ajinomoto.com/stories/the-ajinomoto-groups-unexpected-role-in-semiconductor-manufacturing-…

3 Ajinomoto, Business Briefing: ABF-Based Growth Strategy in ICT, 12 June 2023. Ajinomoto · 12 June 2023 [A] ajinomoto.co.jp/company/en/ir/event/business_briefing/main/01113/teaserItems1/01/linkList/…

Layers of build-up film in one package substrate LAYERS



Source: Ajinomoto business briefing on ABF, June 2023

Glass core substrates

A glass core in place of resin removes most of the swelling problem, which is why chipmakers and substrate makers are both trying it.

- **Intel** announced glass substrates in September 2023 for the latter part of this decade [4].
- **Samsung Electro-Mechanics** signed a joint venture with Sumitomo Chemical in November 2025 to make glass core, with mass production after 2027 [5].

Both programs point past 2027, so glass does not help before then.

The materials underneath

Ajinomoto's film is a small line in a food company's accounts and a chokepoint for every chipmaker. The Healthcare and Others segment that carries electronic materials lifted business profit 45.1 percent to JPY 66.2 billion in the year to March 2026, and Ajinomoto credits the materials for most of that [6]. Capacity comes slowly: about JPY 25 billion of growth investment from fiscal 2023, sized against demand out to 2030 [3].

The core one layer down comes from one supplier too. Nitto Boseki's T-glass is stiff and barely swells when heated, which is what a dense package substrate needs [7]. The company claims an overwhelming advantage in quality and capacity for low-expansion glass, says demand has risen two to three times since its fiscal 2024 to 2027 plan was set, and is spending JPY 80 billion over those four years to catch up [8]. Apple and Qualcomm are among the customers worried about Japanese glass cloth supply [9].

Chinese firms are entering this layer, and Nitto Boseki says so itself: it names Chinese and Taiwanese firms as aggressive fast followers in special glass [8]. At

4 Intel Unveils Industry-Leading Glass Substrates to Meet Demand for.... Intel · 31 January 2025 [A]
download.intel.com/newsroom/archive/2025/en-us-2023-09-18-intel-unveils-industryleading-gl...

5 Samsung Electro-Mechanics Signs MOU with Sumitomo Chemical Group to Establish a Joint Venture for Glass Core Used in Package Substrates. Samsung Electro-Mechanics · 5 November 2025 [A] samsungsem.com/global/newsroom/news/view.do

6 Note: This document has been translated from the Japanese original for reference purposes only. In the event of any. Ajinomoto · 1 May 2026 [A] ajinomoto.co.jp/company/en/ir/library/result/main/014/teaserItems1/0/linkList/06/link/FY25...

3 Ajinomoto, Business Briefing: ABF-Based Growth Strategy in ICT, 12 June 2023. Ajinomoto · 12 June 2023 [A]
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7 T-glass | Electronic Materials Business | Business and Products. Nitto Boseki (Nittobo) [A]
nittobo.co.jp/eng/business/electronicmaterials/t-glass.htm

8 2-4-1, Kojimachi, Chiyoda-ku, Tokyo, 102-8489, Japan. Nitto Boseki (Nittobo) · 16 September 2025 [A]
nittobo.co.jp/eng/ir/pdf/Nittobo_Integrated-Report_2025.pdf

9 Apple and Qualcomm fret over strained supplies of Japan's glass cloth. Nikkei Asia · 14 January 2026 [B]
asia.nikkei.com/business/technology/tech-asia/apple-and-qualcomm-fret-over-strained-suppli...

stake is a chip materials market Nikkei sizes at $73 billion [10].

Who makes it

The plants that build substrates for other firms are an Asian oligopoly, and every large one is expanding. The White House's 2021 hundred-day supply chain review recorded build-up substrate among the materials commenters called vulnerable, and named the makers as Ibiden, Shinko, Nanya, Samsung, Unimicron, Shennan Circuits, Zhuhai Yueya and AKM. None of the eight is American [11].

- **Ibiden** will invest about JPY 500 billion from fiscal 2026 to fiscal 2028, starting with JPY 220 billion at its Gama plant and mass production from fiscal 2027 [12]. In May 2026 management said it typically takes close to 100 percent of a new generation of interposer substrates at launch and loses 20 to 30 points three to six months later. On silicon-bridge substrates, it said, wiring the underside is hard enough that it expects to keep its technical lead through fiscal 2030 [13].
- **Unimicron** took NT$42.9 billion of sales in the second quarter of 2026, 52 percent of it build-up substrates and 61 percent of it AI data center work [14].
- **Samsung Electro-Mechanics** grew package solution sales 37 percent year on year to KRW 771.6 billion in the second quarter, on substrates for AI accelerators and server processors [15].

Shinko has a new owner: Fujitsu sold its 50.02 percent stake to a fund run by the state-backed Japan Investment Corporation [16].

10 China chip material makers battle Japan rivals for $73bn market. Nikkei Asia · 1 July 2026 B
asia.nikkei.com/business/tech/semiconductors/china-chip-material-makers-battle-japan-rival…

11 BUILDING RESILIENT SUPPLY CHAINS,. The White House (Biden administration archive) · 7 June 2021 A
bidenwhitehouse.archives.gov/wp-content/uploads/2021/06/100-day-supply-chain-review-report…

12 Ibiden, Notice Regarding Capital Investment Plan for High-Performance IC Package Substrates. Ibiden · 3 February 2026 A
ibiden.com/company/2026/02/notice-regarding-capital-investment-plan-for-high-performance-i…

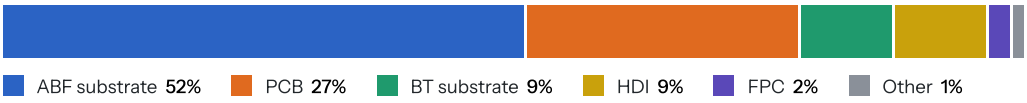
13 Ibiden, questions and answers from the financial presentation for the year ended 31 March 2026. Ibiden · 12 May 2026 A
ibiden.com/ir/items/en_QA_FY25Q4.pdf

14 Unimicron consolidated financial statements, second quarter 2026. Unimicron Technology A
unimicron.com/files/money/Earnings/en/2026-Q2-consolidated-en.pdf

15 Samsung Electro-Mechanics, Q2 2026 business results. Samsung Electro-Mechanics · 30 July 2026 A
m.samsungsem.com/global/newsroom/news/view.do

16 February 17, 2025 Japan Investment Corporation. Japan Investment Corporation · 17 February 2025 A
jiccapital.co.jp/en/news/.assets/E_20250217_JIC_JICC_PressRelease.pdf

Unimicron sales by technology, second quarter 2026 %



Source: Unimicron 2026 Q2 earnings conference

The boards under the GPU

TTM Technologies says it regularly builds printed circuit boards of more than 30 layers and can go past 70 [17]. Those layers carry hundreds of amps to the GPUs and route traffic between them fast enough that how much signal the laminate absorbs becomes a main design constraint.

Data center computing was 24 percent of TTM's $2.91 billion of fiscal 2025 revenue, against 14 percent two years earlier [17] [18]. It names its competition as mostly Chinese and Taiwanese [17]. The alternative is Southeast Asia, where Thailand has built an AI board industry [19].

The chokepoint

Money can expand an oligopoly of substrate plants, and it is being spent. The single points of failure are one layer down: one company's film and one company's glass cloth, both Japanese, both running at full capacity and adding more on building schedules measured in years while AI demand compounds quarterly.

Flatness is the first thing lost as packages grow. TSMC has told investors that its CoWoS packaging roadmap goes beyond 14 times the area a lithography machine can print in one shot [1]. Bigger bodies mean more warping, more layers, more holes that can break the circuit, and lower yield on a part that is already expensive before a die touches it.

Key evaluation criteria

- **Layer count** sets how much current and how many signals the substrate can carry. Each added layer is another chance to lose the part.

17 TTM TECHNOLOGIES INC, Form 10-K annual report for the period ended 2025-12-29 (10-K). U.S. Securities and Exchange Commission (filing by TTM TECHNOLOGIES INC) · 17 February 2026 [A]
sec.gov/Archives/edgar/data/1116942/000119312526051976/ttmi-20251229.htm

18 Revenue from Contract with Customer, Excluding Assessed Tax (us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax) reported by TTM TECHNOLOGIES, INC., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for TTM TECHNOLOGIES, INC.) [A]
data.sec.gov/api/xbrl/companyconcept/CIK0001116942/us-gaap/RevenueFromContractWithCustomer...

19 Thailand carves out a less-glamorous AI niche: printed circuit boards. Nikkei Asia · 3 September 2025 [B]
asia.nikkei.com/business/technology/tech-asia/thailand-carves-out-a-less-glamorous-ai-nich...

1 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. LSEG StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 July 2026 [A]
investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897a...

- **Body size** drives everything else. Once a body is much larger than the dies it carries, warping and handling take over from the wiring in deciding what can be built.
- **Coefficient of thermal expansion**, how much a material swells when heated, must be close enough to silicon that the package survives repeated heating and cooling, which is what T-glass provides and why its supply matters.
- **Dielectric properties**, how much signal the core laminate absorbs, set the speeds the substrate and the board underneath can run.
- **Qualification time** protects the incumbents. Approving a second supplier for a film or a laminate takes years.

CHAPTER
13
STAGE 13 OF 15

Test and Assembly

Assembly and test decide which dies are allowed into a package that costs more than a car. One Japanese firm now sells two thirds of the world's chip testers.

IN PLAIN TERMS

Before a chip can be sold, a machine has to put questions to it and check every answer. While the chips are still on the wafer, thousands of needles press onto small metal pads on each one, send in signals and read what comes back; the tester marks any chip that answers wrong, and that chip goes no further. A modern accelerator glues together a dozen expensive pieces, and one bad piece throws away all of them. So the industry tests sooner, hotter and longer than it used to, to make a weak chip fail on the test bench before it reaches a finished product, and two firms, one Japanese and one American, sell most of the machines that do it.

IN SHORT

Test used to be the cheap step at the end. With a dozen expensive dies in one package, it now protects everything made before it: the market for logic chip testers grew about 68 percent in 2025 [7] and burn-in has started moving back onto the wafer [3]. Advantest has turned a duopoly into a two-thirds share of the machines. The assembly business beneath it stays fragmented, and four of the top ten assembly and test firms are Chinese [19], which is why export controls have little effect on assembly and test.

7 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results briefing, 27 April 2026. Advantest · 27 April 2026 [A] advantest.com/document/en/investors/ir-library/result/JE_BIZ_260427_note.pdf

3 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial Results with Record Quarterly Bookings and $100 Million Effective Backlog. Aehr Test Systems · 14 July 2026 [A] aehr.com/2026/07/aehr-test-systems-reports-fiscal-2026-fourth-quarter-and-full-year-financ…

19 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025 [A] cset.georgetown.edu/article/inside-beijings-chipmaking-offensive

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate No test or assembly tool has only one maker, but switching testers means rebuilding every one of the customer's test programs.	**ASE and Amkor alone had $27.3 billion of revenue in 2025** ASE spent $4.1 billion on machinery and buildings in the first half of 2026

WHO LEADS

- JP **Advantest** 65% of the tester market in 2025
- US **Teradyne** Second in testers; about 80% of the market with Advantest
- TW **ASE** $20.6bn of 2025 revenue, the largest assembly and test contractor
- US **Amkor** $6.71bn of 2025 revenue
- JP **Disco** Leading share in dicing saws, grinders and blades

WHERE IT IS MADE

- TW **Taiwan** ASE, Powertech, KYEC and ChipMOS; the largest assembly and test cluster
- JP **Japan** Advantest testers, Disco dicing and grinding, Micronics Japan probe cards
- US **United States** Teradyne, FormFactor, Cohu, Aehr; Amkor's Arizona packaging plant
- CN **China** Four of the top ten assembly and test firms by 2024 revenue are China-headquartered
- MY **Malaysia** Penang and Kulim; Malaysia ships about 13% of the world's packaged chips

WHY SUBSTITUTION IS POSSIBLE

Advantest and Teradyne sell about 80 percent of the testers between them, but both make machines that customers have approved, and the smaller kinds of test and assembly tool have several makers each. Customers rarely switch, because a new tester means rebuilding every test program, from the first prototype through to the production line. Advantest tells its investors the same thing. A newcomer pays that cost once and takes two to five years. The barrier is low enough that four of the top ten assembly and test firms are already Chinese.

WHERE CHINA STANDS

Assembly and test is the stage where China is most competitive. CSET at Georgetown counts four of the top ten assembly and test firms by 2024 revenue as Chinese. The tools those firms use are far less tightly controlled than lithography or deposition tools.

WHERE THE US STANDS

American firms are strong in test machines and probe cards, the beds of pins that touch each chip on the wafer: Teradyne, FormFactor, Cohu and Aehr. They are weak in assembly. Amkor is building a $7 billion packaging and test campus in Arizona and has a long-term deal with TSMC to fill it.

Semiconductor manufacturing runs to 400 to 600 steps, and Advantest counts test as the only one that puts electricity through the chip [1]. Every stage so far makes dies; the back end of the industry, assembly and test, decides which of them are allowed into a package.

How it works

Each round of test catches failures the round before it cannot see.

- **Wafer sort.** Thousands of needles land on each die while it is still on the wafer, a tester drives patterns in and checks the answers, and the failures go no further.
- **Burn-in.** Run the part hot and under power for hours, so anything that would fail early fails now, before assembly.
- **Final test.** The tester runs the packaged part again, because packaging adds failures of its own.
- **System-level test.** The finished part sits in a socket and runs something close to a real workload, because some failures appear only when everything runs at once.

Proving a die good before it is packaged is now worth far more. Aehr screens for early and hidden failures before dies are cut from the wafer, so a weak processor never reaches high-bandwidth memory and an expensive substrate [2]. Its lead AI customer is moving burn-in off the finished system and onto the wafer, nine 300 mm wafers at a time [3].

Variants and trade-offs

Wafer sort and known-good die

The probe card that holds those needles is the hard part. For a high-bandwidth memory stack it must contact eight, twelve or sixteen dies at speed without degrading the signals [4]. For a large logic die every needle must sit in the same plane, survive repeated touchdowns without drifting, and do it hot. Cards are custom to the device, so each new part means a new card and a new qualification.

Final test, burn-in and system-level test

The economics get worse as packages grow. TSMC has told investors that its CoWoS packaging roadmap runs beyond 14 times the area a lithography machine can print in one shot [5]. Every added die is another chance to lose the whole

1 Investors Guide April 25, 2025. Advantest · 13 May 2025 [A] advantest.com/document/en/investors/ir-library/investors-guide/Investors_Guide_2504E.pdf

2 Aehr Receives $22 Million Follow-On Order for AI Processor Wafer-Level Burn-In Systems. Aehr Test Systems · 12 August 2026 [A] aehr.com/2026/08/aehr-receives-22-million-follow-on-order-for-ai-processor-wafer-level-bur…

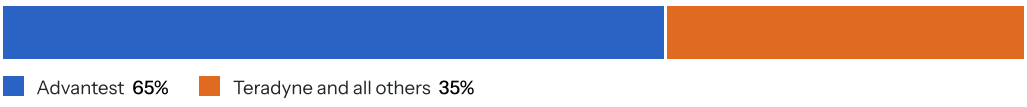
3 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial Results with Record Quarterly Bookings and $100 Million Effective Backlog. Aehr Test Systems · 14 July 2026 [A] aehr.com/2026/07/aehr-test-systems-reports-fiscal-2026-fourth-quarter-and-full-year-financ…

4 FORMFACTOR INC, Form 10-K annual report for the period ended 2025-12-27 (10-K). U.S. Securities and Exchange Commission (filing by FORMFACTOR INC) · 20 February 2026 [A] sec.gov/Archives/edgar/data/1039399/000103939926000009/form-20251227.htm

5 Q2 2026 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. LSEG StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 July 2026 [A] investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2026-08/3e494f0c14dd0890f897a…

assembly, which is why the number of test steps keeps rising. ASE ran 6,797 testers in the second quarter of 2025 and 8,348 a year later [6].

Automated test equipment: share of the ~$9.0bn tester market, 2025 %



Source: Advantest FY2025 results briefing, April 2026

Who makes it

Advantest puts its own share at about 65 percent of a $9.0 billion tester market in 2025: 66 percent of test for logic chips, up ten points in a year, and about 60 percent of memory test [7]. It and Teradyne hold about 80 percent between them [1]. Teradyne grew 13 percent to $3.19 billion for 2025, with fourth-quarter revenue up 44 percent on AI demand [8].

Where the tester money went in calendar 2025 $BN



Source: Advantest FY2025 results briefing, April 2026

Below the two tester makers, the rest of the test and assembly equipment splits into niches, each with its own suppliers.

- **Probe cards.** FormFactor took $785.0 million in fiscal 2025 [9]; Technoprobe grew 15.7 percent to EUR 628.4 million, 38 percent of it from AI, and plans to double capacity by the end of 2027 [10].

6 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for the period ended 2026-07-30. U.S. Securities and Exchange Commission (filing by ASE Technology Holding Co., Ltd.) · 30 July 2026 [A] sec.gov/Archives/edgar/data/1122411/000095010326011351/dp250868_6k.htm

7 Advantest, presentation notes for the FY2025 (year ended 31 March 2026) results briefing, 27 April 2026. Advantest · 27 April 2026 [A] advantest.com/document/en/investors/ir-library/result/JE_BIZ_260427_note.pdf

1 Investors Guide April 25, 2025. Advantest · 13 May 2025 [A] advantest.com/document/en/investors/ir-library/investors-guide/Investors_Guide_2504E.pdf

8 Teradyne Reports Fourth Quarter and Full Year 2025 Results. Teradyne · 2 February 2026 [A] investors.teradyne.com/news-events/press-releases/detail/433/teradyne-reports-fourth-quart…

9 Revenue from Contract with Customer, Excluding Assessed Tax (us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax) reported by FormFactor, Inc., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for FormFactor, Inc.) [A] data.sec.gov/api/xbrl/companyconcept/CIK0001039399/us-gaap/RevenueFromContractWithCustomer…

10 Technoprobe, press release: the Board of Directors approves the draft statutory and consolidated annual report as at 31 December 2025. Technoprobe · 18 March 2026 [A] technoprobe.com/wp-content/uploads/2026/03/PR-FY-2025_.pdf

- **Handlers and burn-in**, the machines that feed parts into a tester and heat them. Cohu took $453.0 million in fiscal 2025 [11]; Aehr did $50.0 million in fiscal 2026 but booked a record $60.7 million in the fourth quarter alone and guides to $130 million to $150 million in fiscal 2027 [3].
- **Dicing and grinding.** Disco leads in wafer grinders and the saws that cut wafers into dies [12], and its newest saw handles pieces up to 400 by 400 mm, for packaging on rectangular panels [13].
- **Bonding.** Kulicke and Soffa did $654.1 million in fiscal 2025 [14], while Besi's second-quarter 2026 revenue rose 68.7 percent to EUR 249.9 million on hybrid bonding and data center demand [15].

Thermocompression bonders, which press memory dies into a stack and place chiplets on substrates, are the most concentrated niche. ASMPT took a repeat order for fifteen chip-to-substrate bonders in December 2025 [16] and, in its 2025 results, put that market at $1.6 billion by 2028, of which it means to hold 35 to 40 percent [17]. See Memory and HBM.

The back end's geography

Outsourced assembly and test, the OSAT business, is Taiwanese with a large Chinese challenger. ASE, the biggest, billed $20.6 billion in 2025 [18]; Georgetown's CSET counts four of the top ten OSATs by 2024 revenue as China-headquartered [19]. Malaysia ships about 13 percent of the world's packaged chips and wants 7 percent of advanced packaging shipments by 2035 [20].

ASE spent $2.7 billion on machinery and $1.4 billion on buildings in the first half of 2026, and says revenue from leading-edge advanced packaging is running

11 Revenue from Contract with Customer, Including Assessed Tax (us-gaap:RevenueFromContractWithCustomerIncludingAssessedTax) reported by COHU, INC., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for COHU, INC.) [A] data.sec.gov/api/xbrl/companyconcept/CIK0000021535/us-gaap/RevenueFromContractWithCustomer...

3 Aehr Test Systems Reports Fiscal 2026 Fourth Quarter and Full Year Financial Results with Record Quarterly Bookings and $100 Million Effective Backlog. Aehr Test Systems · 14 July 2026 [A] aehr.com/2026/07/aehr-test-systems-reports-fiscal-2026-fourth-quarter-and-full-year-financ...

12 DISCO Corporation, The World Leader In Semiconductor Capital Equipment For Cutting, Grinding, Polishing. SemiAnalysis · 19 July 2022 [B] newsletter.semianalysis.com/p/disco-corporation-the-world-leader

13 DISCO develops the DFD6080 fully automatic dicing saw for work sizes up to 400 mm square (最大400mm角のパッケージ切断に対応したダイシングソー「DFD6080」を開発). DISCO Corporation · 15 December 2025 [A] disco.co.jp/eg/news/corp/20251215_1.html

14 Revenues (us-gaap:Revenues) reported by KULICKE AND SOFFA INDUSTRIES, INC., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for KULICKE AND SOFFA INDUSTRIES, INC.) [A] data.sec.gov/api/xbrl/companyconcept/CIK0000056978/us-gaap/Revenues.json

15 BE Semiconductor Industries N.V. Announces Q2-26 and H1-26 Results. BE Semiconductor Industries (Besi) · 23 July 2026 [A] besi.com/investor-relations/press-releases/2025/details-1/be-semiconductor-industries-nv-a...

16 ASMPT Secures Additional Orders for Fifteen Chip-to-Substrate Thermo-Compression Bonding Tools Driven by AI Tailwind. ASMPT · 22 December 2025 [A] asmpt.com/en/investor-relations/news-events/asmpt-secures-additional-orders-for-fifteen-ch...

17 ASMPT Announces 2025 Annual Results AI-Driven Structural Growth Underpins Group Performance. ASMPT · 4 March 2026 [A] asmpt.com/en/investor-relations/news-events/asmpt-announces-2025-annual-results

18 Revenue (ifrs-full:Revenue) reported by ASE Technology Holding Co., Ltd., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for ASE Technology Holding Co., Ltd.) [A] data.sec.gov/api/xbrl/companyconcept/CIK0001122411/ifrs-full/Revenue.json

19 Inside Beijing's Chipmaking Offensive. Center for Security and Emerging Technology (CSET) · 14 July 2025 [A] cset.georgetown.edu/article/inside-beijings-chipmaking-offensive

20 Advanced Packaging: How Malaysia is Packaging the Future of AI. Malaysian Investment Development Authority · 1 July 2026 [A] mida.gov.my/advanced-packaging-how-malaysia-is-packaging-the-future-of-ai

ahead of its $3.5 billion guidance for the year [21]. Powertech aims to run the first panel-level packaging line for AI chips in 2027 [22].

Amkor took $6.71 billion of revenue in 2025 [23], and $1.90 billion in the second quarter of 2026 alone, up 26 percent year on year [24]. It has expanded its Arizona campus to a $7 billion investment [25]. Under a partnership signed in June 2026 [26] it will run packaging and test there for TSMC, whose Phoenix fabs sit next door [27].

Testers installed at ASE, the largest assembly and test contractor TESTERS



Source: ASE Technology Holding, second quarter 2026 earnings release (SEC Form 6-K)

The chokepoint

Unlike lithography, no test or assembly tool has a single supplier. Advantest argues that switching tester vendor means rebuilding the whole development, evaluation and production test environment, which is why share moves slowly [1].

Assembly and test is also the part of the chain where China is strongest, because assembly, dicing, bonding and test tools are not covered by the export controls on lithography and deposition. ASE and Amkor billed $27.3 billion between them in 2025, against industry sales of $791.7 billion [28], a small share of the money for a step every chip has to pass through.

Key evaluation criteria

- **Test coverage** is the share of hidden failures caught before they reach a package that cannot be taken apart.
- **Test time per part** sets throughput and cost, and large dies, memory stacks and chiplets all push it up.

21 ASE Technology Holding Co., Ltd., Form 6-K report of foreign private issuer for the period ended 2026-07-30. U.S. Securities and Exchange Commission (filing by ASE Technology Holding Co., Ltd.) · 30 July 2026 [A] sec.gov/Archives/edgar/data/1122411/000095010326011353/dp250875_6k.htm

22 Powertech eyes world's first panel-level packaging for AI chips in 2027. Nikkei Asia · 27 August 2026 [B] asia.nikkei.com/business/tech/semiconductors/powertech-eyes-world-s-first-panel-level-pack...

23 Revenue from Contract with Customer, Excluding Assessed Tax (us-gaap:RevenueFromContractWithCustomerExcludingAssessedTax) reported by AMKOR TECHNOLOGY, INC., XBRL company concept data. U.S. Securities and Exchange Commission (XBRL data for AMKOR TECHNOLOGY, INC.) [A] data.sec.gov/api/xbrl/companyconcept/CIK0001047127/us-gaap/RevenueFromContractWithCustomer...

24 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-07-27 (8-K). U.S. Securities and Exchange Commission (filing by AMKOR TECHNOLOGY, INC.) · 27 July 2026 [A] sec.gov/Archives/edgar/data/1047127/000104712726000043/amkr6302026erex-991.htm

25 Amkor Technology Breaks Ground on New Semiconductor Advanced Packaging and Test Campus in Arizona; Expands Investment to $7 Billion. Amkor Technology [A] ir.amkor.com/news-releases/news-release-details/amkor-technology-breaks-ground-new-semicon...

26 TSMC and Amkor Technology Announce Long Term Partnership to Accelerate Advanced Packaging in the United States. Amkor Technology [A] ir.amkor.com/news-releases/news-release-details/tsmc-and-amkor-technology-announce-long-te...

27 Amkor and TSMC to Expand Partnership and Collaborate on Advanced Packaging in Arizona. TSMC · 4 October 2024 [A] pr.tsmc.com/english/news/3174

1 Investors Guide April 25, 2025. Advantest · 13 May 2025 [A] advantest.com/document/en/investors/ir-library/investors-guide/Investors_Guide_2504E.pdf

28 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025 - Semiconductor Industry Association. Semiconductor Industry Association · 6 February 2026 [A] semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025

- **Probe card flatness and life** decide what wafer sort costs. Every needle must stay in the same plane across a full die and hold that over a long life.
- **Thermal control** matters more every generation. A kilowatt-class accelerator has to be held at temperature in the socket while it is tested at speed.
- **Capacity and geography** are the practical constraint. Taiwan, China and Malaysia hold most of the floor space, and new capacity takes years.

CHAPTER
14
STAGE 14 OF 15

Systems and Networking

A finished package is useless until it is bolted to a baseboard, fed a thousand amps and wired to seventy-one other packages. Many firms can do the assembly; few can make the lasers for the optical links.

IN PLAIN TERMS

One AI chip is not enough to train a model. So seventy-two of them are wired together in a rack, a cabinet the size of a wardrobe, and run as one computer. Each chip sits on its own board and draws more than a thousand amps, several times the current a whole house is wired to carry. The chips have to swap results with each other all the time, and a copper wire can carry those signals only a few meters before they fade, so the longer links send them as light down glass fiber. Many companies can build the rack. Only a few can make the lasers that send the light, and that is the narrow point.

IN SHORT

At the rack, the hard parts are power and light. Assembly is competitive and can move countries in a year. The control points are NVLink-class scale-up switching, which one firm still owns [3], and indium phosphide lasers, where Nvidia paid $4 billion for capacity it could not order [17] and 70 percent of the feedstock is Chinese [16].

3 NVLink & NVLink Switch for Advanced Multi-GPU Communication. Nvidia · 20 April 2026 A nvidia.com/en-us/data-center/nvlink

17 NVIDIA Announces Strategic Partnership With Lumentum to Develop State-of-the-Art Optics Technology. Nvidia · 2 March 2026 A nvidianews.nvidia.com/news/nvidia-announces-strategic-partnership-with-lumentum-to-develop...

16 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 A pubs.usgs.gov/periodicals/mcs2026/mcs2026-indium.pdf

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Rated on the two most concentrated parts: Nvidia alone sells NVLink, the link that joins the accelerators in a rack, and few firms make the lasers for the optical links.	Nvidia put $2B each into Coherent and Lumentum in March 2026 to buy indium phosphide laser capacity

WHO LEADS

- US **Nvidia** NVLink and NVSwitch for links inside a rack; InfiniBand and Spectrum-X Ethernet between racks
- US **Broadcom** Switch chips sold to any buyer (Tomahawk, Jericho) and most custom AI accelerators
- TW **Foxconn** Largest AI rack assembler; builds in Taiwan, Mexico, Texas
- CN **InnoLight** High-speed optical modules; plants in Suzhou, Taiwan and Thailand
- US **Coherent** Indium phosphide lasers and data center transceivers

WHERE IT IS MADE

- US **United States** Switch and accelerator silicon, indium phosphide lasers, system design
- TW **Taiwan** Contract design and rack assembly; high-layer-count circuit boards
- CN **China** Optical module assembly and test; most of the world's indium
- TH **Thailand** Chinese module makers' offshore transceiver plants
- MX **Mexico** Rack integration for the North American market

WHY SUBSTITUTION IS POSSIBLE

Rack assembly is easy to replace: if Foxconn stopped, Quanta and Wistron would take over the volume within a few quarters. The two concentrated parts are inside the rack, and a fix is under way for each. UALink is a published rival to NVLink, running 200 Gb/s per lane and joining up to 1,024 accelerators. For the lasers, Nvidia put $2 billion each into Coherent and Lumentum to secure supply that an ordinary order could not. Any newcomer with money would have to do the same, over two to five years.

WHERE CHINA STANDS

China holds two parts of the optical link that the United States does not. Chinese firms assemble the high-speed optical modules in volume, mostly in plants in Thailand and Suzhou. China also produces about 70 percent of the world's indium, the raw material for the lasers inside those modules.

WHERE THE US STANDS

American firms make the switch chips, the links that join accelerators inside a rack, and the lasers, but do almost none of the rack assembly. Nvidia's partners began building racks in Houston and Dallas in 2025. In July 2026 the Federal Communications Commission barred new approvals for imported equipment containing parts from firms on its Covered List.

A package computes nothing until it is built into a machine, and building the machine is a separate industry from making the package.

How it works

An accelerator ships as a module, the chip on a small board with its power supply and memory stacks. Nvidia calls its version SXM, and the open-standards equivalent is the Open Compute Project's accelerator module. Eight bolt onto a baseboard that carries their power and the switch chips, which pass data between any two modules at full speed.

Rack-scale systems drop the baseboard. An Nvidia GB300 NVL72 holds 18 compute trays plus nine switch trays, 72 GPUs in all, and draws up to 142 kW, more than ten times a conventional enterprise rack [1]; Rubin keeps that shape [2]. Inside the rack the trays reach each other over copper. Beyond it, every link runs on glass fiber with an optical transceiver at each end, where a laser turns the electrical signal into pulses of light and a detector at the far end turns the pulses back into current.

Every AI cluster runs two networks that are not interchangeable. Scale-up binds a few dozen accelerators tightly enough that they can read each other's memory, at terabytes per second; scale-out joins thousands of those groups at hundreds of gigabits per second. A byte is eight bits, so the inner network is tens of times faster than the outer one. A model too big for one accelerator gets split across the scale-up group, so that group's bandwidth sets the size of model that can be trained.

Variants and trade-offs

Scale-up: NVLink, UALink and Google's torus

The designs differ in how many accelerators can share one pool of memory.

- **NVLink and NVSwitch.** Nvidia's per-GPU bandwidth went from 900 GB/s in NVLink 4 to 1,800 in NVLink 5 and 3,600 in NVLink 6; the rack's own network moves 130 TB/s on Blackwell and 260 TB/s on Rubin [3].
- **UALink.** The open answer runs 200 Gb/s per lane, four lanes to a station for 800 Gb/s, and can address up to 1,024 endpoints in one group [4].
- **Google's 3D torus.** Ironwood, the seventh generation of Google's own TPU accelerator, wires 64 chips to a rack and uses optical switches to join cubes into superpods of 9,216 chips, trading latency for the ability to route around a failed rack in software [5].

1 System Hardware & Components. Nvidia [A] docs.nvidia.com/enterprise-reference-architectures/nvl72-ai-factory/latest/components.html

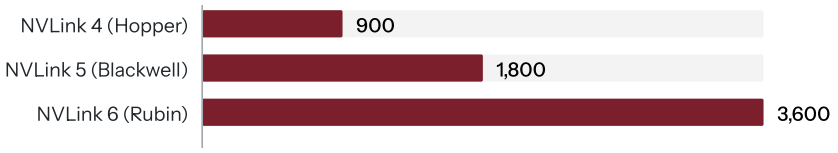
2 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for Vera Rubin. Nvidia · 13 October 2025 [A] blogs.nvidia.com/blog/gigawatt-ai-factories-ocp-vera-rubin

3 NVLink & NVLink Switch for Advanced Multi-GPU Communication. Nvidia · 20 April 2026 [A] nvidia.com/en-us/data-center/nvlink

4 UALink™ 200G 1.0 Specification Overview. UALink Consortium [A] ualinkconsortium.org/blog/ualink-200g-1-0-specification-overview-802

5 Inside the Ironwood TPU codesigned AI stack. Google Cloud · 6 November 2025 [A] cloud.google.com/blog/products/compute/inside-the-ironwood-tpu-codesigned-ai-stack

NVLink bandwidth per GPU by generation GB/S

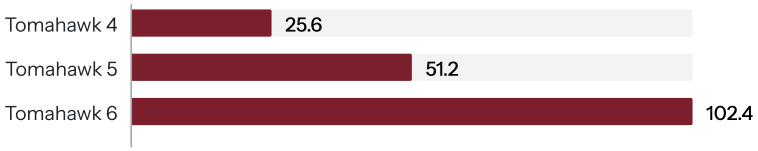


Source: NVIDIA NVLink product page, 2026

Scale-out: InfiniBand versus Ethernet

InfiniBand, the specialized network built for supercomputers, carried nearly all scale-out traffic when AI clusters were small and alike. Ethernet overtook it on raw capacity first: Broadcom doubled the bandwidth of the switch chips it sells to all comers every generation, from 25.6 Tbps on one chip in Tomahawk 4 [6] to 51.2 in Tomahawk 5 [7] and 102.4 in Tomahawk 6 [8]. Then the Ultra Ethernet Consortium published its 1.0 specification in June 2025, standardizing the load balancing, congestion control and lossless behavior InfiniBand had from the start [9].

Broadcom Tomahawk switching capacity on one chip TBPS



Source: Broadcom product releases for Tomahawk 4, 5 and 6

Nvidia sells both networks, telling investors that a $75.2 billion data center quarter came from Blackwell 300 plus demand for "InfiniBand, Spectrum-X Ethernet, and NVLink" [10], because large buyers run both networks.

Optics: pluggable, then co-packaged

Every link longer than a few meters is optical, and a cluster needs several transceivers, the modules that turn electricity into light and back, for each accelerator. Their power cost is why both switch vendors are moving the optics onto the switch package itself: Broadcom's Tomahawk 6 Davisson cuts optical interconnect power by about 70 percent, more than 3.5 times better than plug-in modules [11], and Nvidia's Spectrum-X Photonics reaches 512 ports of 800 gigabits during 2026 using four times fewer lasers [12].

6 Broadcom Ships Tomahawk 4, Industry's Highest Bandwidth Ethernet Switch Chip at 25.6 Terabits per Second. Broadcom [A] broadcom.com/company/news/product-releases/52756

7 Broadcom Ships Tomahawk 5, Industry's Highest Bandwidth Switch Chip to Accelerate AI/ML Workloads. Broadcom [A] investors.broadcom.com/news-releases/news-release-details/broadcom-ships-tomahawk-5-indust…

8 Broadcom Now Shipping World's First 102.4 Tbps Switch in Production Volume. Broadcom [A] broadcom.com/company/news/product-releases/64031

9 Ultra Ethernet Consortium (UEC) Launches Specification 1.0 Transforming Ethernet for AI and HPC at Scale - Ultra Ethernet Consortium. Ultra Ethernet Consortium · 11 June 2025 [A] ultraethernet.org/ultra-ethernet-consortium-uec-launches-specification-1-0-transforming-et…

10 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-04-26 (10-Q). U.S. Securities and Exchange Commission (filing by NVIDIA CORP) · 20 May 2026 [A] sec.gov/Archives/edgar/data/1045810/000104581026000052/nvda-20260426.htm

11 Broadcom Announces Tomahawk® 6 – Davisson, the Industry's First 102.4-Tbps Ethernet Switch with Co-Packaged Optics. Broadcom [A] broadcom.com/company/news/product-releases/63626

12 NVIDIA Announces Spectrum-X Photonics, Co-Packaged Optics Networking Switches to Scale AI Factories to Millions of GPUs. Nvidia · 18 March 2025 [A] nvidianews.nvidia.com/news/nvidia-spectrum-x-co-packaged-optics-networking-switches-ai-fac…

Who makes it

Optical modules are assembled where labor and land are cheap, and the Chinese makers went offshore early. InnoLight has run a Thai plant since 2019 alongside Suzhou and Taiwan [13], and Eoptolink assembles in Rayong and Chonburi [14]. The American rules now follow the firm, wherever its plants sit: the Federal Communications Commission's July 2026 order bars new authorizations for devices that contain logic chips from firms on its Covered List, and the record for that rule names optical transceivers as one such component [15].

The lasers inside those modules come from far fewer firms. Indium phosphide is the only practical material for the lasers in 800-gigabit and 1.6-terabit optics, and its feedstock is concentrated: China accounts for about 70 percent of world indium production [16]. Nvidia put $2 billion into Lumentum and $2 billion into Coherent in March 2026, buying research, future capacity and access rights [17] [18].

Custom accelerator and switch silicon is a US duopoly. Broadcom sells the Ethernet switch chips other firms build their switches around, and designs most of the custom accelerators, including Google's TPUs and Meta's own; its AI semiconductor revenue reached $16.7 billion in the third quarter of fiscal 2026, up 221 percent [19]. The other custom-silicon designer is Marvell, whose five-year agreement with AWS, announced in December 2024, covers custom AI chips alongside the optical and networking parts that go with them [20].

Physical assembly is done by Taiwanese firms. Foxconn booked NT$8.1 trillion of revenue in 2025, up 18 percent, with cloud and networking among the drivers [21]. Assembly is moving closer to the customer: Nvidia's partners commissioned more than a million square feet of US manufacturing space in 2025, Foxconn in Houston and Wistron in Dallas [22].

The chokepoint

Several capable contract manufacturers on three continents screw racks together, and the power shelves, copper busbars, cables and connectors have many suppliers. If Foxconn stopped tomorrow, Quanta and Wistron would absorb the volume in a couple of quarters.

13 InnoLight Technology, company overview page. InnoLight Technology [A] innolight.com/about

14 Eoptolink Technology, contacts page. Eoptolink Technology [A] eoptolink.com/about-us/contacts

15 Federal Communications Commission FCC-26-50. U.S. Federal Communications Commission [A] docs.fcc.gov/public/attachments/FCC-26-50A1.pdf

16 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 [A] pubs.usgs.gov/periodicals/mcs2026/mcs2026-indium.pdf

17 NVIDIA Announces Strategic Partnership With Lumentum to Develop State-of-the-Art Optics Technology. Nvidia · 2 March 2026 [A] nvidianews.nvidia.com/news/nvidia-announces-strategic-partnership-with-lumentum-to-develop...

18 NVIDIA and Coherent Announce Strategic Partnership to Develop Optics Technology to Scale Next-Generation Data Center Architecture. Nvidia · 2 March 2026 [A] nvidianews.nvidia.com/news/nvidia-and-coherent-announce-strategic-partnership-to-develop-o...

19 Broadcom Inc., Form 8-K current report for the period ended 2026-09-02 (8-K). U.S. Securities and Exchange Commission (filing by Broadcom Inc.) · 2 September 2026 [A] sec.gov/Archives/edgar/data/1730168/000173016826000076/avgo-08022026x8kxex99.htm

20 Marvell Technology, Inc., Form 8-K current report for the period ended 2024-12-02 (8-K). U.S. Securities and Exchange Commission (filing by Marvell Technology, Inc.) · 2 December 2024 [A] sec.gov/Archives/edgar/data/1835632/000183563224000193/final2024_12x02xmarvell-.htm

21 Hon Hai Technology Group (Foxconn) Announces FY2025 & 4Q25 Financial Results - Hon Hai Technology Group. Hon Hai Precision Industry (Foxconn) · 16 March 2026 [A] honhai.com/en-us/press-center/press-releases/latest-news/1978

22 NVIDIA to Manufacture American-Made AI Supercomputers in US for First Time. Nvidia · 14 April 2025 [A] blogs.nvidia.com/blog/nvidia-manufacture-american-made-ai-supercomputers-us

The parts that cannot be replaced that fast are the components that go into the rack.

- **Indium phosphide laser capacity.** The shortage is in material and fab space: Nvidia's answer was to buy into two laser makers [18], and the indium those lasers need is 70 percent Chinese [16].
- **NVSwitch.** Nobody else sells a scale-up switch of NVLink's bandwidth to all comers, which is what UALink exists to change [4].
- **High-layer-count circuit boards.** The boards, and the low-loss laminate they are built from, are concentrated in Taiwan and Japan; see Substrates and PCBs.

Optical module assembly is a separate case: concentrated, but not technically hard. Where the modules get built follows cost and policy, and it can change in a year or two.

Key evaluation criteria

- **Scale-up domain size** is how many accelerators share one pool of memory: up to 1,024 endpoints in the UALink spec [4] and 9,216 for an Ironwood superpod [5].
- **Bandwidth per accelerator** limits how far one model can be split across chips. NVLink 6 is 3,600 GB/s [3].
- **Picojoules per bit** is the energy cost of moving data. Optics dominate network power at 800 gigabits and above, and co-packaging is more than 3.5 times better than plug-in modules [11].
- **Serviceability** falls with co-packaged optics, which trade field-replaceable modules for a switch that goes back whole when a channel fails.
- **Supply concentration** is in the components: many firms assemble the modules; far fewer can make the lasers inside them.

18 NVIDIA and Coherent Announce Strategic Partnership to Develop Optics Technology to Scale Next-Generation Data Center Architecture. Nvidia · 2 March 2026 [A] nvidianews.nvidia.com/news/nvidia-and-coherent-announce-strategic-partnership-to-develop-o…

16 Mineral Commodity Summaries 2026. U.S. Geological Survey · 5 February 2026 [A] pubs.usgs.gov/periodicals/mcs2026/mcs2026-indium.pdf

4 UALink™ 200G 1.0 Specification Overview. UALink Consortium [A] ualinkconsortium.org/blog/ualink-200g-1-0-specification-overview-802

5 Inside the Ironwood TPU codesigned AI stack. Google Cloud · 6 November 2025 [A] cloud.google.com/blog/products/compute/inside-the-ironwood-tpu-codesigned-ai-stack

3 NVLink & NVLink Switch for Advanced Multi-GPU Communication. Nvidia · 20 April 2026 [A] nvidia.com/en-us/data-center/nvlink

11 Broadcom Announces Tomahawk® 6 – Davisson, the Industry's First 102.4-Tbps Ethernet Switch with Co-Packaged Optics. Broadcom [A] broadcom.com/company/news/product-releases/63626

CHAPTER
15
STAGE 15 OF 15

Data Centers and Power

The chips are no longer the slowest part of a data center buildout. A large transformer takes three years, turbine output does not reach 30 GW a year until 2030, and Texas has fifty times more large loads, mostly data centers, waiting to connect than it has approved to switch on.

IN PLAIN TERMS

A data center's job is to feed electricity to the machines inside it and carry away the heat they give off. One rack of AI chips, a cabinet the size of a wardrobe, draws 142 kilowatts, and every watt of it comes back out as heat, more than air can carry away, so water is piped through the rack instead. A large training site already holds a hundred thousand chips, and the newest sites are planned in gigawatts. One gigawatt is the whole output of a large power station, and OpenAI's planned sites add up to more than nine of them by 2029. Getting that much electricity to the door is harder than putting up the building, because the transformers and heavy switches that connect a site to the grid, and the turbines that make the power, come from a few suppliers and take years to arrive.

IN SHORT

The rack became the unit of AI compute at 142 kW and is heading for a megawatt by 2027 [4], which forces liquid cooling and an 800-volt bus on every large operator [3]. Nine vendors sell the cooling equipment, so buying it is a question of price [3]. The power equipment comes with queues: transformers take three years [1], six years of turbine output is already booked [9], and Texas has fifty times more large load in the queue than it has approved to switch on [13]. When an AI buildout slips, the delay is usually in the substation yard.

4 NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories. Nvidia · 20 May 2025 [A] developer.nvidia.com/blog/nvidia-800-v-hvdc-architecture-will-power-the-next-generation-of...

3 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for Vera Rubin. Nvidia · 13 October 2025 [A] blogs.nvidia.com/blog/gigawatt-ai-factories-ocp-vera-rubin

1 U.S. Department of Energy Large Power Transformer Resilience Report to Congress, July 2024. U.S. Department of Energy · 10 July 2024 [A] energy.gov/sites/default/files/2024-10/EXEC-2022-001242%20-%20Large%20Power%20Transformer%...

9 GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission (filing by GE Vernova Inc.) · 22 July 2026 [A] sec.gov/Archives/edgar/data/1996810/000199681026000147/gevpressrelease2q26.htm

13 Electric Reliability Council of Texas, ERCOT update to the Senate Committee on Business and Commerce, 29 July 2026. Electric Reliability Council of Texas · 29 July 2026 [A] ercot.com/files/docs/2026/07/29/ERCOT-Senate-July-29-Panel-1-Assessing-The-Grid.pdf

CHOKEPOINT CARD

HIGH CONCENTRATION

MODERATELY HARD TO SUBSTITUTE

CONCENTRATION	SUBSTITUTABILITY	PRICE OR MARKET SIZE
High	Moderate Several approved vendors sell every piece of power and cooling equipment, but a transformer takes three years to arrive and GE Vernova's gas turbines have orders for about six years ahead.	**Transformer lead times of 36 months, against under a year before 2020** GE Vernova's gas backlog and reserved slots grew from 100 GW to 116 GW in 2026

WHO LEADS

- US **GE Vernova** 116 GW of gas equipment backlog and reserved factory slots at mid-2026
- CH **Hitachi Energy** Large power transformers and high-voltage switchgear
- DE **Siemens Energy** Grid technologies, transformers, turbines
- US **Vertiv** Data center power and cooling systems; NVIDIA gigawatt AI factory partner
- FR **Schneider Electric** Electrical distribution and cooling; partner in NVIDIA's 800-volt DC designs

WHERE IT IS MADE

- US **United States** Most new AI capacity; ERCOT and PJM carry the load growth
- CH **Switzerland** Hitachi Energy and ABB transformer and switchgear engineering
- DE **Germany** Siemens Energy grid technologies and turbines
- TW **Taiwan** Delta Electronics power shelves, busbars and thermal modules
- JP **Japan** Hitachi group transformer manufacturing

WHY SUBSTITUTION IS POSSIBLE

Nothing in a data center's power and cooling has to be invented. Nine firms sell the power and cooling equipment for Nvidia's gigawatt data center designs, so a buyer can get the equipment and negotiate the price. Money cannot shorten the wait. A large power transformer now takes 36 months to arrive, against under a year before 2020. GE Vernova has close to six years of gas turbine orders waiting. New capacity is therefore two to five years away.

WHERE CHINA STANDS

China has spare generating capacity, a large domestic industry making transformers and switchgear, and can connect new sites quickly. Its AI buildout is limited by the supply of accelerators, and it has power to spare.

WHERE THE US STANDS

The United States has the demand and the money but cannot shorten the wait for equipment. Transformers and turbines both take years to arrive. Most large power transformers are imported, and no restarted nuclear reactor is running yet.

For thirty years the semiconductor industry set the pace and everything downstream followed. A wafer clears a leading-edge fab in months; a large power transformer takes three years, and before 2020 the same order took under one [1].

How it works

A rack is now the unit of AI compute. A GB300 NVL72 draws up to 142 kW through eight 33 kW power shelves, the rack's own power supplies, more than ten times a conventional enterprise rack in the same floor area [2].

Between the grid and the rack, a substation steps the high voltage down with transformers. Switchgear, the building's heavy circuit breakers, can cut any part of the site off in a fault. Battery-backed supplies hold the load for the seconds it takes backup generators to start. Distribution boards split the feed among the rows of racks. At the back of each rack a busbar, a solid copper bar doing a job a cable would melt at, carries the current down to the trays. Every one of those is heavy equipment with its own lead time.

Heat leaves in water. Cold plates on the GPU packages pass it to a building loop that releases it outdoors, and Nvidia's Vera Rubin racks run 45 °C supply water, warm enough that many sites can skip chillers [3].

Nvidia is also moving rack distribution to 800 volts of direct current. Raising the voltage cuts the current for the same power, and heat in copper comes from current, so the same conductor carries over 150 percent more power and the 200 kg busbars feeding a single rack are no longer needed [3].

Variants and trade-offs

Cooling

- **Air.** Heat exchangers in the rack door work at moderate densities but cannot carry away the heat of an AI rack.
- **Direct-to-chip liquid.** The default for anything Blackwell-class or later; every new AI hall is now plumbed for it.
- **Immersion.** Higher density still, but hard to service and barely used at scale.

Cooling is not the bottleneck. Nvidia names nine firms supplying data center power and cooling systems for its gigawatt reference designs, among them ABB, Eaton, Hitachi Energy, Mitsubishi Electric, Schneider Electric, Siemens and Vertiv [3].

1 U.S. Department of Energy Large Power Transformer Resilience Report to Congress, July 2024. U.S. Department of Energy · 10 July 2024 [A] energy.gov/sites/default/files/2024-10/EXEC-2022-001242%20-%20Large%20Power%20Transformer%...

2 System Hardware & Components. Nvidia [A] docs.nvidia.com/enterprise-reference-architectures/nvl72-ai-factory/latest/components.html

3 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for Vera Rubin. Nvidia · 13 October 2025 [A] blogs.nvidia.com/blog/gigawatt-ai-factories-ocp-vera-rubin

Rack power

Nvidia's 800-volt DC architecture is drawn for racks from 100 kW to more than a megawatt, with megawatt racks from 2027, because the 54-volt distribution in today's racks hits physical limits past about 200 kW [4].

A megawatt rack changes the busbar, the floor loading, the coolant flow, the leak detection and the stored energy that carries the rack through a dip. Nvidia's Rubin racks hold 20 times more stored energy, to steady the draw as tens of thousands of GPUs enter and leave a training step together [3].

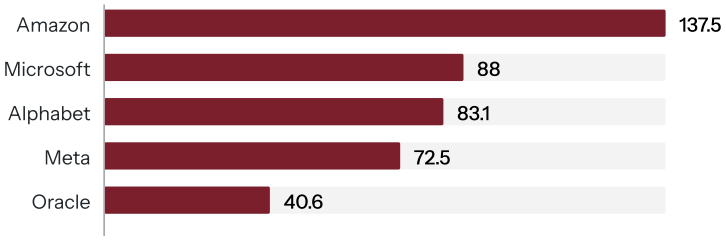
Where the electrons come from

- **Grid interconnection.** The slow path, and the one every announced campus is queued in.
- **Behind-the-meter gas**, generation on the site itself. The fast path, which is why campuses are sited next to gas.
- **Nuclear.** Almost entirely prospective. Constellation agreed to sell the output of the 835 MW Crane Clean Energy Center to Microsoft before restart work began [5]. The plant shut in 2019 and is restarting on a $1 billion federal loan closed in November 2025, still subject to Nuclear Regulatory Commission approval [6]. No restarted reactor is serving data centers yet.

Who makes it

The largest data center operators pay for the buildout. Alphabet, Amazon, Meta, Microsoft and Oracle spent $448.3 billion of capital in 2025, and on that trend reach $770 billion in 2026 [7].

Capital spending by the five largest US data center operators, 2025 $B



Source: Epoch AI, hyperscaler capex trend, February 2026, from company filings

Campuses are now described in gigawatts: OpenAI's Stargate portfolio targets more than 9 GW across US sites by 2029, with Abilene, Texas going from 0.3 GW to 1.2 GW by late 2026 [8].

4 NVIDIA 800 VDC Architecture Will Power the Next Generation of AI Factories. Nvidia · 20 May 2025 [A] developer.nvidia.com/blog/nvidia-800-v-hvdc-architecture-will-power-the-next-generation-of…

3 NVIDIA, Partners Drive Next-Gen Efficient Gigawatt AI Factories in Buildup for Vera Rubin. Nvidia · 13 October 2025 [A] blogs.nvidia.com/blog/gigawatt-ai-factories-ocp-vera-rubin

5 Constellation Ahead of Schedule for Launch of Crane Clean Energy Center. Constellation Energy · 19 February 2025 [A] constellationenergy.com/news/2025/constellation-ahead-of-schedule-for-launch-of-crane-clea…

6 Energy Department Closes Loan to Restart Nuclear Power Plant in Pennsylvania. U.S. Department of Energy · 17 November 2025 [A] energy.gov/articles/energy-department-closes-loan-restart-nuclear-power-plant-pennsylvania

7 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI · 26 February 2026 [A] epoch.ai/data-insights/hyperscaler-capex-trend

8 OpenAI Stargate: where the US sites stand. Epoch AI · 17 April 2026 [A] epoch.ai/publications/openai-stargate-where-the-us-sites-stand

The equipment vendors are the constraint. GE Vernova, Siemens Energy, Hitachi Energy, ABB, Eaton and Schneider Electric supply the turbines, transformers and switchgear. GE Vernova's gas equipment backlog and reserved factory slots went from 100 GW to 116 GW inside 2026, with data center orders past $5 billion by mid-year, more than double the whole of 2025 [9].

GE Vernova annual gas turbine output, actual and planned GW/YR

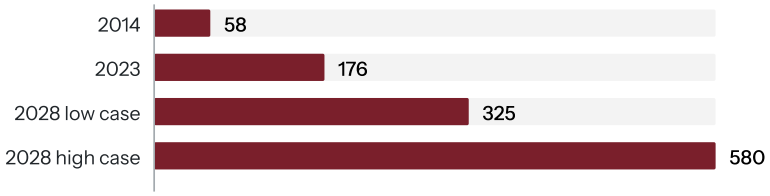


Source: GE Vernova second-quarter 2026 results, July 2026

Capacity and demand

US data centers used 176 TWh in 2023, 4.4 percent of national electricity, and Lawrence Berkeley National Laboratory projects 325 to 580 TWh by 2028, between 6.7 percent and 12 percent of the total [10]. The Energy Information Administration expects national electricity use to rise 1 percent in 2026 and 3 percent in 2027, the fourth straight annual increase and the strongest four-year run since 2000 [11].

US data center electricity use, actual and 2028 projection range TWH



Source: Lawrence Berkeley National Laboratory, December 2024

Chip efficiency keeps the low case plausible: Google's Ironwood TPU delivers twice the performance per watt of its predecessor [12]. The energy per token, where a token is the chunk of text a model reads or writes, falls while total energy rises, because the number of tokens rises faster.

The chokepoint

Announced capacity and energized capacity are a long way apart. ERCOT, the grid operator for most of Texas, was tracking about 474 GW of large loads seeking a

9 GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission (filing by GE Vernova Inc.) · 22 July 2026 [A] sec.gov/Archives/edgar/data/1996810/000199681026000147/gevpressrelease2q26.htm

10 Berkeley Lab Report Evaluates Increase in Electricity Demand from Data Centers. Lawrence Berkeley National Laboratory · 15 January 2025 [A] newscenter.lbl.gov/2025/01/15/berkeley-lab-report-evaluates-increase-in-electricity-demand...

11 EIA Press Release (01/13/2026): EIA forecasts strongest four-year growth in U.S. electricity demand since 2000, fueled by data centers. U.S. Energy Information Administration [A] eia.gov/pressroom/releases/press582.php

12 Inside the Ironwood TPU codesigned AI stack. Google Cloud · 6 November 2025 [A] cloud.google.com/blog/products/compute/inside-the-ironwood-tpu-codesigned-ai-stack

connection in June 2026, roughly 90 percent of it data centers [13]. Only 9,042 MW had been approved to switch on [14].

Behind that queue sit the equipment lead times.

- **Large power transformers.** Thirty-six-month lead times are commonly quoted and the worst reach 60, against under a year before the pandemic; 82 percent of the units put into US service in 2019 were imported [1].
- **Gas turbines.** GE Vernova will build 20 GW of turbines this year and is working toward 30 GW a year in 2030, against a backlog of 116 GW, close to six years of output [9].
- **Firm new generation**, the kind that runs whatever the weather. The one reactor restart with a data center buyer behind it, the 835 MW Crane plant, shut in 2019 and is still waiting on its license [6].

The chip chain moves faster: foundry and memory capacity have repeatedly expanded within a couple of years of the order. Money buys silicon capacity faster than it buys power capacity, and that difference decides how much of the 2026 capital spending turns into working compute.

Key evaluation criteria

- **Time to power** counts the months from signed lease to energized megawatt, and it is the number developers compete on.
- **Transformer and turbine lead time** covers the two queues that set the schedule, at three years [1] and six years of booked output [9].
- **Rack density supported** decides what an existing hall can take. An air-cooled one cannot host a 142 kW NVL72 without a full mechanical retrofit [2].
- **Firmness of supply** separates a signed interconnection agreement from a delivered transformer.
- **Energy per token** is the only efficiency measure that matters commercially. It keeps falling, but the number of tokens rises faster [12].

13 Electric Reliability Council of Texas, ERCOT update to the Senate Committee on Business and Commerce, 29 July 2026. Electric Reliability Council of Texas · 29 July 2026 [A] ercot.com/files/docs/2026/07/29/ERCOT-Senate-July-29-Panel-1-Assessing-The-Grid.pdf

14 Electric Reliability Council of Texas, Large Load Interconnection Status Update, 13 March 2026. Electric Reliability Council of Texas · 13 March 2026 [A] ercot.com/files/docs/2026/03/12/March-TAC-Report.pdf

1 U.S. Department of Energy Large Power Transformer Resilience Report to Congress, July 2024. U.S. Department of Energy · 10 July 2024 [A] energy.gov/sites/default/files/2024-10/EXEC-2022-001242%20-%20Large%20Power%20Transformer%...

9 GE Vernova Inc., Form 8-K current report for the period ended 2026-07-22 (8-K). U.S. Securities and Exchange Commission (filing by GE Vernova Inc.) · 22 July 2026 [A] sec.gov/Archives/edgar/data/1996810/000199681026000147/gevpressrelease2q26.htm

6 Energy Department Closes Loan to Restart Nuclear Power Plant in Pennsylvania. U.S. Department of Energy · 17 November 2025 [A] energy.gov/articles/energy-department-closes-loan-restart-nuclear-power-plant-pennsylvania

2 System Hardware & Components. Nvidia [A] docs.nvidia.com/enterprise-reference-architectures/nvl72-ai-factory/latest/components.html

12 Inside the Ironwood TPU codesigned AI stack. Google Cloud · 6 November 2025 [A] cloud.google.com/blog/products/compute/inside-the-ironwood-tpu-codesigned-ai-stack

CHAPTER 16
THE ECONOMICS

The Economics of AI Chips

Inside an AI accelerator the silicon is the cheap part. Memory and packaging are most of the cost.

IN PLAIN TERMS

A bill of materials is the list of parts inside a product and what each one costs to build. For an AI accelerator, the processor itself, the chip that does the calculating, is the cheap part, about a seventh of the total. The stacked memory beside it is nearly half, and the packaging that wires the two together costs more than the chip. Three firms in the world make that memory and one Taiwanese firm does that packaging, so the costliest parts are also the scarcest, and that is where a government's rules bite hardest.

IN SHORT

On a B200 the logic dies are 14 percent of the build, memory 45, and packaging with its scrap another 33 [1]. Nvidia's 75 percent gross margin is the most visible number in the chain [6], and a custom chip can undercut it by buying the same wafers and the same memory. The lines underneath are harder to cut: memory stacking yield, CoWoS interposer yield and the megawatts do not get cheaper as volume rises.

Building a Blackwell B200 costs about $6,400. Nvidia sells it for $30,000 to $40,000 [1]. That $6,400 splits into stages that are expensive because of physics, because capacity is short, or because one firm sets the price.

How it works

Three numbers set the cost of a die, the finished rectangle of circuitry cut out of a wafer.

- **Wafer price.** TSMC does not publish it. Epoch AI's model of Blackwell's custom TSMC 4NP process [2] assumes about $17,000 for a 300 mm wafer [1].
- **Dies per wafer.** A wafer is a circle and dies are rectangles, so the edge of the disc goes to waste: an 800 mm2 die, near the largest a scanner can print in one shot, fits 68 times.
- **Yield.** Defects land at random, so a bigger die is likelier to catch one. Epoch AI models the Blackwell die at 40 to 70 percent good, centered on 60 [1].

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI · 10 December 2025 [A] epoch.ai/data-insights/b200-cost-breakdown

6 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia · 26 August 2026 [A] nvidianews.nvidia.com/news/nvidia-announces-financial-results-for-second-quarter-fiscal-2027

2 NVIDIA Blackwell Architecture. Nvidia · 18 December 2025 [A] nvidia.com/en-us/data-center/technologies/blackwell-architecture

Those three numbers put the two dies inside a B200 at about $900, 14 percent of the build [1]. Doubling the wafer price moves the whole bill by a few hundred dollars.

Variants and trade-offs

Memory is the largest line

The 192 GB of high-bandwidth memory beside those dies, sold as HBM3E, is the single largest line: about $2,900, 45 percent of the build, at $14 to $17 per gigabyte [1]. Nvidia controls that line least of all: three firms make the memory, and they sell every stack they can build.

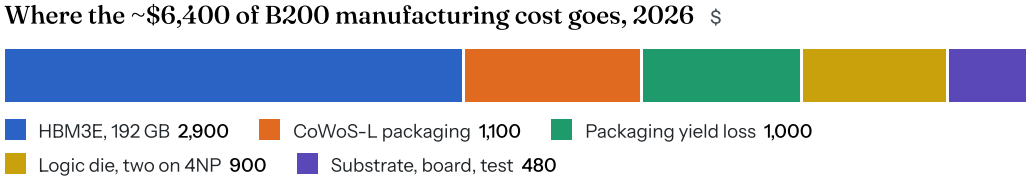
Packaging, substrate and test

CoWoS-L, the TSMC process that mounts dies and memory on one carrier and wires them together, adds about $1,100, more than the logic dies it carries [1]. Packaging yield runs 65 to 95 percent, and a package that fails after the memory goes on throws away the dies and the memory with it, adding roughly $1,000 to every B200 that ships [1]. Substrate, board, test and assembly add $480 [1]. See Advanced packaging for why packaging capacity runs out first.

The B200 bill of materials

LINE	ESTIMATE	RANGE
HBM3E, 192 GB in eight stacks	$2,900	$2,800-3,100
CoWoS-L packaging	$1,100	$1,000-1,200
Packaging yield loss	$1,000	$430-1,700
Logic die, two on TSMC 4NP	$900	$720-1,200
Substrate, board, test, other	$480	$370-600
Manufacturing cost	~$6,400	$5,700-7,300
Selling price	$30,000-40,000	

Epoch AI models each line as a range, deriving the packaging line from TSMC's advanced packaging revenue and Nvidia's share of CoWoS capacity [1]. Nothing of that quality exists for the H100 or Rubin.



Source: Epoch AI B200 cost model, from a $17,000 4NP wafer, 40–70% die yield, $14–17 per GB of memory, and a packaging line derived from TSMC's advanced packaging revenue

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI · 10 December 2025 A epoch.ai/data-insights/b200-cost-breakdown

From package to rack to cluster

At rack scale the number that matters is the cost of a GPU-hour. For an operator that owns the machine, SemiAnalysis puts a Vera Rubin NVL72 at $3.57, against $1.84 for a GB200 and $2.36 for a GB300 [3]. Each generation costs its owner more per hour than the last, and buyers pay because throughput rises faster than the rate.

One level up, a 100,000 H100 cluster costs over $4 billion in servers, draws about 150 MW, uses 1.59 TWh a year, pays $123.9 million for power and needs 98,304 optical transceivers [4].

Who captures the value

FIRM	STAGE	QUARTER ENDED	Gross margin
Micron	HBM and DRAM	May 2026	84.6%
Nvidia	Accelerator design	July 2026	75.0%
TSMC	Wafers and CoWoS	June 2026	67.7%
KLA	Inspection	June 2026	61.4%
ASML	Lithography	June 2026	54.0%
Lam Research	Etch	June 2026	51.7%
Applied Materials	Deposition	August 2026	50.3%
Ibiden	Substrates	June 2026	36.4%
ASE Technology	OSAT	June 2026	21.0%
Amkor	OSAT	June 2026	16.8%

Every figure is the firm's own reported gross margin [5] [6] [7] [8] [9] [10] [11] [12] [13] [14].

3 Vera Rubin NVL72 vs GB200 NVL72? Inference TCO & Architecture Analysis. SemiAnalysis · 23 July 2026 [B] inferencex.semianalysis.com/blog/vera-rubin-nvl72-vs-gb200-nvl72-inference

4 100,000 H100 Clusters: Power, Network Topology, Ethernet vs InfiniBand, Reliability, Failures, Checkpointing. SemiAnalysis · 17 June 2024 [B] newsletter.semianalysis.com/p/100000-h100-clusters-power-network

5 Micron Technology, Inc. Reports Record Results for the Third Quarter of Fiscal 2026. Micron Technology · 24 June 2026 [A] investors.micron.com/news/press-release/2026/Micron-Technology-Inc--Reports-Record-Results...

6 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia · 26 August 2026 [A] nvidianews.nvidia.com/news/nvidia-announces-financial-results-for-second-quarter-fiscal-2027

7 TSMC Reports Second Quarter EPS of NT$27.25. TSMC · 16 July 2026 [A] pr.tsmc.com/english/news/3326

8 KLA CORPORATION REPORTS FISCAL 2026 FOURTH QUARTER AND FULL YEAR RESULTS. KLA Corporation · 28 July 2026 [A] ir.kla.com/news-events/press-releases/detail/518/kla-corporation-reports-fiscal-2026-fourt...

9 ASML reports €9.3 billion total net sales and €2.9 billion net income in Q2 2026. ASML · 15 July 2026 [A] asml.com/en/news/press-releases/2026/q2-2026-financial-results

10 Lam Research Corporation Reports Financial Results for the Quarter Ended June 28, 2026. Lam Research · 29 July 2026 [A] newsroom.lamresearch.com/2026-07-29-Lam-Research-Corporation-Reports-Financial-Results-for...

11 Applied Materials Announces Third Quarter 2026 Results. Applied Materials · 13 August 2026 [A] ir.appliedmaterials.com/news-releases/news-release-details/applied-materials-announces-thi...

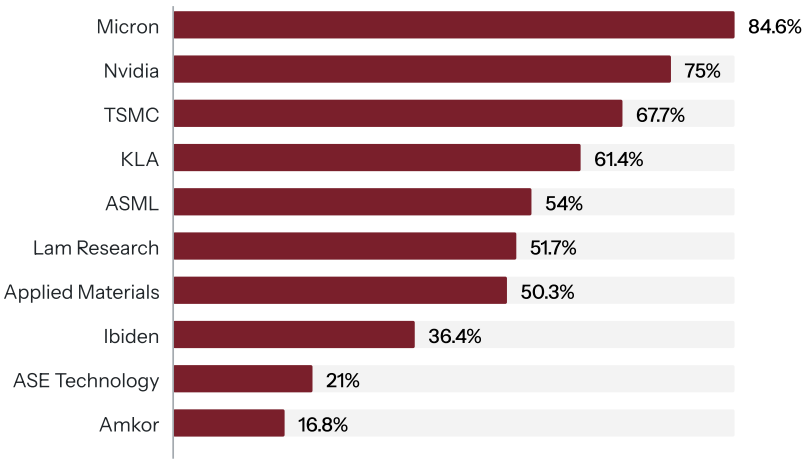
12 Note : This document has been translated from a part of the Japanese original for reference purposes only.. Ibiden · 3 August 2026 [A] ibiden.com/ir/items/tannshinn2026Q1.pdf

13 ASE Technology Holding Co., Ltd. Reports Its Unaudited Consolidated Financial Results for the Second Quarter of 2026. PR Newswire · 30 July 2026 [A] prnewswire.com/news-releases/ase-technology-holding-co-ltd-reports-its-unaudited-consolida...

14 AMKOR TECHNOLOGY, INC., Form 8-K current report for the period ended 2026-07-27 (8-K). U.S. Securities and Exchange Commission (filing by AMKOR TECHNOLOGY, INC.) · 27 July 2026 [A] sec.gov/Archives/edgar/data/0001047127/000104712726000043/amkr6302026erex-991.htm

SK hynix publishes no gross margin. Its operating margin hit 76 percent in the June quarter, on revenue up 257 percent in a year [15].

Gross margin, each firm's most recent reported quarter %



Source: Each firm's own quarterly earnings release; quarters end between May and August 2026 (TSMC shown)

Design, inspection and leading-edge wafers earn gross margins of 50 to 75 percent, substrates 36, assembly and test 17 to 21. In 2026 memory passed all of them: Micron earned 39.8 percent for the year to August 2025 [16] and 84.6 percent in the quarter to May 2026 [5]. The margin moved up the chain, to Nvidia's suppliers.

The two bodies that count this industry disagree. World Semiconductor Trade Statistics forecast $772 billion of global chip sales for 2025 [17]; the Semiconductor Industry Association counted $791.7 billion [18]. The forecasters put 2026 at $975 billion, with memory and logic each growing more than 30 percent [17]. Below those totals sit $135.1 billion of equipment billings [19] and $448.3 billion of 2025 capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle, on track for $770 billion in 2026 [20].

15 SK hynix Announces 2Q26 Financial Results. SK hynix [A] news.skhynix.com/en/q2-2026-business-results

16 Micron Technology, Inc. Reports Results for the Fourth Quarter and Full Year of Fiscal 2025. Micron Technology · 23 September 2025 [A] investors.micron.com/news/press-release/2025/Micron-Technology-Inc--Reports-Results-for-th...

5 Micron Technology, Inc. Reports Record Results for the Third Quarter of Fiscal 2026. Micron Technology · 24 June 2026 [A] investors.micron.com/news/press-release/2026/Micron-Technology-Inc--Reports-Record-Results...

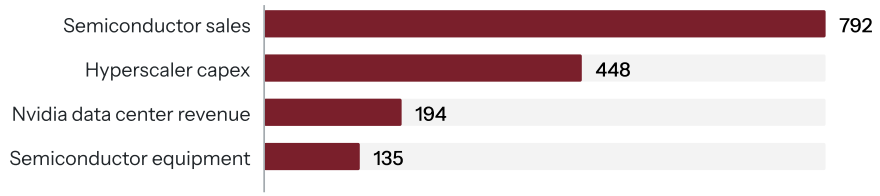
17 Global Semiconductor Market Approaches $1T in 2026. World Semiconductor Trade Statistics [A] wsts.org/76/103/Global-Semiconductor-Market-Approaches-1T-in-2026

18 Global Annual Semiconductor Sales Increase 25.6% to $791.7 Billion in 2025 - Semiconductor Industry Association. Semiconductor Industry Association · 6 February 2026 [A] semiconductors.org/global-annual-semiconductor-sales-increase-25-6-to-791-7-billion-in-2025

19 SEMI Reports Global Semiconductor Equipment Billings Reached $135 Billion in 2025, Up 15% Year-on-Year. SEMI · 7 April 2026 [A] semi.org/en/SEMI-Reports-Global-Semiconductor-Equipment-Billings-Reached-135-Billion-in-2025

20 Hyperscaler capex has quadrupled since GPT-4's release. Epoch AI · 26 February 2026 [A] epoch.ai/data-insights/hyperscaler-capex-trend

Size of each layer of the chain, 2025 $B



Source: Semiconductor Industry Association on chip sales; Epoch AI on capital spending at Alphabet, Amazon, Meta, Microsoft and Oracle; Nvidia FY2026 results; SEMI on equipment billings

Where the money goes

Per unit of compute, logic keeps getting cheaper and everything wrapped around it gets more expensive. Packaging and its scrap cost more than twice the logic dies [1], and a GB300 NVL72 rack already draws 142 kW [21].

STAGE	CLASS	WHY
EUV lithography	Structurally expensive	One supplier; price set by twenty years of optics development.
Masks and pellicles	Structurally expensive	Scales with layer count; every design pays again.
Metrology and inspection	Structurally expensive	Inspection time rises faster than throughput.
HBM	Structurally expensive	Stacking and through-silicon vias cut good die per wafer.
Advanced packaging, CoWoS-L	Structurally expensive	65 to 95 percent packaging yield; scrap destroys dies and memory.
Power and cooling plant	Structurally expensive	Priced by grid queues and turbine lead times.
Leading-edge logic wafers	Compressible with scale	Defect density and utilization improve through a ramp.
Design and EDA	Compressible with scale	Engineering cost is spread across volume.
Deposition and etch tools	Compressible with scale	Credible second sources for most deposition and etch steps.
ABF substrates	Compressible with scale	A laminate line, buildable at scale.
Optical transceivers	Compressible with scale	Copper and co-packaged optics cap the price.
Test and assembly (OSAT)	Already commoditized	17 to 21 percent margins; sold on price.
Wafers and bulk gases	Already commoditized	Multi-supplier, long-qualified, thin margins.
Rack integration (ODM)	Already commoditized	Low-teens margins on parts the customer specifies.

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI · 10 December 2025 [A] epoch.ai/data-insights/b200-cost-breakdown

21 System Hardware & Components. Nvidia [A] docs.nvidia.com/enterprise-reference-architectures/nvl72-ai-factory/latest/components.html

Key evaluation criteria

- **Cost per good die.** A 30 percent wafer price rise at constant yield is survivable. A ten-point yield drop on an 800 mm2 die is not.
- **Memory dollars per gigabyte, and gigabytes per part.** The largest line on the bill, set by three firms.
- **Packaging yield.** Every point lost throws away finished dies and the memory attached to them [1].
- **Share of the bill the vendor does not control.** Nvidia sets its own margin but buys memory, packaging and substrates at another firm's price.
- **Rack watts.** Past about 140 kW a rack, power and cooling capex grows faster than accelerator capex [21].
- **Compressibility class.** Whether the line falls when volume doubles. If it does not, capacity is the constraint.

1 NVIDIA's B200 costs around $6,400 to produce. Epoch AI · 10 December 2025 A epoch.ai/data-insights/b200-cost-breakdown

21 System Hardware & Components. Nvidia A docs.nvidia.com/enterprise-reference-architectures/nvl72-ai-factory/latest/components.html

CHAPTER 17
THE CONTEST

Geopolitics

Every stage in this chain sits in someone's jurisdiction. Since 2018 the United States has been converting that fact into policy, and China has been building substitutes.

IN PLAIN TERMS

An export control is a rule that makes a sale need the government's permission. The government names an item and a buyer, and from then on its own companies must hold a license before they ship that item to that buyer. The American government stretches the rule to cover goods made abroad with American technology. Few governments can use such a rule to much effect, because it works only when their own firms sell something the buyer cannot get elsewhere. The machines, software and chemicals that make an advanced chip come from a short list of firms in the United States, the Netherlands, Japan and South Korea. China can copy the simpler machines within a few years. It cannot yet copy the machine that prints the finest circuits, the chemistry that machine needs, or high-bandwidth memory, the stacked memory beside an AI chip.

IN SHORT

One or two countries hold a dozen points in the chain, and export controls are built on that. They have held China below the leading edge, where it still cannot get extreme-ultraviolet machines, their resist or high-bandwidth memory. They have not stopped substitution beneath it, where domestic tools now take a third of China's own market. The Netherlands and Japan paid for that, their toolmakers' China share of sales down from 41 to 33 percent at ASML and from 47 to 27 percent at Tokyo Electron. What began as a small yard is now tariffs, equity stakes and a cut of licensed sales, and none of it has moved leading-edge logic or packaging out of Taiwan.

3,223 WORDS / 14 MIN

Washington's control regime and Beijing's answer to it are both unfinished.

The shape of the contest

In 2022 Washington changed the goal from staying a generation ahead to "as large a lead as possible", sold as a small yard with a high fence [1]. The October rules that followed aimed to push China backwards [2]. Licensing once asked who was buying and why; country-wide prohibition asks only where the item is going.

Chokepoints and who holds them

SEGMENT	LEADER	SHARE	BASIS
EUV lithography	Netherlands	100%	revenue, 2025
DUV lithography	Netherlands	95%	revenue, 2025
Metrology and inspection	United States	72%	revenue, 2025
Deposition	United States	60%	revenue, 2025
Etch and clean	United States	53%	revenue, 2025
Photoresist	Japan	78.4%	revenue, 2023
Silicon wafers	Japan	53%	revenue, 2023
Leading-edge logic, 5 nm and below	Taiwan	72%	capacity, 2025, site estimate
Foundry, all nodes	Taiwan	78%	market share, 2025
HBM	South Korea	82.5%	revenue, 2025: SK hynix 63.2, Samsung 19.3, Micron 17.4
AI accelerator design	United States	93%	revenue, 2025
Mature logic, 28 nm and above	China	38%	capacity, 2025, site estimate

- **Equipment.** CSET's ETO Supply Chain Explorer, TechInsights 2025 revenue by parent-company headquarters [3].
- **Materials.** Country shares of photoresist and silicon wafers as Japan's Ministry of Economy, Trade and Industry published them in December 2025, from Fuji Keizai's materials survey, by parent-company headquarters [4].

1 Where the Chips Fall: U.S. Export Controls Under the Biden Administration from 2022 to 2024. Center for Strategic and International Studies · 12 December 2024 [A] csis.org/analysis/where-chips-fall-us-export-controls-under-biden-administration-2022-2024

2 Choking off China's Access to the Future of AI. Center for Strategic and International Studies · 11 October 2022 [A] csis.org/analysis/choking-chinas-access-future-ai

3 ETO Chip Explorer source data: provision.csv. Center for Security and Emerging Technology (CSET) / Emerging Technology Observatory [A] raw.githubusercontent.com/georgetown-cset/eto-chip-explorer/main/data/provision.csv

4 METI, Semiconductor and digital industry strategy: future direction (半導体・デジタル産業戦略の今後の方向性), 23 December 2025. Ministry of Economy, Trade and Industry (Japan) · 23 December 2025 [A] meti.go.jp/policy/mono_info_service/joho/conference/semicon_digital/0014/handeji14-4.pdf

Japan held 76.7 percent of photoresist in 2021 and 78.4 in 2023, and 52.4 percent of wafers against 53.0, so neither position is moving.

- **Taiwan.** The 78 percent is the Taiwan Semiconductor Industry Association's 2025 count, as the Stimson Center gives it [5]. The 72 percent is a Chip Supply Chain estimate for capacity at 5 nm and below, carried forward from the 2022 industry baseline because nobody publishes a current measurement [6]. The 92 percent quoted more often is a February 2024 US International Trade Commission figure, relayed by that same Stimson piece, on a wider reading of "most advanced" [5].
- **Memory.** SK hynix's US listing prospectus splits 2025 high-bandwidth memory revenue between SK hynix at 63.2 percent, Samsung at 19.3 and Micron at 17.4, so Korea holds 82.5; by the first quarter of 2026 Korea was down to 76.9 [7]. The underlying figures are IDC's, relayed by the prospectus.
- **Design.** CSIS [8].
- **China's mature nodes.** The 38 percent is a Chip Supply Chain reading of logic capacity at 28 nm and above. It sits between the 2022 industry measurement [6] and CSIS's finding that China now holds about half of global mature-node capacity, on a broader definition that includes discrete, analog and power chips [8].

Advanced packaging is missing because nobody publishes a capacity split and TSMC will not give one. No country is self-sufficient, so each side's controls can hurt the other.

The US export-control campaign, 2018 to today

Commerce built the regime one rule at a time, each closing a gap the last one left.

- **17 August 2020, the Foreign Direct Product Rule.** The Entity List, Commerce's roster of buyers that need a license, cut Huawei and 68 affiliates off from US technology in May 2019 [9], so Huawei bought foreign chips instead. Commerce then extended the rule to any foreign item made with US technology and bound for Huawei [10]. That reach makes American controls extraterritorial.

5 Why Taiwan Fears ‘America First’ Risks Eroding Its ‘Silicon Shield’. Stimson Center · 10 October 2025 [A] stimson.org/2025/why-taiwan-fears-america-first-risks-eroding-its-silicon-shield

6 R AJ VA RA DA RA JA N / I ACOB KOCH -W E SE R / CH RI S RI CH A RD / JOSE P H FI TZ GE RA L D /. Semiconductor Industry Association · 9 May 2024 [A] semiconductors.org/wp-content/uploads/2024/05/Report_Emerging-Resilience-in-the-Semiconduc…

7 SK hynix Inc., Form 424B4 prospectus for the period ended 2026-07-10 (424(B)(4)). U.S. Securities and Exchange Commission (filing by SK hynix Inc.) · 10 July 2026 [A] sec.gov/Archives/edgar/data/0002120882/000119312526299963/d32785d424b4.htm

8 China’s Localization Drive in Semiconductors Gains Impetus from Allied Chip Export Controls. Center for Strategic and International Studies · 24 March 2026 [A] csis.org/analysis/chinas-localization-drive-semiconductors-gains-impetus-allied-chip-expor…

9 Addition of Entities to the Entity List. Federal Register (Commerce Department; Industry and Security Bureau) · 21 May 2019 [A] federalregister.gov/documents/2019/05/21/2019-10616/addition-of-entities-to-the-entity-list

10 Addition of Huawei Non-U.S. Affiliates to the Entity List, the Removal of Temporary General License, and Amendments to General Prohibition Three (Foreign-Produced Direct Product Rule). Federal Register (Commerce Department; Industry and Security Bureau) · 20 August 2020 [A] federalregister.gov/documents/2020/08/20/2020-18213/addition-of-huawei-non-us-affiliates-t…

- **7 October 2022, the first country-wide rules.** Two new export classifications, 3A090 and 3B090, capped exportable AI chips and set fab thresholds at 16 nm FinFET logic, 18 nm half-pitch DRAM and 128-layer NAND [2]. They also barred US persons from servicing advanced Chinese fabs, emptying them of American engineers within days [11].
- **17 October 2023, the density fix.** Chip designers had tuned interconnect speed to fall under the 2022 threshold while keeping the compute, so Commerce's Bureau of Industry and Security (BIS) dropped that parameter for two others: total processing performance and performance density. A data-center chip now needs a license at 4,800 points of that performance score, or at 1,600 with a density of 5.92 [12].
- **2 December 2024, the widest rule.** 24 equipment types, three software tools, the first controls on high-bandwidth memory and 140 Entity List additions [13].
- **15 January 2025, AI Diffusion.** A rule requiring a license for the top band of accelerators, 3A090.a and 4A090.a, to any destination worldwide, then reopening it by exception: License Exception AIA to the nineteen destinations named in supplement no. 5 to part 740, License Exception LPP in capped volumes elsewhere, nothing to China [14]. BIS said on 13 May that it would not enforce the framework, two days before the compliance date [15]. It withdrew none of the text. The worldwide license requirement, License Exceptions AIA, ACM and LPP, supplement no. 5 and the 790 million TPP per-country allocation all stand in the Code today, and BIS says it enforces the requirement only against Country Groups D:1, D:4 and D:5 and against firms headquartered in D:5 or Macau [16] [17]. No tiered replacement has appeared; bilateral deals took the place of the caps.

On 9 April 2025 the government told Nvidia it needed a license to ship the H20 to China, to the D:5 group of arms-embargoed countries, and to any company headquartered or ultimately owned there, so no Chinese buyer could route the order through a foreign subsidiary [18]. Nvidia took a $4.5 billion charge on stock it could no longer sell. Licenses came back in August at a price: US officials, the filing says, "have expressed an expectation that the USG will receive 15 percent of

2 Choking off China's Access to the Future of AI. Center for Strategic and International Studies · 11 October 2022 [A] csis.org/analysis/choking-chinas-access-future-ai

11 Implementation of Additional Export Controls: Certain Advanced Computing and Semiconductor Manufacturing Items; Supercomputer and Semiconductor End Use; Entity List Modification. Federal Register (Commerce Department; Industry and Security Bureau) · 13 October 2022 [A] federalregister.gov/documents/2022/10/13/2022-21658/implementation-of-additional-export-co…

12 Explainer: The Commerce Department's October 2023 Export Control Update. Center for Security and Emerging Technology (CSET) · 4 December 2023 [A] cset.georgetown.edu/article/bis-2023-update-explainer

13 Commerce Strengthens Export Controls to Restrict China's Capability to Produce Advanced Semiconductors for Military Applications. U.S. Bureau of Industry and Security · 2 December 2024 [A] bis.gov/press-release/commerce-strengthens-export-controls-restrict-chinas-capability-prod…

14 Framework for Artificial Intelligence Diffusion. Federal Register (Commerce Department; Industry and Security Bureau) · 15 January 2025 [A] federalregister.gov/documents/2025/01/15/2025-00636/framework-for-artificial-intelligence-…

15 Department of Commerce Announces Rescission of Biden-Era Artificial Intelligence Diffusion Rule, Strengthens Chip-Related Export Controls. U.S. Bureau of Industry and Security · 13 May 2025 [A] bis.gov/press-release/department-commerce-announces-rescission-biden-era-artificial-intell…

16 Enhanced Favorable Treatment for the United Arab Emirates Under the Export Administration Regulations. Federal Register (Commerce Department; Industry and Security Bureau) · 14 July 2026 [A] federalregister.gov/documents/2026/07/14/2026-14132/enhanced-favorable-treatment-for-the-u…

17 15 CFR 742.6, Regional stability. Electronic Code of Federal Regulations [A] ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-742/section-742.6

18 NVIDIA CORP, Form 8-K current report for the period ended 2025-04-09 (8-K). U.S. Securities and Exchange Commission (filing by NVIDIA CORP) · 15 April 2025 [A] sec.gov/Archives/edgar/data/1045810/000104581025000082/nvda-20250409.htm

the revenue generated from licensed H20 sales", though no regulation codifies it [19].

Early 2026 set the current settlement.

- **13 January 2026, effective 15 January.** BIS announced on 13 January that it would move the H200 and AMD MI325X to case-by-case review, with third-party testing and a cap at 50 percent of what the exporter ships to US customers [20]. The rule, at 91 FR 1685, took effect on publication two days later [21].
- **15 January 2026.** A national-security proclamation under Section 232 put a 25 percent tariff on the same class of chips, then exempted almost every American use, from data centers to startups [22].
- **12 February 2026.** Applied Materials paid $252 million, the statutory maximum, over $126 million of ion implanters routed through Korea to an Entity-Listed Chinese customer [23].

Eight rule changes that shaped the regime

- 2019-05-16
  Huawei and 68 affiliates added to the Entity List
- 2022-10-07
  First country-wide controls on AI chips and fab tools
- 2023-10-17
  Performance density added; A800 and H800 captured
- 2024-12-02
  24 tool types, HBM controls, 140 entity listings
- 2025-01-15
  AI Diffusion rule published, three country tiers
- 2025-05-13
  AI Diffusion rescinded
- 2025-08-11
  US officials seek 15% of licensed H20 sales
- 2026-01-13
  H200 and MI325X case-by-case review announced, effective 15 January

Source: BIS press releases and Federal Register rules

19 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2025-07-27 (10-Q). U.S. Securities and Exchange Commission (filing by NVIDIA CORP) · 27 August 2025 [A] sec.gov/Archives/edgar/data/1045810/000104581025000209/nvda-20250727.htm

20 Department of Commerce Revises License Review Policy for Semiconductors Exported to China. U.S. Bureau of Industry and Security · 13 January 2026 [A] bis.gov/press-release/department-commerce-revises-license-review-policy-semiconductors-exp…

21 Revision to License Review Policy for Advanced Computing Commodities. Federal Register (Commerce Department; Industry and Security Bureau) · 15 January 2026 [A] federalregister.gov/documents/2026/01/15/2026-00789/revision-to-license-review-policy-for-…

22 Adjusting Imports of Semiconductors, Semiconductor Manufacturing Equipment, and Their Derivative Products Into the United States. Federal Register (Executive Office of the President) · 20 January 2026 [A] federalregister.gov/documents/2026/01/20/2026-01052/adjusting-imports-of-semiconductors-se…

23 Applied Materials to Pay $252 Million Penalty to BIS for Illegally Exporting Semiconductor Manufacturing Equipment. U.S. Bureau of Industry and Security · 11 February 2026 [A] bis.gov/press-release/applied-materials-pay-252-million-penalty-bis-illegally-exporting-se…

Cloud and remote access

Renting a chip is not exporting it. An export, at 15 CFR 734.13, is a shipment or transmission out of the United States or a release of technology or source code to a foreign person, so a Singapore data center selling time on an H200 already installed there moves nothing across a border [25].

The American rules reach that trade through the firms holding the chips. A January 2026 license for China or Macau is conditioned on the applicant getting the consignee's know-your-customer procedures and its list of Infrastructure-as-a-Service customers in Belarus, China, Cuba, Iran, Macau, North Korea, Russia and Venezuela, "necessary to prevent unauthorized remote access from end users described in paragraph (dd)(1)(iv)", and on a promise that no algorithm trained on the chips is served back to those users [26].

The diffusion rule addressed the other end, the output. It classified frontier model weights as ECCN 4E091, warned American cloud providers that training such a model for the US subsidiary of a foreign-headquartered customer raises a red flag that the weights will leave without a license, and barred its validated end users from training one abroad, while saying that API access to a model, and rented capacity for inference, "are not prohibited" [14].

So the controls reach the chip, frontier model weights, and the promises a licensed consignee signs. Compute sold by the hour on chips lawfully installed abroad sits outside them.

Allies: the Netherlands, Japan, Korea and Europe

Unilateral tool controls would only transfer the sales, so the campaign depends on The Hague and Tokyo.

25 **15 CFR 734.13 -- Export.**. Electronic Code of Federal Regulations A ecfr.gov/current/title-15/subtitle-B/chapter-VII/subchapter-C/part-734/section-734.13

26 **Federal Register, Volume 91 Issue 10 (Thursday, January 15, 2026).** U.S. Government Publishing Office A govinfo.gov/content/pkg/FR-2026-01-15/html/2026-00789.htm

14 **Framework for Artificial Intelligence Diffusion.** Federal Register (Commerce Department; Industry and Security Bureau) · 15 January 2025 A federalregister.gov/documents/2025/01/15/2025-00636/framework-for-artificial-intelligence-...

- **The Netherlands.** The Dutch began licensing advanced deep-ultraviolet tools on 1 September 2023, widened that in September 2024 [27] and reached measurement and inspection tools on 1 April 2025 [28].
- **Japan.** Controls on advanced manufacturing equipment took effect on 23 July 2023 and apply to all destinations and name no country [29].
- **South Korea.** BIS revoked the Validated End-User status that let Samsung, SK hynix and Intel run their China fabs without individual licenses, effective 31 December 2025. It will license those fabs to keep running, but not to expand or upgrade [30].

On 30 September 2025 a BIS rule extended Entity List restrictions to any subsidiary half-owned by a listed firm [31]. The same day the Dutch economy minister invoked the Goods Availability Act against Nexperia, the Nijmegen chipmaker that supplies European car lines, citing governance failures that risked moving technology out of Europe [32]. Neither lasted: BIS suspended its rule for a year on 10 November [33] and the minister suspended his order on 19 November [34]. Europe holds a chokepoint at the leading edge and none at the mature nodes its car industry runs on.

The MATCH Act, introduced on 8 April 2026, would ban sales of immersion deep-ultraviolet scanners to China, name CXMT, Hua Hong, Huawei, SMIC and YMTC in statute, and give allies 150 days to match before the US extends the Foreign Direct Product Rule alone [35].

27 The Netherlands expands export control measure for advanced semiconductor manufacturing equipment. Government of the Netherlands · 6 September 2024 A government.nl/latest/news/2024/09/06/the-netherlands-expands-export-control-measure-advanc…

28 Klever: export controls on advanced semiconductor manufacturing equipment to be tightened. Government of the Netherlands · 15 January 2025 A government.nl/latest/news/2025/01/15/klever-export-controls-on-advanced-semiconductor-manu…

29 CSIS Translation: Updated Japanese Export Controls on High-Performance Semiconductor Manufacturing Equipment. Center for Strategic and International Studies · 18 July 2023 A csis.org/analysis/csis-translation-updated-japanese-export-controls-high-performance-semic…

30 Revocation of Validated End-User Authorizations in the People's Republic of China. Federal Register (Commerce Department; Industry and Security Bureau) · 2 September 2025 A federalregister.gov/documents/2025/09/02/2025-16735/revocation-of-validated-end-user-autho…

31 Expansion of End-User Controls To Cover Affiliates of Certain Listed Entities. Federal Register (Commerce Department; Industry and Security Bureau) · 30 September 2025 A federalregister.gov/documents/2025/09/30/2025-19001/expansion-of-end-user-controls-to-cove…

32 Minister of Economic Affairs invokes Goods Availability Act. Government of the Netherlands · 12 October 2025 A government.nl/latest/news/2025/10/12/minister-of-economic-affairs-invokes-goods-availabili…

33 One Year Suspension of Expansion of End-User Controls for Affiliates of Certain Listed Entities. Federal Register (Commerce Department; Industry and Security Bureau) · 12 November 2025 A federalregister.gov/documents/2025/11/12/2025-19846/one-year-suspension-of-expansion-of-en…

34 Update on invoking Goods Availability Act. Government of the Netherlands · 19 November 2025 A government.nl/documents/2025/11/19/update-on-invoking-goods-availability-act

35 Risch, Ricketts, Kim Introduce MATCH Act; Level the Global Playing Field for U.S. Tech. U.S. Senate Committee on Foreign Relations · 8 April 2026 A foreign.senate.gov/press/rep/release/risch-ricketts-kim-introduce-match-act-level-the-glob…

China's response

Five bodies run Beijing's side. The Ministry of Industry and Information Technology (MIIT) sets substitution targets, and in 2024 reportedly told Chinese telecom operators to strip foreign chips out of their networks by 2027 [8]. The National Development and Reform Commission (NDRC) plans the compute build-out and co-issued the national computing-power network plan [36]. The Ministry of Commerce (MOFCOM) writes and administers the export controls, including the mineral rules below [37]. The State Administration for Market Regulation (SAMR) is the antitrust authority, and opened its Nvidia investigation in December 2024 [38]. The Cyberspace Administration of China (CAC) runs the security reviews, and in September 2025 told firms including ByteDance and Alibaba to cancel Nvidia orders [39].

The retaliation itself targets the inputs. Micron was first, failing a CAC review on 21 May 2023 and losing its critical infrastructure buyers [40]. Mineral controls followed.

- **3 December 2024.** Notice 46, from MOFCOM, banned exports of gallium, germanium, antimony and superhard materials to the United States, one day after the American chip rules [41].
- **4 April 2025.** Announcement 18 put seven medium and heavy rare earths under licensing, from samarium to yttrium, as metal, oxide, compound or finished magnet [37].
- **9 October 2025.** A further announcement reached foreign-made goods containing Chinese rare earths worth 0.1 percent or more of their value,

8 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip Export Controls. Center for Strategic and International Studies · 24 March 2026 A csis.org/analysis/chinas-localization-drive-semiconductors-gains-impetus-allied-chip-expor...

36 China accelerates building of national computing power network. The State Council of the People's Republic of China · 27 December 2023 A english.www.gov.cn/news/202312/27/content_WS658b72afc6d0868f4e8e28ba.html

37 Announcement No.18 of 2025 of The Ministry of Commerce and The General Administration of Customs of The People's Republic of China Announcing the decision to implement export control on some medium and heavy rare earth related items. Ministry of Commerce of the People's Republic of China A english.mofcom.gov.cn/Policies/AnnouncementsOrders/art/2025/art_0dd87cbee7b045bf93fabe6ab2...

38 Cyber Week in Review: December 13, 2024. Council on Foreign Relations · 13 December 2024 A cfr.org/articles/cyber-week-review-december-13-2024

39 Beijing blocks Nvidia sales in vote of confidence for Chinese chipmakers, the White House strikes a deal for 10 percent of Intel, and Commerce voids a $7.4B CHIPS Act grant. Center for Security and Emerging Technology (CSET) · 18 September 2025 A cset.georgetown.edu/newsletter/september-18-2025

40 Micron Aggression: The Right Response to Beijing's Ban on the U.S. Chipmaker. Center for Strategic and International Studies · 22 June 2023 A csis.org/analysis/micron-aggression-right-response-beijings-ban-us-chipmaker

41 Ministry of Commerce Notice 2024 No. 46: Notice Concerning Strengthening Controls on Exports of Relevant Dual-Use Items to the United States. Center for Security and Emerging Technology (CSET) · 3 December 2024 A cset.georgetown.edu/publication/china-rare-earth-export-ban

putting a Chinese export license between a foreign firm and its own product [42].

- **November 2025.** After Trump and Xi met at the Asia-Pacific summit, the ministry suspended the gallium ban for a year [43].

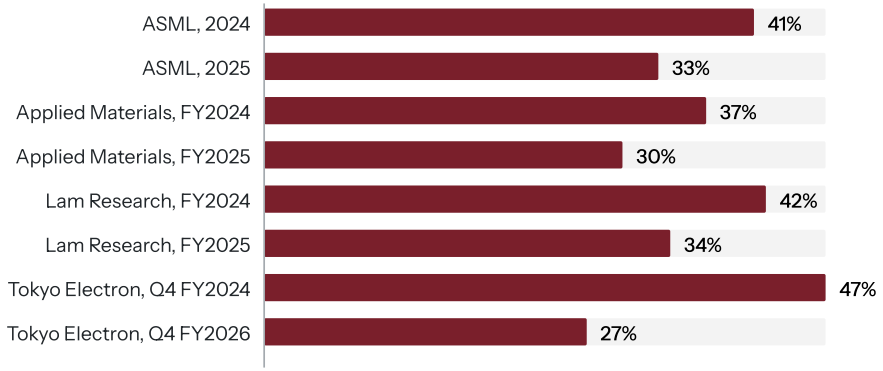
Nothing was repealed; the gallium ban is paused and the licensing regimes stand.

The blunter instrument is demand. Chinese regulators discouraged customers from buying, importing or using Nvidia's data-center products, and endorsed performance-per-watt and bandwidth standards for accelerators in Chinese data centers [44]. When Washington licensed H200 sales in February 2026, Beijing restricted them, and Nvidia's shipments came to less than 1 percent of data-center revenue [44]. That quarter's data-center revenue was $89.0 billion, and the next quarter's guidance assumed no China compute revenue [45].

Chinese tool vendors held 10 to 15 percent of their home market before 2022, took 25 percent in 2024 and 35 percent in 2025, and hold about 40 percent of domestic etch and deposition [8].

Substitution stalls at the leading edge. SMIC prints 7 nm-class logic on older deep-ultraviolet scanners, exposing each layer several times because it cannot buy an extreme-ultraviolet scanner, and Huawei shipped about 805,000 Ascend units in 2025 on that silicon. The binding constraint is memory. A stockpile of roughly 13 million memory stacks, most bought before the December 2024 rules took effect, supports about 1.6 million Ascend 910C packages; CXMT, China's own memory maker, should reach about 2 million stacks in 2026, enough for 250,000 to 300,000 more [46].

China's share of revenue at the four largest chip-tool makers, peak year and latest %



Source: ASML Q4 2025 investor presentation, Applied Materials and Lam Research 10-K filings, Tokyo Electron Q4 FY2026 results

42 Ministry of Commerce Notice 2025 No. 61: Announcement of the Decision to Implement Controls on Exports of Rare Earth-Related Items to Foreign Countries. Center for Security and Emerging Technology (CSET) · 9 October 2025 [A] cset.georgetown.edu/publication/mofcom-notice-2025-61

43 The U.S.-China Trade Truce Has Not Solved the Gallium Problem. Center for Strategic and International Studies · 11 May 2026 [A] csis.org/analysis/us-china-trade-truce-has-not-solved-gallium-problem

44 NVIDIA CORP, Form 10-Q quarterly report for the period ended 2026-07-26 (10-Q). U.S. Securities and Exchange Commission (filing by NVIDIA CORP) · 26 August 2026 [A] sec.gov/Archives/edgar/data/1045810/000104581026000075/nvda-20260726.htm

45 NVIDIA Announces Financial Results for Second Quarter Fiscal 2027. Nvidia · 26 August 2026 [A] nvidianews.nvidia.com/news/nvidia-announces-financial-results-for-second-quarter-fiscal-2027

8 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip Export Controls. Center for Strategic and International Studies · 24 March 2026 [A] csis.org/analysis/chinas-localization-drive-semiconductors-gains-impetus-allied-chip-expor...

46 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The Bottleneck. SemiAnalysis · 8 September 2025 [B] newsletter.semianalysis.com/p/huawei-ascend-production-ramp

Industrial policy: CHIPS, Rapidus, Big Fund and the rest

The CHIPS and Science Act authorized about $52.7 billion [8]. By July 2025 $30.9 billion of direct funding had reached 19 companies for 40 projects, a median 14.2 percent of each project's capital cost [48].

On 22 August 2025 the government converted $5.7 billion of unpaid Intel grants and $3.2 billion of Secure Enclave defense money into a 9.9 percent stake worth $8.9 billion [49].

Tariffs became the second instrument. The 2026 US-Taiwan trade agreement cuts the tariff on Taiwanese goods to 15 percent in exchange for at least $250 billion of Taiwanese investment in US chip production [50]. On 16 July 2026 TSMC added $100 billion for four more Arizona fabs at 2 nm and below, taking that campus to $265 billion [51], more than the whole CHIPS Act authorization.

Europe's first Chips Act produced no leading-edge fab, and the Chips Act 2.0 proposal adopted on 3 June 2026 widens state aid and shifts the emphasis from supply to demand without changing that [52]. Japan chose a national champion instead: Rapidus verified 2 nm gate-all-around transistors on its Chitose pilot line in 2025 and raised another 267.6 billion yen, about $1.7 billion, targeting mass production in 2027 [53].

China spends more, and differently. The latest round of its National Integrated Circuit Industry Investment Fund, the Big Fund, reached $47 billion [54]. CSIS puts cumulative state funding since 2014 at $150 billion, roughly triple the CHIPS authorization [8]. The money has produced a domestic tool industry and no leading-edge node.

8 China's Localization Drive in Semiconductors Gains Impetus from Allied Chip Export Controls. Center for Strategic and International Studies · 24 March 2026 A csis.org/analysis/chinas-localization-drive-semiconductors-gains-impetus-allied-chip-expor...

48 GAO-26-107882, SEMICONDUCTORS: Information on Projects Funded to Strengthen U.S. Supply Chain. U.S. Government Accountability Office A files.gao.gov/reports/GAO-26-107882/index.html

49 Intel and Trump Administration Reach Historic Agreement to Accelerate American Technology and Manufacturing Leadership. Intel · 22 August 2025 A intc.com/news-events/press-releases/detail/1748/intel-and-trump-administration-reach-histo...

50 U.S.-Taiwan Trade Agreement Leaves Major Questions Open. Council on Foreign Relations · 13 February 2026 A cfr.org/articles/u-s-taiwan-trade-agreement-leaves-major-questions-open

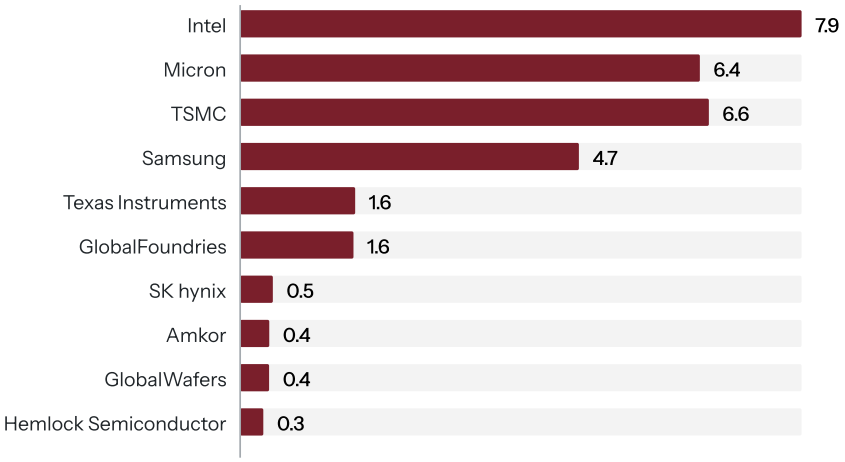
51 TSMC Announces Additional $100 Billion Investment In Arizona. City of Phoenix · 16 July 2026 A phoenix.gov/newsroom/ced-news/tsmc-announces-additional--100-billion-investment-in-arizona...

52 Proposal for the Chips Act 2.0. European Commission A digital-strategy.ec.europa.eu/en/library/proposal-chips-act-20

53 Rapidus Secures 267.6 Billion Yen in Funding from Japan Government and Private Sector Companies This strategic funding plan will enable Rapidus to steadily progress from its current R&D phase to mass production of 2nm logic semiconductors by 2027 - Information - Rapidus Corporation. Rapidus A rapidus.inc/en/news_topics/information/rapidus-secures-267-6-billion-yen-in-funding-from-j...

54 China's Evolving Industrial Policy for AI. RAND Corporation · 26 June 2025 A rand.org/pubs/perspectives/PEA4012-1.html

CHIPS Act direct funding awarded, by firm, as of July 2025 $B



Source: GAO-26-107882, Semiconductors: Information on Projects Funded to Strengthen U.S. Supply Chain

Taiwan

Taiwan is the concentration risk that no policy has reduced. Its foundries hold 78 percent of the world's contract chipmaking on the industry association's 2025 count, and 2 nm runs nowhere else [5]. At 5 nm and below the island holds an estimated 72 percent of capacity in 2025, a Chip Supply Chain estimate [6]. The packaging is on the island too. Asked in October 2025 how much CoWoS capacity he would add, TSMC's chairman said only that it was "working very hard to narrow the gap between the demand and supply", with wafer and packaging capacity both "very tight" [55].

The silicon shield argument holds that this concentration deters invasion: destroying TSMC would cost an aggressor more than the island is worth. The counterargument is that staying irreplaceable gives Taipei reason to slow the diversification Washington wants. Taiwan bars the overseas production of its most advanced chips, so what TSMC builds in the United States runs a generation behind what it builds at home, and nobody expects Arizona to reach 2 nm before 2028 [5].

Arizona narrows the gap in wafers, and TSMC has two advanced packaging fabs planned there [55]. But a fab without its Taiwanese mask shops, chemical suppliers and trained crews is not a substitute. The shield is real and weakening, and diversification is not keeping up.

5 Why Taiwan Fears 'America First' Risks Eroding Its 'Silicon Shield'. Stimson Center · 10 October 2025 [A] stimson.org/2025/why-taiwan-fears-america-first-risks-eroding-its-silicon-shield

6 R AJ VA RA DA RA JA N / I ACOB KOCH -W E SE R / CH RI S RI CH A RD / JOSE P H FI TZ GE RA L D /. Semiconductor Industry Association · 9 May 2024 [A] semiconductors.org/wp-content/uploads/2024/05/Report_Emerging-Resilience-in-the-Semiconduc…

55 Q3 2025 Taiwan Semiconductor Manufacturing Co Ltd Earnings Call (Chinese, English) — edited transcript. Refinitiv StreetEvents (transcript of a TSMC earnings call), via TSMC · 16 October 2025 [A] investor.tsmc.com/english/encrypt/files/encrypt_file/reports/2025-10/6860312f04fd291d0f26b…

Where the chips go: the Gulf, Southeast Asia, and smuggling

The Gulf has been let inside the controls. In November 2025 Commerce authorized G42 of the United Arab Emirates and HUMAIN of Saudi Arabia to buy up to 35,000 Nvidia GB300s each under security and reporting conditions [56]. On 10 July 2026 it moved the Emirates from Country Groups D:3 and D:4 into A:5, the tier that holds Washington's close allies, and named the Emirati government alongside G42 and the US hyperscalers as consignees who can take controlled AI chips and systems with no quantity cap. Licenses survive for the top-tier items, 3A090.a and 4A090.a, so the favorable treatment is tied to the named buyers [16]. Saudi Arabia still works license by license.

Southeast Asia is where chips get past the controls. Since 14 July 2025 Malaysia has required a permit to export, tranship or transit any high-performance US-origin AI chip, plus 30 days' notice under its Strategic Trade Act [57]. On 25 March 2026 the Justice Department charged three people over about $170 million of servers bought through Thai front companies for delivery to China [58].

The House select committee on China puts the median estimate at 140,000 chips smuggled to Chinese entities in 2024 [59]. Congress's answer, the Chip Security Act, would require location verification on controlled chips within 180 days of enactment [60]. The House Foreign Affairs Committee passed it 42-0 on 26 March 2026 [61]; it has not become law.

Who gains and who pays

- **Taiwan and Korea.** Nothing in the policy mix moves leading-edge logic, memory stacking or advanced packaging off an island and a peninsula before 2030; reshoring adds capacity elsewhere without subtracting it there.
- **The Netherlands and Japan.** Their tool makers build the controlled machines, so they gain a say in who gets what and lose the business that paid for it: Tokyo Electron's China share fell from 47.4 to 26.8 percent of quarterly sales in two years [62].
- **The United States.** It holds 93 percent of accelerator design revenue and will keep it. It has built no wafer, memory or assembly industry, and $30.9 billion of grants will not.

56 Statement on UAE and Saudi Chip Exports. U.S. Department of Commerce · November 2025 A commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports

16 Enhanced Favorable Treatment for the United Arab Emirates Under the Export Administration Regulations. Federal Register (Commerce Department; Industry and Security Bureau) · 14 July 2026 A federalregister.gov/documents/2026/07/14/2026-14132/enhanced-favorable-treatment-for-the-u…

57 EXPORT, TRANSSHIPMENT AND TRANSIT OF HIGH-PERFORMANCE AI CHIPS OF US ORIGIN. Ministry of Investment, Trade and Industry (Malaysia) · 14 July 2025 A miti.gov.my/miti/resources/Media%20Release/%5BFINAL%5D_MITI_Press_Stmt_Malaysia_Regulates_…

58 Chinese National and Two U.S. Citizens Charged with Conspiring to Smuggle Artificial Intelligence Technology to China. U.S. Department of Justice · 25 March 2026 A justice.gov/opa/pr/chinese-national-and-two-us-citizens-charged-conspiring-smuggle-artific…

59 Protecting U.S. Tech: China Committee and Bipartisan, Bicameral Leaders Unite to Stop CCP AI Chip Smuggling. U.S. House Select Committee on the Chinese Communist Party · 7 July 2025 A chinaselectcommittee.house.gov/media/press-releases/protecting-us-tech-china-committee-and…

60 S.1705 – Chip Security Act, 119th Congress (2025-2026), bill text. U.S. Congress (Congress.gov, Library of Congress) A congress.gov/bill/119th-congress/senate-bill/1705/text

61 UNANIMOUS: Bipartisan Huizenga Legislation to Curb AI Chip Smuggling Passes Committee. U.S. House of Representatives (Office of Rep. Bill Huizenga) · 26 March 2026 A huizenga.house.gov/news/documentsingle.aspx

62 Investor Relations / April 30, 2026. Tokyo Electron · 30 April 2026 A tel.com/ir/library/report/pjuomj00000000tf-att/fy26q4transcript-e.pdf

- **China.** Mature nodes, sub-leading-edge tools, DRAM and NAND are going domestic on a visible timetable. Extreme-ultraviolet machines, their resist and high-bandwidth memory at scale are not, and stockpile arithmetic caps Ascend output near a million units a year.

The Gulf gains without making anything: named Emirati buyers take advanced compute on terms European ones do not get. And Washington's instruments now reach past licensing: a 25 percent tariff on advanced chips [22], a 9.9 percent federal stake in Intel [49], an expected 15 percent cut of licensed H20 sales [19] and a case-by-case queue for the H200 [21].

What to watch

- **November 2026.** Whether China renews the one-year suspension of its gallium ban or lets it expire [43].
- **The MATCH Act's 150-day deadline.** If it passes, allied divergence on deep-ultraviolet tools gets a statutory deadline [35].
- **CXMT's memory ramp.** Domestic memory caps Chinese accelerator volume before logic wafers do [46].
- **SMIC's advanced-node ramp.** 45,000 wafers a month at end-2025, an estimated 60,000 in 2026 and 80,000 in 2027, every one of them multiply exposed because SMIC has no extreme-ultraviolet machine [46]. Yield is the open question.
- **Taiwan's overseas-production ban against its $250 billion investment promise.** The two contradict each other, and one will be dropped.

22 Adjusting Imports of Semiconductors, Semiconductor Manufacturing Equipment, and Their Derivative Products Into the United States. Federal Register (Executive Office of the President) · 20 January 2026 [A] federalregister.gov/documents/2026/01/20/2026-01052/adjusting-imports-of-semiconductors-se...

49 Intel and Trump Administration Reach Historic Agreement to Accelerate American Technology and Manufacturing Leadership. Intel · 22 August 2025 [A] intc.com/news-events/press-releases/detail/1748/intel-and-trump-administration-reach-histo...

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21 Revision to License Review Policy for Advanced Computing Commodities. Federal Register (Commerce Department; Industry and Security Bureau) · 15 January 2026 [A] federalregister.gov/documents/2026/01/15/2026-00789/revision-to-license-review-policy-for-...

43 The U.S.-China Trade Truce Has Not Solved the Gallium Problem. Center for Strategic and International Studies · 11 May 2026 [A] csis.org/analysis/us-china-trade-truce-has-not-solved-gallium-problem

35 Risch, Ricketts, Kim Introduce MATCH Act; Level the Global Playing Field for U.S. Tech. U.S. Senate Committee on Foreign Relations · 8 April 2026 [A] foreign.senate.gov/press/rep/release/risch-ricketts-kim-introduce-match-act-level-the-glob...

46 Huawei Ascend Production Ramp: Die Banks, TSMC Continued Production, HBM is The Bottleneck. SemiAnalysis · 8 September 2025 [B] newsletter.semianalysis.com/p/huawei-ascend-production-ramp

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Made by **Yashvardhan Sharma** with Claude. Inspired by Humanity's Last Machine and CSET's supply-chain work. Corrections and questions: contact@chipsupplychain.org.

UPDATED 21 SEPTEMBER 2026

A company filings, governments, standards bodies, universities and research institutions; B SemiAnalysis, Reuters, the Financial Times and Nikkei Asia.

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September 2026. https://chipsupplychain.org/ (accessed 2026-09-21).
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  note   = {Accessed 2026-09-21}
}
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